

Quantitative-Analytic Reports on Chess  
Berliner's games,  
5th World Ch. ICCF 1965

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# Chapter 1

## Abramson Moisey vs Berliner Hans J (USA)

### 1.1 Game Information

|                |                       |
|----------------|-----------------------|
| <b>Event</b>   | Ch World (corr.)      |
| <b>Site</b>    | corr.                 |
| <b>Date</b>    | 1965.??.??            |
| <b>Round</b>   | 5                     |
| <b>White</b>   | Abramson Moisey       |
| <b>Black</b>   | Berliner Hans J (USA) |
| <b>Result</b>  | 0-1                   |
| <b>Opening</b> | A41 – Unknown         |

### 1.2 Player Statistics

#### 1.2.1 Abramson Moisey (White)

- Total moves: 36
- Accuracy: 86.7%
- Average centipawn loss: 28.5
- Best/Excellent moves: 8 / 7
- Good moves: 11
- Inaccuracies: 4
- Mistakes: 5
- Blunders: 1

#### 1.2.2 Berliner Hans J (USA) (Black)

- Total moves: 36
- Accuracy: 94.7%

- Average centipawn loss: 10.8
- Best/Excellent moves: 16 / 6
- Good moves: 8
- Inaccuracies: 6
- Mistakes: 0
- Blunders: 0

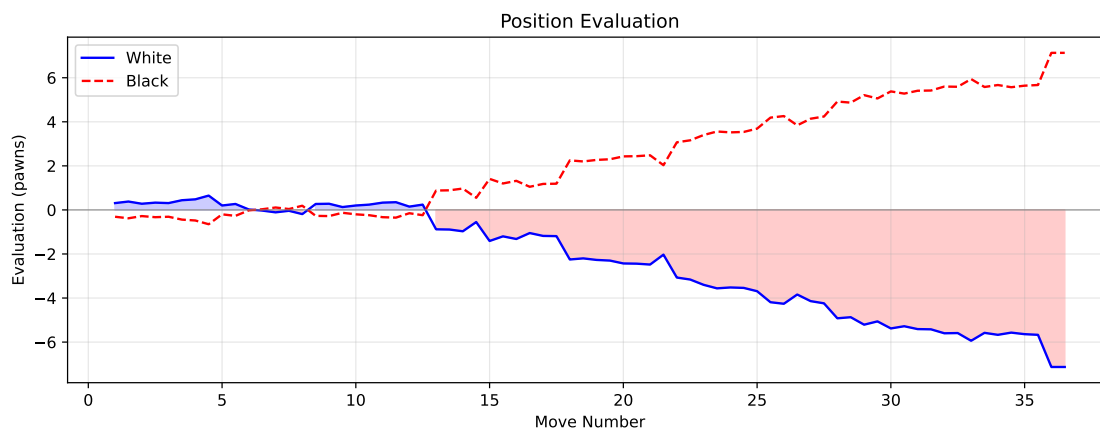
### 1.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

#### 1.3.1 Position Evaluation Over Time

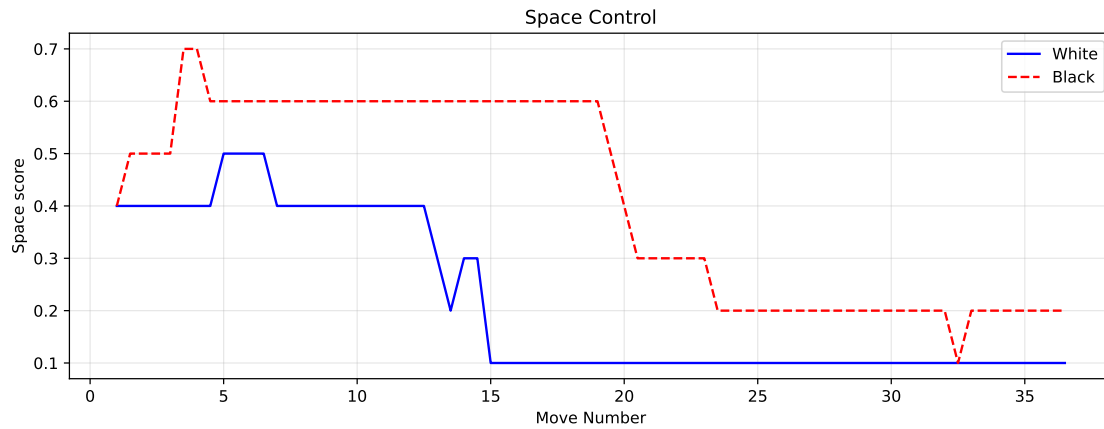
**Game Character Classification.** Evaluation range: minimum  $m_1 = -7.13$ , maximum  $m_2 = +0.65$ , spread  $d = m_2 - m_1 = 7.78$ .

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.



#### 1.3.2 Space Control

- White average: 0.22 (range: 0.10 to 0.50)
- Black average: 0.41 (range: 0.10 to 0.70)



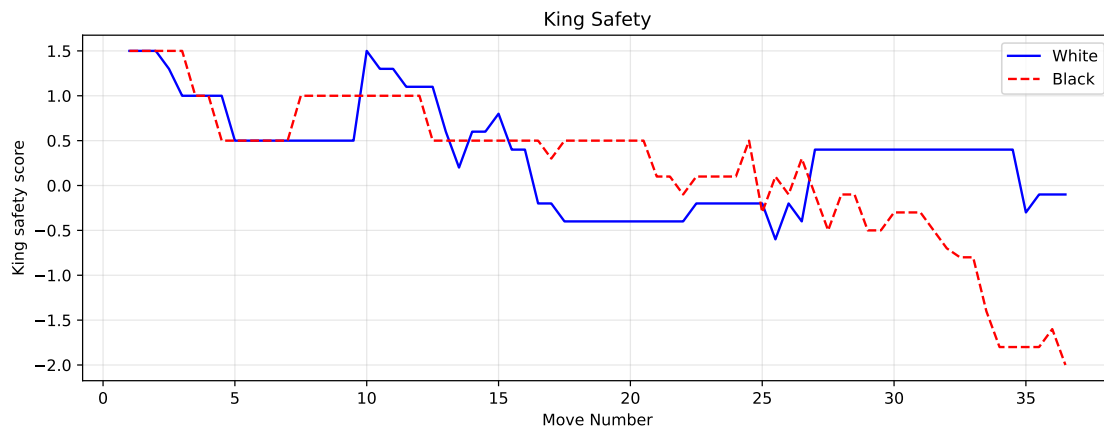
### 1.3.3 Mobility (Piece Activity)

- White average: 1.93 (range: 0.90 to 2.65)
- Black average: 1.84 (range: 1.00 to 2.60)



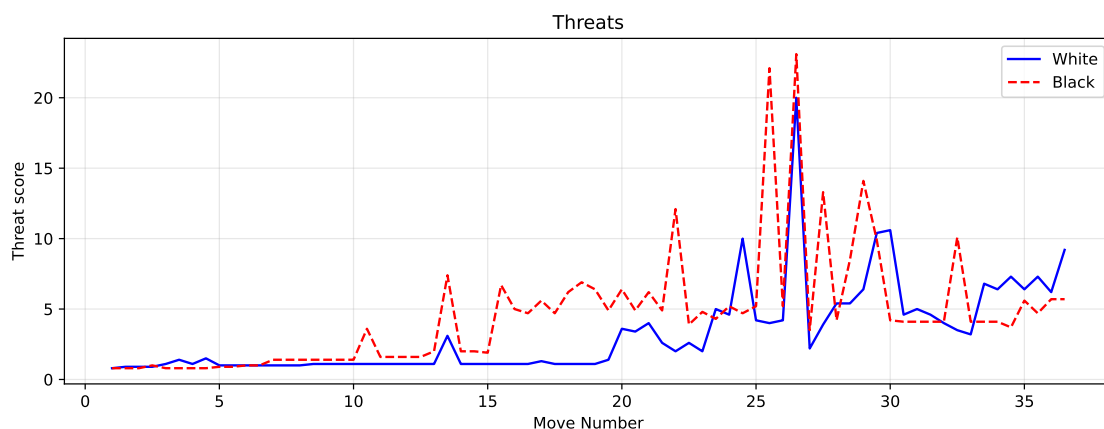
### 1.3.4 King Safety

- White average: 0.34
- Black average: 0.20



### 1.3.5 Threats

- White average: 3.26 (range: 0.80 to 20.00)
- Black average: 4.60 (range: 0.80 to 23.10)
- Average advantage: -1.35 (Black maintained more threats)



## 1.4 Annotated Game

1. Nf3 g6 2. c4?! Bg7 3. d4 d6 4. Nc3 e5 5. e3?! Nd7 6. a3 Ngf6 7. b4 O-O 8. Bb2 c6?! 9. Be2! Re8 10. O-O e4 11. Nd2! Nf8 12. Re1 h5 13. f4? *White loses 112cp. Consider instead: a4, d5, b5.*

13...exf3! 14. Bxf3! N8h7?! 15. e4? *White loses 87cp. Consider instead: d5, Qb3.*

15...Ng4 16. Nf1 Qh4?! 17. Qd2 Ng5 18. Re2? *White loses 119cp. Consider instead: Bxg4.*

18...Be6! 19. Qd3 c5 20. bxc5 dxc5! 21. Nd5! cxd4?! 22. Bxg4? *White loses 111cp. Consider instead: Bxd4.*

22...Qxg4! 23. Rae1 b5! 24. c5! Bxd5?! 25. exd5! Rxe2! 26. Qxe2 Qxe2?! 27. Rxe2! d3! 28. Rd2? *White loses 64cp. Consider instead: Rf2, Rd2.*

28...Ne4! 29. Bxg7?! Nxd2 30. Bd4 Ne4 31. c6 Kf8! 32. Be5 f6 33. Bf4?! Ke7 34. Ne3 d2! 35. Kf1 Nc3! 36. d6+?? *White loses 150cp. Consider instead: Bc7, g4, Kf2.*

36...Ke6!

0-1

## 1.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

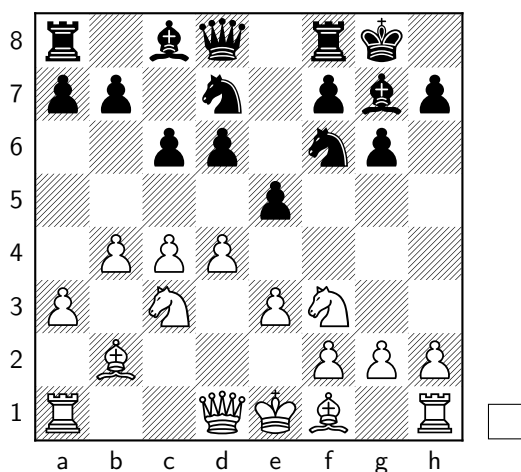
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 8...c6 ★

*Biggest evaluation swing(s) (46cp)*

Evaluation: +0.27



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.40  | 0.60  |
| Mobility (MG) | 4.60  | 4.00  |
| King Safety   | 0.50  | 1.00  |
| Threats       | 1.10  | 1.40  |

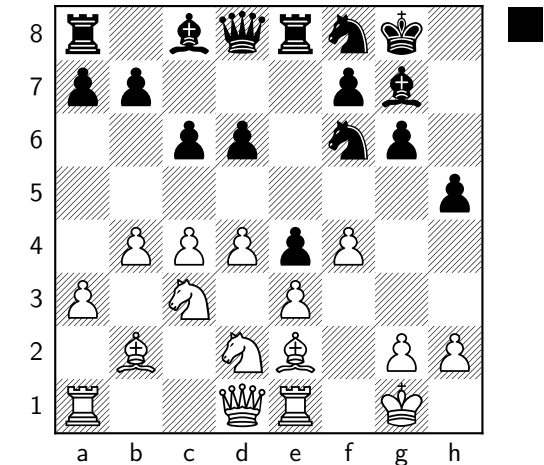
Instead of c6: exd4, Qe7, Re8

Best continuation: Be2 exd4 Nxd4

### Position 2: After 13. f4

*Critical moment*

Evaluation: -0.88



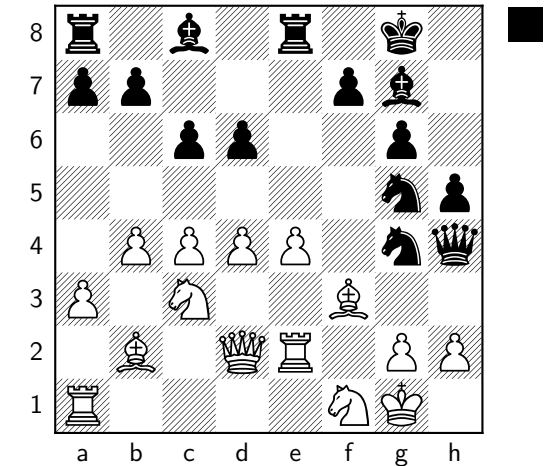
| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.60  |
| Mobility (MG) | 4.30  | 4.50  |
| King Safety   | 0.60  | 0.50  |
| Threats       | 1.10  | 2.00  |

Instead of f4: a4, d5, b5 Best continuation: exf3 Bxf3 a5

Position 3: After 18. Re2

Critical moment

Evaluation: -2.25

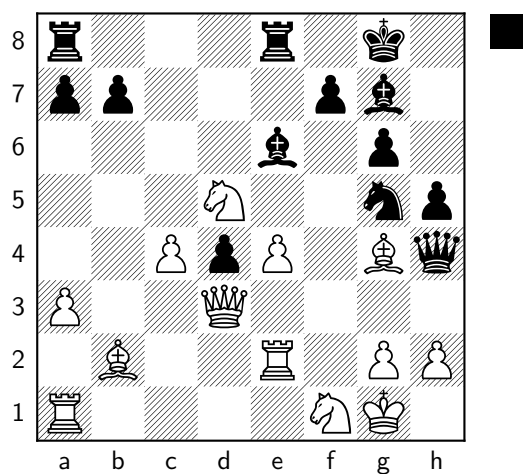


| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.10  | 0.60  |
| Mobility (MG) | 4.70  | 4.10  |
| King Safety   | -0.40 | 0.50  |
| Threats       | 1.10  | 6.20  |

Instead of Re2: Bxg4 Best continuation: Be6 Qd3 c5

**Position 4: After 22. Bxg4***Critical moment*

Evaluation: -3.07



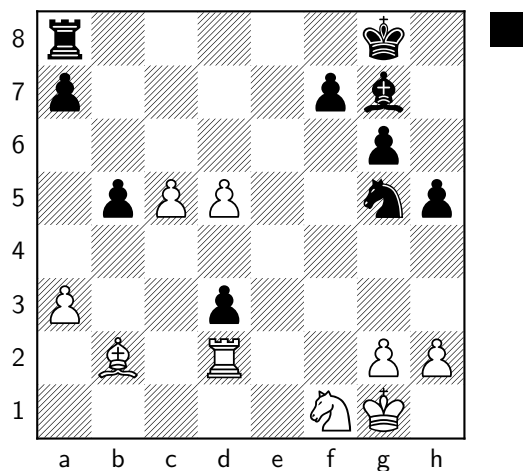
| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.10  | 0.30  |
| Mobility (MG) | 4.90  | 4.20  |
| King Safety   | -0.40 | -0.10 |
| Threats       | 2.00  | 12.10 |

Instead of Bxg4: Bxd4

Best continuation: Qxg4 Ree1 b5

**Position 5: After 28. Rd2 ★***Biggest evaluation swing(s) (68cp)*

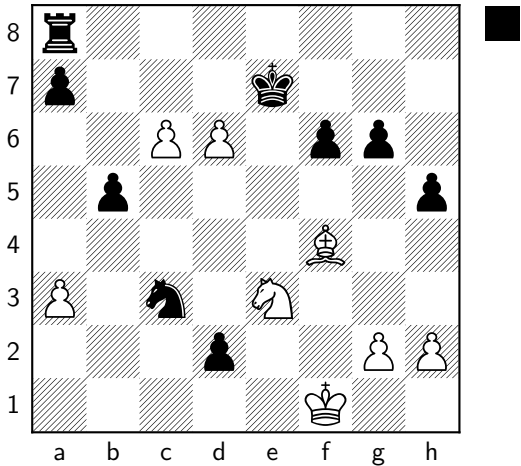
Evaluation: -4.92



| Metric          | White | Black                            |
|-----------------|-------|----------------------------------|
| Space           | 0.10  | 0.20                             |
| Mobility (MG)   | 2.30  | 2.60                             |
| King Safety     | 0.40  | -0.10                            |
| Threats         | 5.40  | 4.20                             |
| Instead of Rd2: | Rf2   | Best continuation: Ne4 Bxg7 Nxd2 |

Position 6: After 36. d6+

White blunders (-146cp)  
Evaluation: -7.13



| Metric          | White        | Black                        |
|-----------------|--------------|------------------------------|
| Space           | 0.10         | 0.20                         |
| Mobility (MG)   | 1.80         | 2.60                         |
| King Safety     | -0.10        | -1.60                        |
| Threats         | 6.20         | 5.70                         |
| Instead of d6+: | Bc7, g4, Kf2 | Best continuation: Ke6 g3 g5 |

1.6 Fireteam Index Prediction

Tracking: Berliner Hans J (USA) (Black)

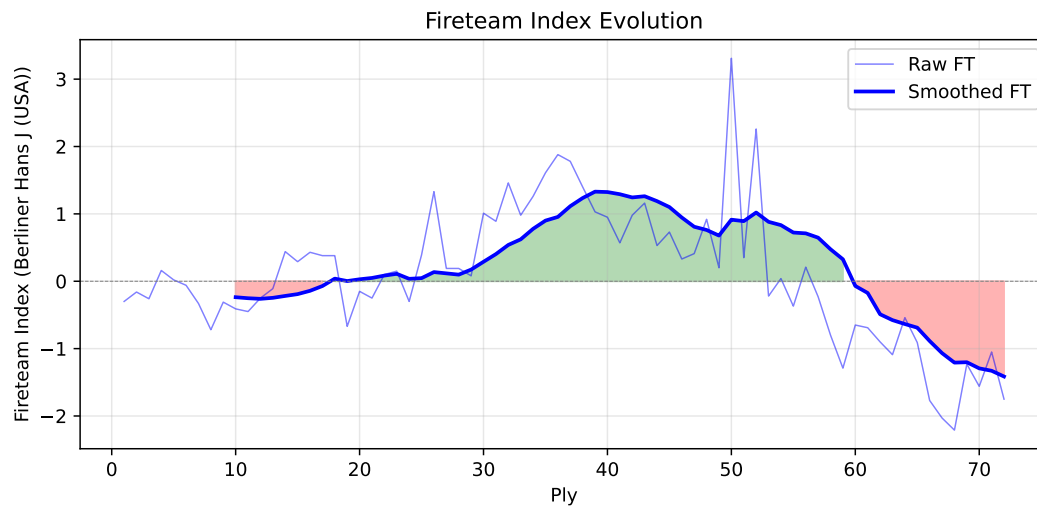
| FTI Variant | Algorithm | Prediction | Max Streak |
|-------------|-----------|------------|------------|
| Standard    | Per-ply   | WIN        | 28 plies   |
| Standard    | Windowed  | WIN        | 42 plies   |
| Truncated   | Per-ply   | WIN        | 29 plies   |
| Truncated   | Windowed  | WIN        | 50 plies   |

Actual result: 0-1 (Berliner Hans J (USA)'s outcome: WIN)

Standard: Per-ply ✓ Windowed ✓

Truncated: Per-ply ✓ Windowed ✓





## 1.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 710.4 seconds



## Chapter 2

# Roman Zinovievich Altshuler vs Hans Berliner

### 2.1 Game Information

|                |                                       |
|----------------|---------------------------------------|
| <b>Event</b>   | 5th Correspondence World Championship |
| <b>Site</b>    | ICCF                                  |
| <b>Date</b>    | 1965.07.01                            |
| <b>Round</b>   | 10                                    |
| <b>White</b>   | Roman Zinovievich Altshuler           |
| <b>Black</b>   | Hans Berliner                         |
| <b>Result</b>  | 0-1                                   |
| <b>Opening</b> | B04 – Unknown                         |

### 2.2 Player Statistics

#### 2.2.1 Roman Zinovievich Altshuler (White)

- Total moves: 25
- Accuracy: 87.0%
- Average centipawn loss: 28.0
- Best/Excellent moves: 7 / 3
- Good moves: 9
- Inaccuracies: 1
- Mistakes: 4
- Blunders: 1

#### 2.2.2 Hans Berliner (Black)

- Total moves: 25
- Accuracy: 93.0%

- Average centipawn loss: 14.5
- Best/Excellent moves: 11 / 3
- Good moves: 6
- Inaccuracies: 4
- Mistakes: 1
- Blunders: 0

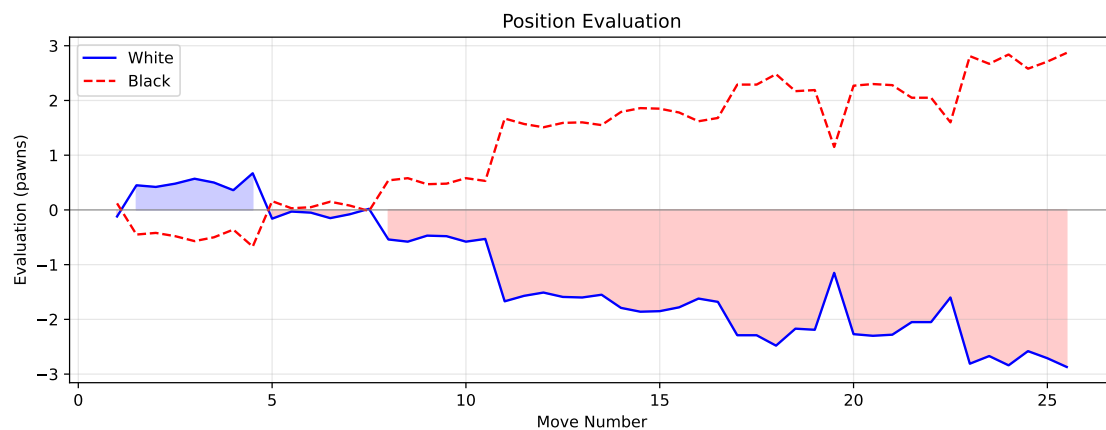
## 2.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 2.3.1 Position Evaluation Over Time

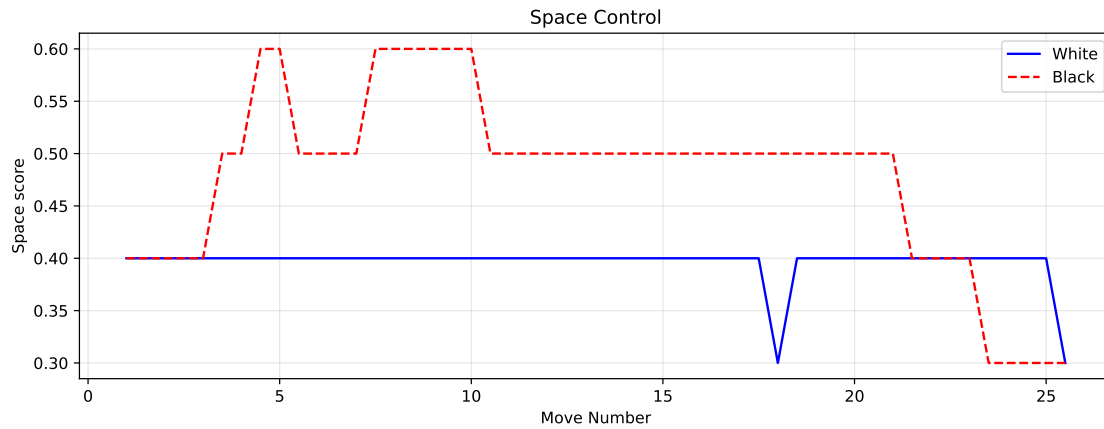
**Game Character Classification.** Evaluation range: minimum  $m_1 = -2.87$ , maximum  $m_2 = +0.67$ , spread  $d = m_2 - m_1 = 3.54$ .

This game is classified as **tactical** (normal). A tactical game with moderate swings.



### 2.3.2 Space Control

- White average: 0.40 (range: 0.30 to 0.40)
- Black average: 0.48 (range: 0.30 to 0.60)



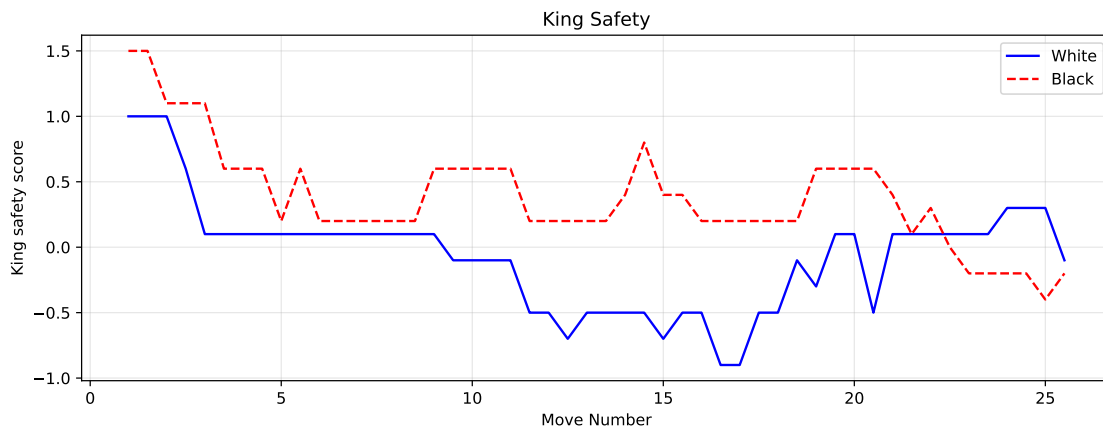
### 2.3.3 Mobility (Piece Activity)

- White average: 2.13 (range: 1.80 to 2.50)
- Black average: 2.33 (range: 1.60 to 3.00)



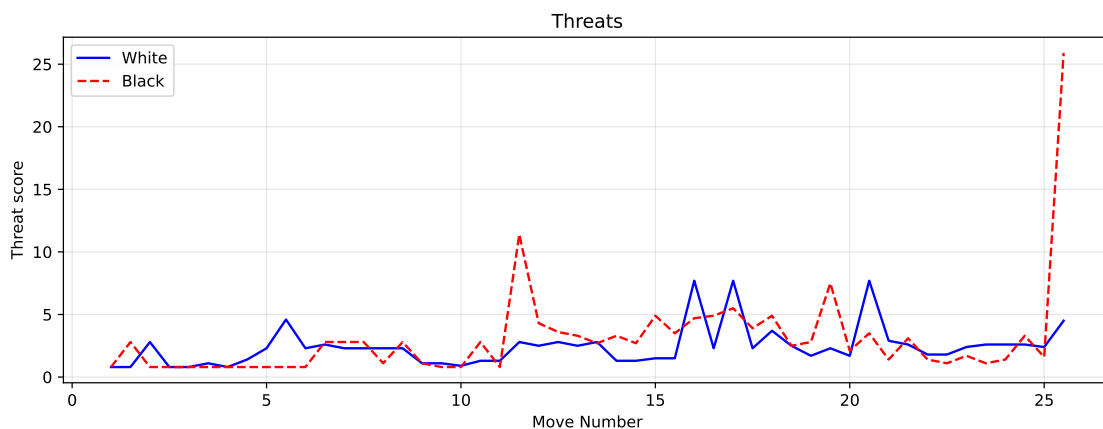
### 2.3.4 King Safety

- White average: -0.06
- Black average: 0.38



### 2.3.5 Threats

- White average: 2.42 (range: 0.80 to 7.70)
- Black average: 3.07 (range: 0.80 to 25.90)
- Average advantage: -0.65 (Black maintained more threats)



## 2.4 Annotated Game

1. e4 Nf6?! 2. e5 Nd5 3. d4 d6 4. Nf3 g6?! 5. Ng5? *White loses 80cp. Consider instead: Bc4, c4, h3.*

5...dxe5 6. dxe5 Bg7 7. Bc4! c6 8. Qe2?! h6! 9. Nf3! Bg4! 10. Nbd2 b5 11. Bb3? *White loses 116cp. Consider instead: Bxd5.*

11...Nf4! 12. Qf1! Qa5! 13. c3 Nd7! 14. Bc2 Nxe5! 15. Nd4! Qc7?! 16. Ne4! Nc4! 17. h3? *White loses 68cp. Consider instead: f3.*

17...Bd7! 18. g3 Ne6 19. Nf3 Ne5? *Black loses 111cp. Consider instead: Nd6, c5, O-O.*

20. Qe2? *White loses 109cp. Consider instead: Nxe5.*

20...Nxf3+! 21. Qxf3 f5 22. Nd2 O-O?! 23. Nb3?? *White loses 125cp. Consider instead: Qe2.*

23...c5! 24. O-O! c4 25. Nd4! Nxd4!

0-1

## 2.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

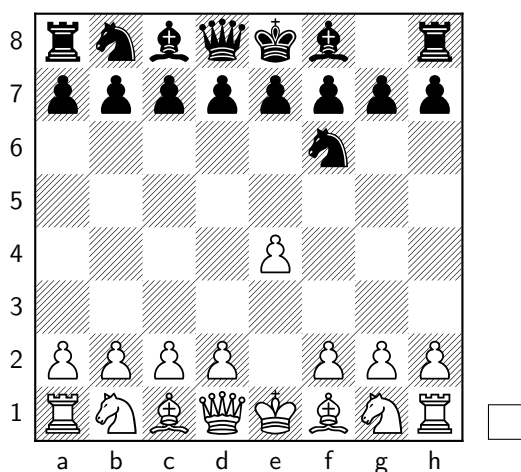
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 1...Nf6 ★

*Biggest evaluation swing(s) (57cp)*

Evaluation: +0.45



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.40  | 0.40  |
| Mobility (MG) | 3.90  | 3.80  |
| King Safety   | 1.00  | 1.50  |
| Threats       | 0.80  | 2.80  |

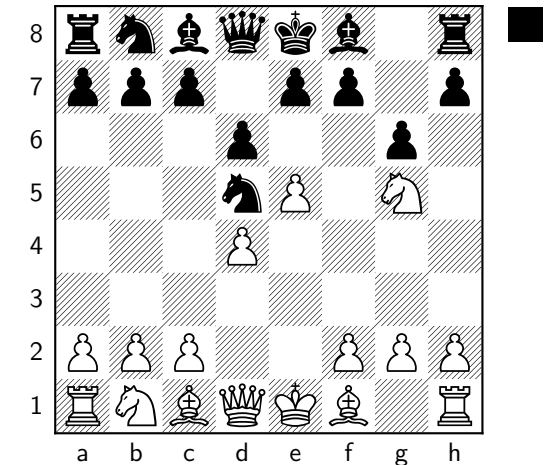
Instead of Nf6: e5, c5, c6

Best continuation: e5 Nd5 c4

### Position 2: After 5. Ng5 ★

*Biggest evaluation swing(s) (83cp)*

Evaluation: -0.16



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.40  | 0.60  |
| Mobility (MG) | 4.20  | 3.50  |
| King Safety   | 0.10  | 0.20  |
| Threats       | 2.30  | 0.80  |

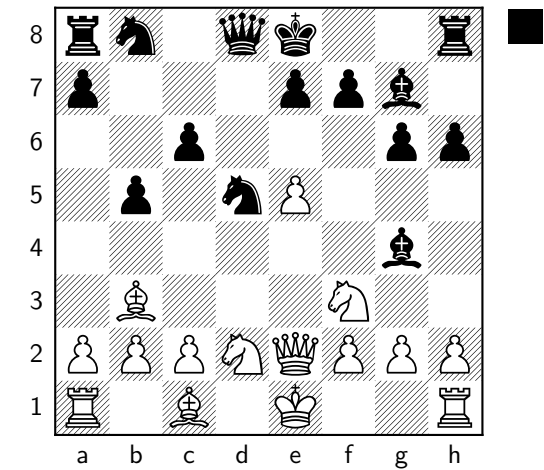
Instead of Ng5: Bc4, c4, h3

Best continuation: dxe5 dxe5 Bg7

Position 3: After 11. Bb3

Critical moment

Evaluation: -1.67



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.40  | 0.50  |
| Mobility (MG) | 4.30  | 4.30  |
| King Safety   | -0.10 | 0.60  |
| Threats       | 1.30  | 0.80  |

Instead of Bb3: Bxd5

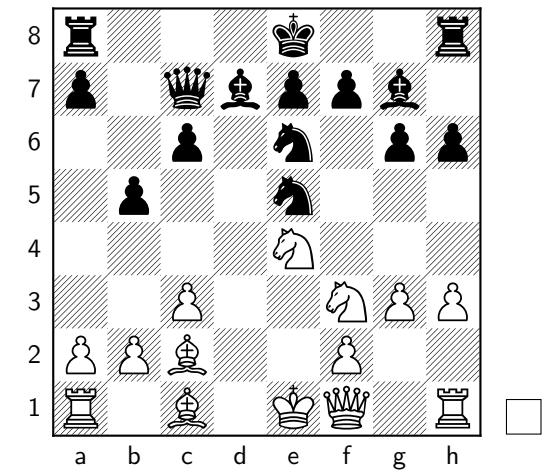
Best continuation: Nf4 Qf1 Qa5



Position 4: After 19...Ne5

Critical moment

Evaluation: -1.15



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.40  | 0.50  |
| Mobility (MG) | 4.70  | 5.50  |
| King Safety   | 0.10  | 0.60  |
| Threats       | 2.30  | 7.50  |

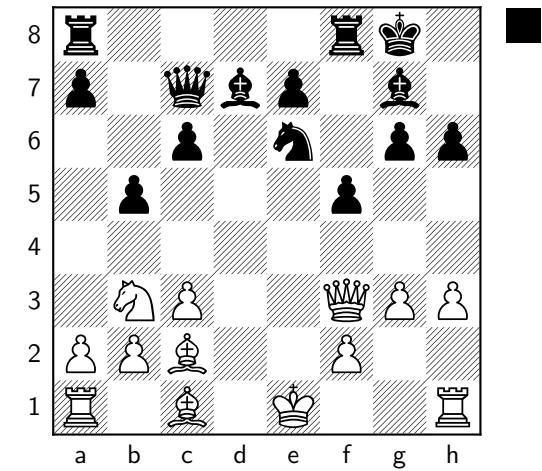
Instead of Ne5: Nd6, c5, O-O

Best continuation: Nxe5 Qxe5 Be3

Position 5: After 23. Nb3

White blunders (-121cp)

Evaluation: -2.81



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.40  | 0.40  |
| Mobility (MG) | 4.30  | 5.20  |
| King Safety   | 0.10  | -0.20 |
| Threats       | 2.40  | 1.70  |

Instead of Nb3: Qe2

Best continuation: c5 O-O c4

## 2.6 Fireteam Index Prediction

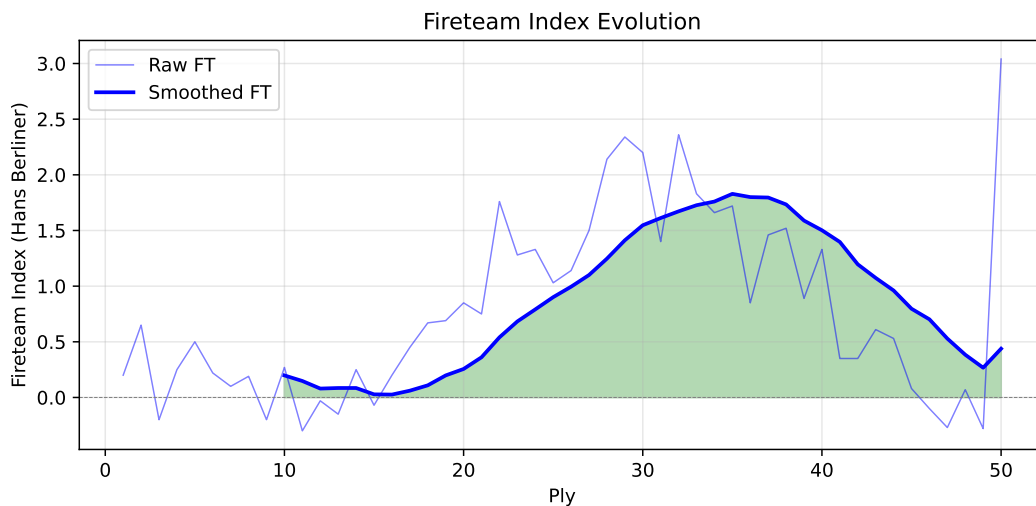
Tracking: **Hans Berliner** (Black)

| FTI Variant | Algorithm | Prediction | Max Streak |
|-------------|-----------|------------|------------|
| Standard    | Per-ply   | WIN        | 29 plies   |
| Standard    | Windowed  | WIN        | 34 plies   |
| Truncated   | Per-ply   | DRAW       | 9 plies    |
| Truncated   | Windowed  | WIN        | 26 plies   |

Actual result: 0-1 (Hans Berliner's outcome: WIN)

Standard: Per-ply ✓ Windowed ✓

Truncated: Per-ply ✗ Windowed ✓



## 2.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 437.6 seconds

## Chapter 3

# Berliner Hans J (USA) vs Borisenko Georgy K (UZB)

### 3.1 Game Information

|                |                                 |
|----------------|---------------------------------|
| <b>Event</b>   | Ch World (corr.)                |
| <b>Site</b>    | corr.                           |
| <b>Date</b>    | 1965.??.??                      |
| <b>Round</b>   | 16                              |
| <b>White</b>   | Berliner Hans J (USA)           |
| <b>Black</b>   | Borisenko Georgy K (UZB) (2354) |
| <b>Result</b>  | 1-0                             |
| <b>Opening</b> | D86 – Unknown                   |

### 3.2 Player Statistics

#### 3.2.1 Berliner Hans J (USA) (White)

- Total moves: 17
- Accuracy: 96.3%
- Average centipawn loss: 7.5
- Best/Excellent moves: 7 / 1
- Good moves: 7
- Inaccuracies: 2
- Mistakes: 0
- Blunders: 0

#### 3.2.2 Borisenko Georgy K (UZB) (Black)

- Total moves: 16
- Accuracy: 91.0%

- Average centipawn loss: 18.9
- Best/Excellent moves: 4 / 1
- Good moves: 7
- Inaccuracies: 3
- Mistakes: 1
- Blunders: 0

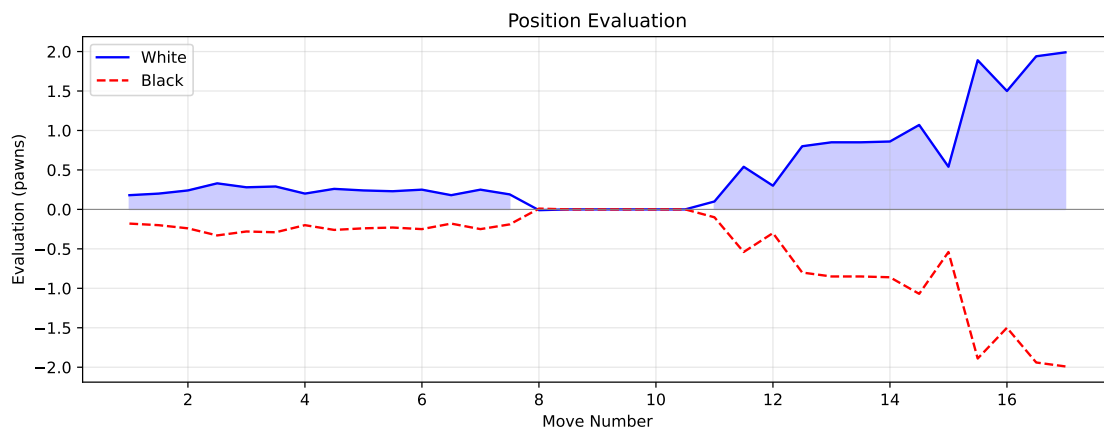
### 3.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

#### 3.3.1 Position Evaluation Over Time

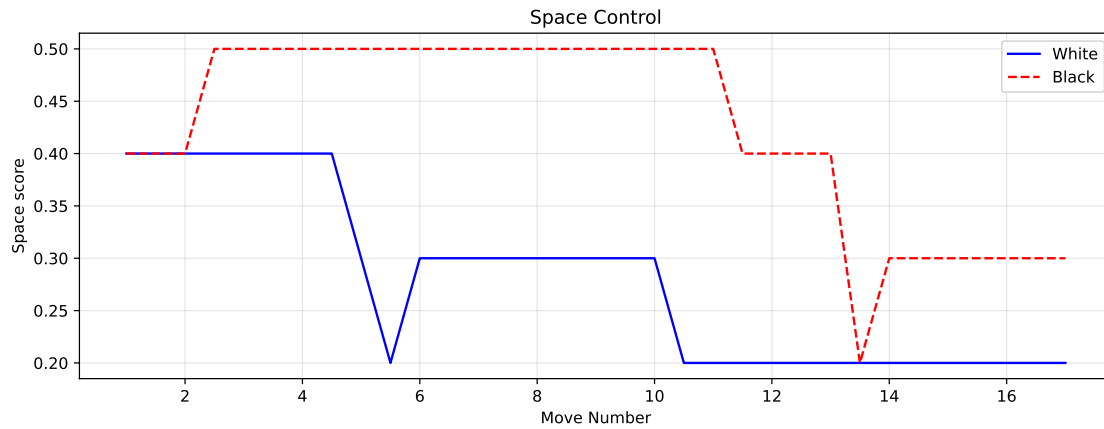
**Game Character Classification.** Evaluation range: minimum  $m_1 = -0.01$ , maximum  $m_2 = +1.99$ , spread  $d = m_2 - m_1 = 2.00$ .

This game is classified as **tense** (one-sided). A tense, one-sided game. White maintained the advantage throughout.



#### 3.3.2 Space Control

- White average: 0.28 (range: 0.20 to 0.40)
- Black average: 0.43 (range: 0.20 to 0.50)



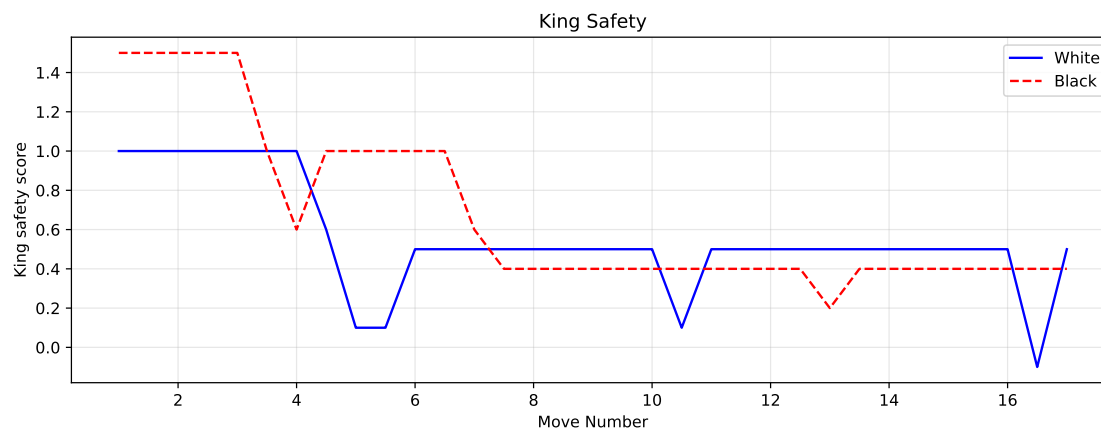
### 3.3.3 Mobility (Piece Activity)

- White average: 2.26 (range: 1.90 to 2.55)
- Black average: 1.93 (range: 1.60 to 2.20)



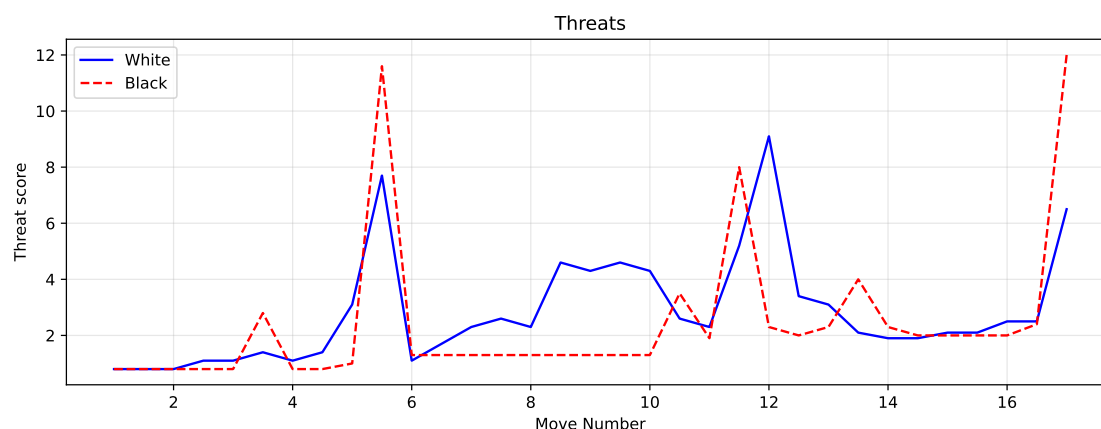
### 3.3.4 King Safety

- White average: 0.55
- Black average: 0.68



### 3.3.5 Threats

- White average: 2.86 (range: 0.80 to 9.10)
- Black average: 2.47 (range: 0.80 to 12.00)
- Average advantage: 0.39 (White maintained more threats)



## 3.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Bc4! O-O! 8. h4 c5! 9. Ne2! Nc6 10. Be3! cxd4! 11. cxd4! b5?! 12. Bd5 Bd7?! 13. h5! e6 14. Bb3! Na5 15. Qd2?! Nb7? *Black loses 112cp. Consider instead: Nxb3, Re8, Rc8.*

16. Rc1?! Rc8?! 17. Rxc8!

1-0

## 3.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

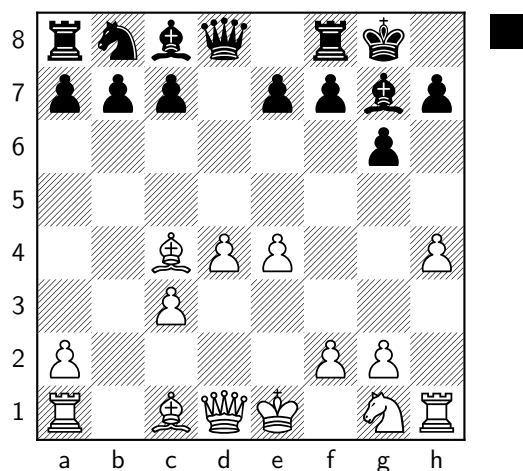
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 8. h4 ★

*Biggest evaluation swing(s) (20cp)*

Evaluation: -0.01



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.50  |
| Mobility (MG) | 4.70  | 3.50  |
| King Safety   | 0.50  | 0.40  |
| Threats       | 2.30  | 1.30  |

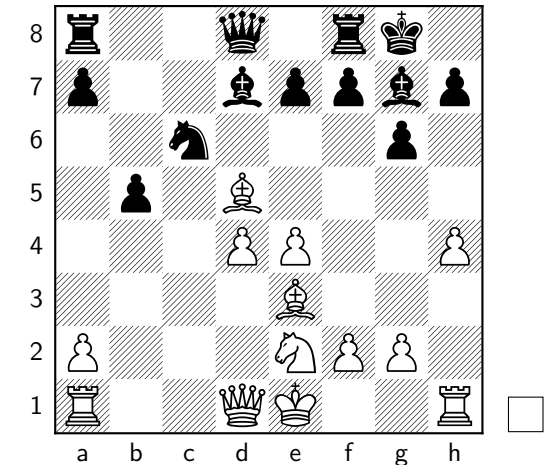
Instead of h4: Be3, Ne2

Best continuation: c5 Ne2 cxd4

### Position 2: After 12...Bd7 ★

*Biggest evaluation swing(s) (50cp)*

Evaluation: +0.80



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.20  | 0.40  |
| Mobility (MG) | 4.70  | 4.40  |
| King Safety   | 0.50  | 0.40  |
| Threats       | 3.40  | 2.00  |

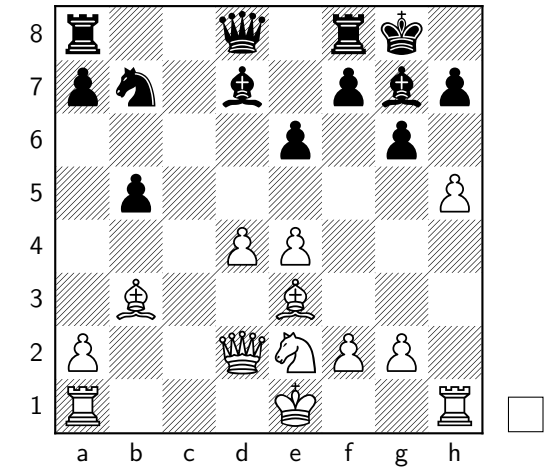
Instead of Bd7: Qa5+, Bb7, Qd7

Best continuation: h5 e6 Bb3

Position 3: After 15...Nb7

Critical moment

Evaluation: +1.89



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.20  | 0.30  |
| Mobility (MG) | 4.60  | 3.90  |
| King Safety   | 0.50  | 0.40  |
| Threats       | 2.10  | 2.00  |

Instead of Nb7: Nxb3, Re8, Rc8

Best continuation: h6 Bh8 Rc1



### 3.6 Fireteam Index Prediction

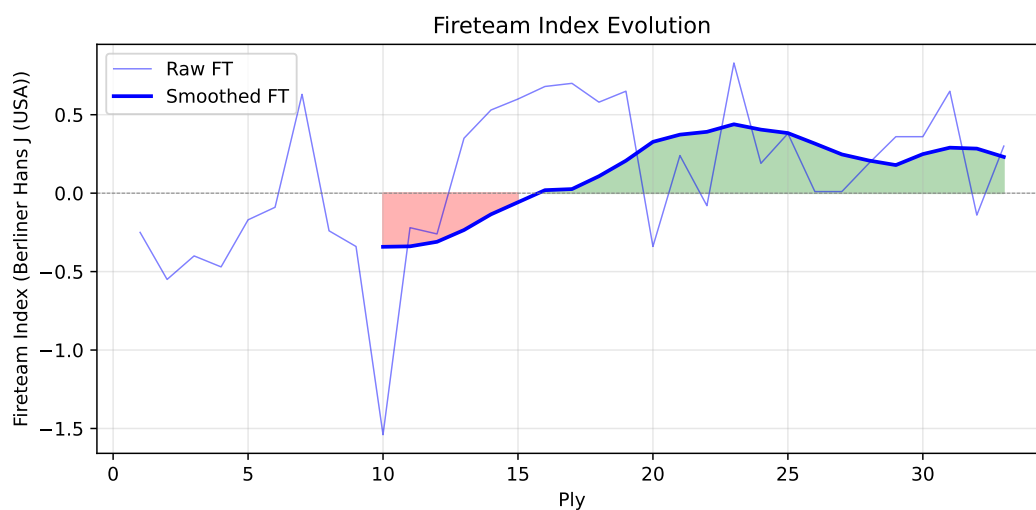
Tracking: **Berliner Hans J (USA)** (White)

| FTI Variant | Algorithm | Prediction | Max Streak |
|-------------|-----------|------------|------------|
| Standard    | Per-ply   | DRAW       | 9 plies    |
| Standard    | Windowed  | WIN        | 17 plies   |
| Truncated   | Per-ply   | DRAW       | 3 plies    |
| Truncated   | Windowed  | WIN        | 16 plies   |

Actual result: 1-0 (Berliner Hans J (USA)'s outcome: WIN)

Standard: Per-ply ✗ Windowed ✓

Truncated: Per-ply ✗ Windowed ✓



### 3.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 223.2 seconds



## Chapter 4

# Heinz De Carbonnel vs Hans Berliner

### 4.1 Game Information

|                |                                       |
|----------------|---------------------------------------|
| <b>Event</b>   | 5th Correspondence World Championship |
| <b>Site</b>    | ICCF                                  |
| <b>Date</b>    | 1965.07.01                            |
| <b>Round</b>   | 4                                     |
| <b>White</b>   | Heinz De Carbonnel                    |
| <b>Black</b>   | Hans Berliner                         |
| <b>Result</b>  | 0-1                                   |
| <b>Opening</b> | E80 – Unknown                         |

### 4.2 Player Statistics

#### 4.2.1 Heinz De Carbonnel (White)

- Total moves: 46
- Accuracy: 63.2%
- Average centipawn loss: 91.8
- Best/Excellent moves: 13 / 5
- Good moves: 12
- Inaccuracies: 6
- Mistakes: 6
- Blunders: 4

#### 4.2.2 Hans Berliner (Black)

- Total moves: 46
- Accuracy: 93.1%
- Average centipawn loss: 14.4

- Best/Excellent moves: 22 / 7
- Good moves: 8
- Inaccuracies: 7
- Mistakes: 1
- Blunders: 1

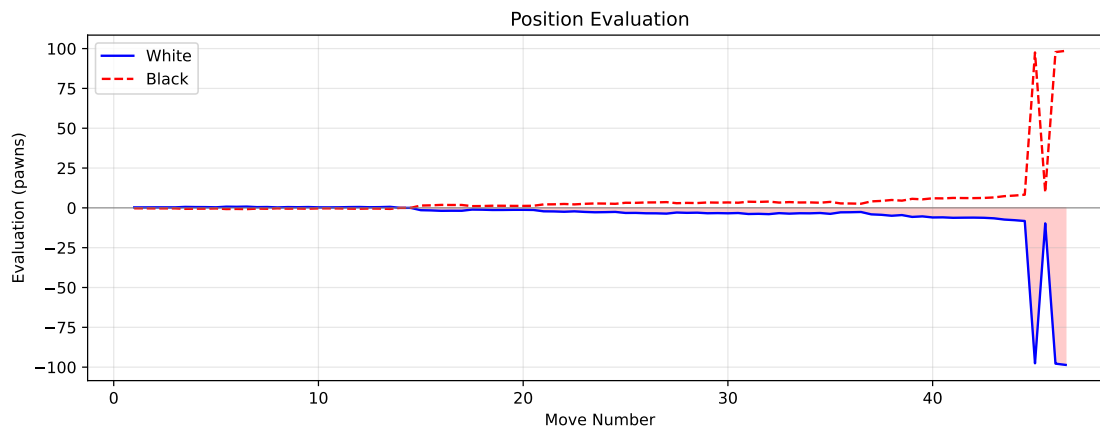
### 4.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

#### 4.3.1 Position Evaluation Over Time

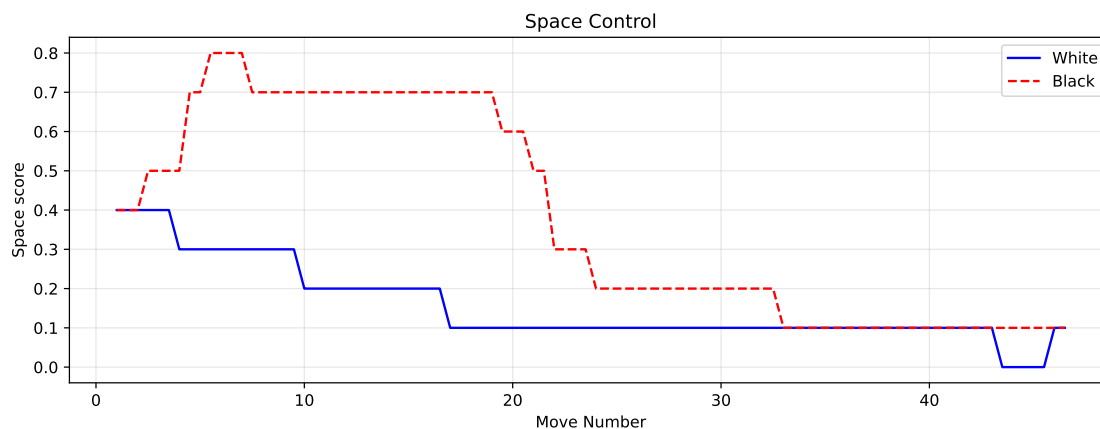
**Game Character Classification.** Evaluation range: minimum  $m_1 = -15.00$ , maximum  $m_2 = +0.79$ , spread  $d = m_2 - m_1 = 15.79$ .

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.



#### 4.3.2 Space Control

- White average: 0.16 (range: 0.00 to 0.40)
- Black average: 0.38 (range: 0.10 to 0.80)



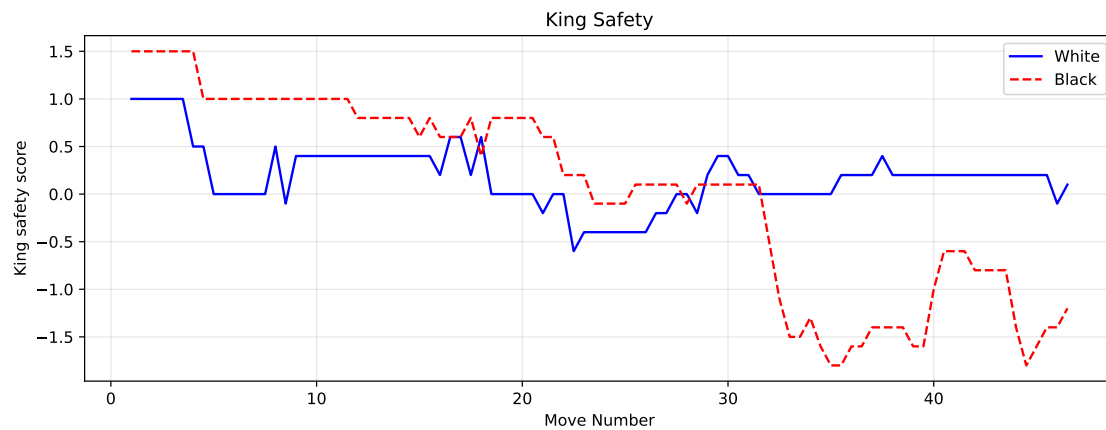
### 4.3.3 Mobility (Piece Activity)

- White average: 2.23 (range: 1.65 to 2.95)
- Black average: 2.12 (range: 1.55 to 2.95)



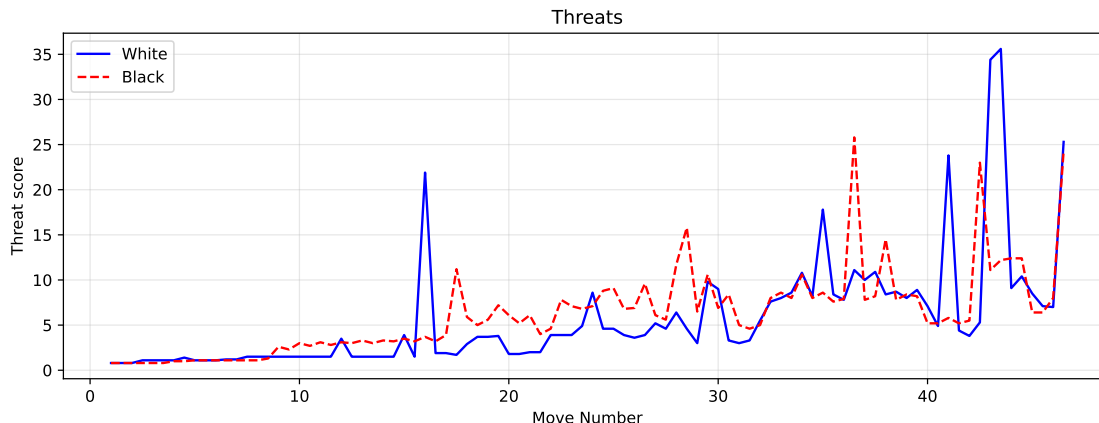
### 4.3.4 King Safety

- White average: 0.18
- Black average: 0.04



### 4.3.5 Threats

- White average: 5.72 (range: 0.80 to 35.60)
- Black average: 6.27 (range: 0.80 to 25.80)
- Average advantage: -0.55 (Black maintained more threats)



## 4.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 Bg7 4. e4 d6 5. f3 c6?! 6. Be3 a6 7. Qd2?! b5 8. O-O-O Qa5 9. Kb1 Nbd7 10. g4 Rb8! 11. h4! h5 12. g5! Nh7! 13. Rh2! Nh8 14. Rc1? *White loses 64cp. Consider instead: Rf2.*

14...Nb6 15. cxb5?? *White loses 146cp. Consider instead: d5.*

15...axb5! 16. Nd1?! Qa8! 17. f4 Nc4? *Black loses 65cp. Consider instead: Nfd7, d5, Nc4.*

18. Bxc4! bxc4! 19. Rf2 c5 20. Nc3! Ne6! 21. dxc5? *White loses 105cp. Consider instead: Nge2, Nf3.*

21...Bb7! 22. cxd6! Bxe4+?! 23. Ka1! O-O! 24. dxe7 Rfe8 25. Qd1? *White loses 89cp. Consider instead: Qd7.*

25...Rxe7! 26. Nf3 Rd8! 27. Nd2 Nd4?! 28. Bxd4 Bxd4 29. Ndx4?! Bxf2 30. Qc2! Be3! 31. Rf1?! Rd3! 32. Nf6+! Kg7?! 33. f5! Re5 34. fxg6! fxg6?! 35. Qh2 Qb8?? *Black loses 146cp. Consider instead: Bd4.*

36. Rb1! Rb5?! 37. Qh1?? *White loses 158cp. Consider instead: Qg2.*

37...Bd4! 38. Nxb5? *White loses 63cp. Consider instead: Nfe4, Qe1, Nxb5.*

38...Qxb5! 39. Qh2? *White loses 74cp. Consider instead: Qe1, Qh2.*

39...Qb7?! 40. Ne8+? *White loses 95cp. Consider instead: Re1, Qc2.*

40...Kh8! 41. Nd6?! Qd5! 42. Qf4 Rf3! 43. Qe4?! c3! 44. Qe8+ Kh7! 45. Qe7+?? *White loses 1500cp. Consider instead: Re1.*

45...Bg7! 46. bxc3?? *White loses 1500cp.*

46...Qxd6!

0-1

## 4.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

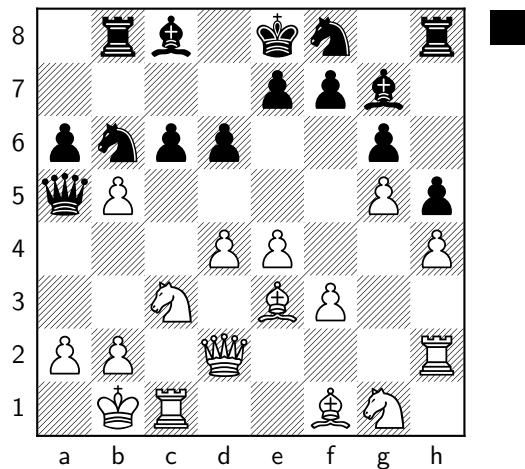
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 15. cxb5

White blunders (-146cp)

Evaluation: -1.46



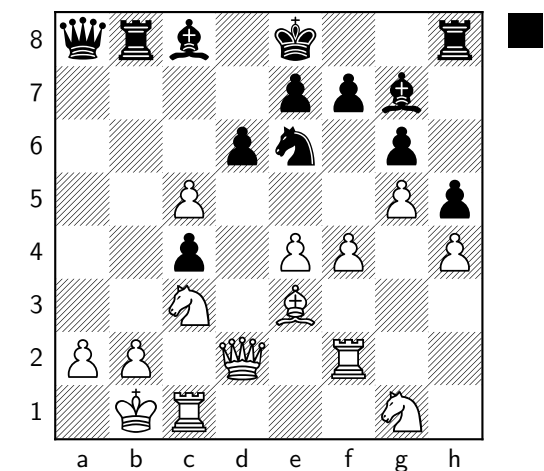
| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.20  | 0.70  |
| Mobility (MG) | 5.40  | 3.50  |
| King Safety   | 0.40  | 0.60  |
| Threats       | 3.90  | 3.50  |

Instead of cxb5: d5 Best continuation: axb5 Rf2 Ne6

Position 2: After 21. dxc5 ★

Biggest evaluation swing(s) (85cp)

Evaluation: -2.17



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.10  | 0.50  |
| Mobility (MG) | 4.70  | 4.00  |
| King Safety   | -0.20 | 0.60  |
| Threats       | 2.00  | 6.10  |

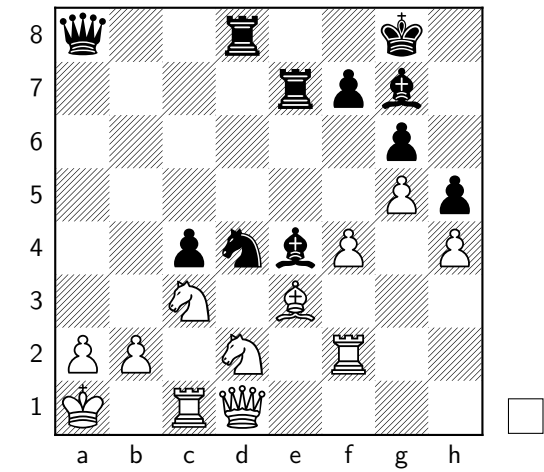
Instead of dxc5: Nge2, Nf3

Best continuation: Bb7 Ka1 Bxe4

Position 3: After 27...Nd4 ★

*Biggest evaluation swing(s) (69cp)*

Evaluation: -2.92



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.10  | 0.20  |
| Mobility (MG) | 4.60  | 5.80  |
| King Safety   | 0.00  | 0.10  |
| Threats       | 4.60  | 5.60  |

Instead of Nd4: Rd3, Bd3

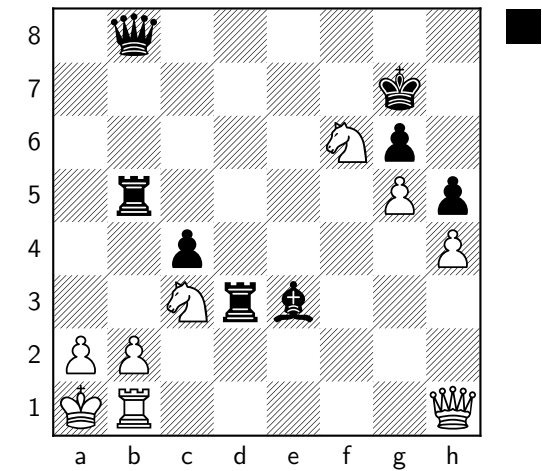
Best continuation: Bxd4 Bxd4 Ndxex4



Position 4: After 37. Qh1

White blunders (-148cp)

Evaluation: -4.07



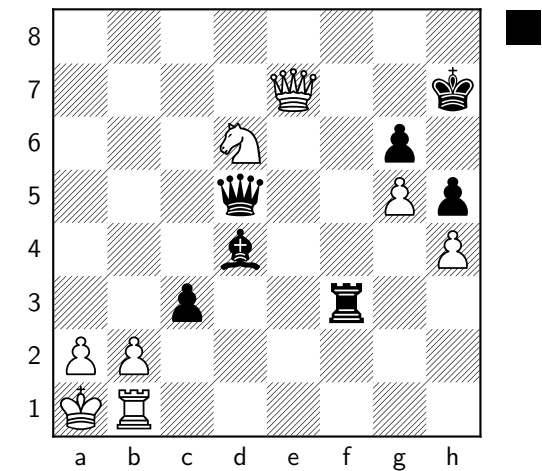
| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.10  | 0.10  |
| Mobility (MG) | 4.30  | 4.80  |
| King Safety   | 0.20  | -1.40 |
| Threats       | 10.00 | 7.80  |

Instead of Qh1:    Qg2                      Best continuation: Bd4 Nfe4 Rb7

Position 5: After 45. Qe7+

White blunders (-8936cp)

Evaluation: -97.60



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.00  | 0.10  |
| Mobility (MG) | 3.90  | 4.10  |
| King Safety   | 0.20  | -1.60 |
| Threats       | 8.50  | 6.40  |

Instead of Qe7+: Re1

Best continuation: Bg7 bxc3 Qxd6

## 4.6 Fireteam Index Prediction

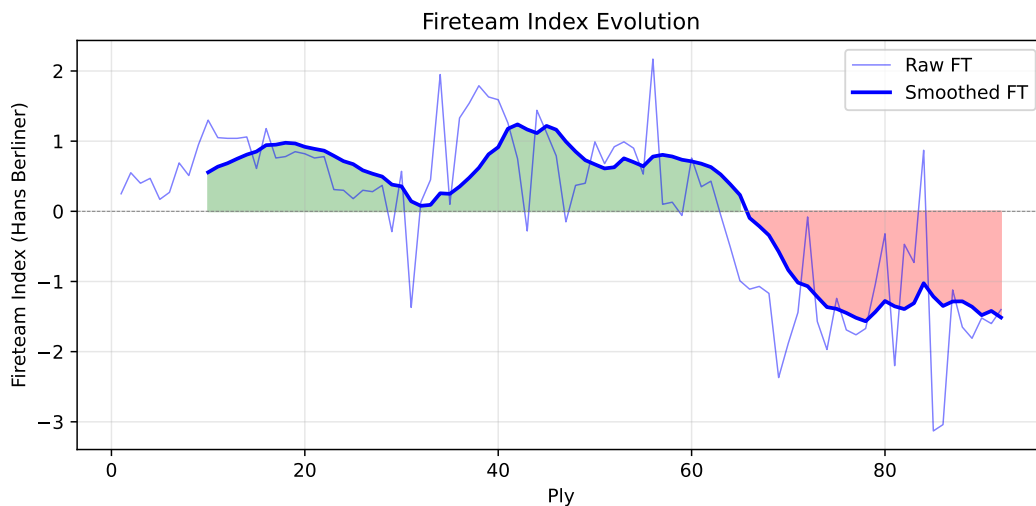
Tracking: **Hans Berliner** (Black)

| FTI Variant | Algorithm | Prediction | Max Streak |
|-------------|-----------|------------|------------|
| Standard    | Per-ply   | WIN        | 12 plies   |
| Standard    | Windowed  | WIN        | 49 plies   |
| Truncated   | Per-ply   | WIN        | 15 plies   |
| Truncated   | Windowed  | WIN        | 52 plies   |

Actual result: 0-1 (Hans Berliner's outcome: WIN)

Standard: Per-ply ✓ Windowed ✓

Truncated: Per-ply ✓ Windowed ✓



## 4.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 1939.1 seconds

## Chapter 5

# Lucius Endzelins vs Hans Berliner

### 5.1 Game Information

|                |                                       |
|----------------|---------------------------------------|
| <b>Event</b>   | 5th Correspondence World Championship |
| <b>Site</b>    | ICCF                                  |
| <b>Date</b>    | 1965.07.01                            |
| <b>Round</b>   | 6                                     |
| <b>White</b>   | Lucius Endzelins                      |
| <b>Black</b>   | Hans Berliner                         |
| <b>Result</b>  | 1/2-1/2                               |
| <b>Opening</b> | E98 – Unknown                         |

### 5.2 Player Statistics

#### 5.2.1 Lucius Endzelins (White)

- Total moves: 21
- Accuracy: 95.6%
- Average centipawn loss: 8.9
- Best/Excellent moves: 9 / 4
- Good moves: 6
- Inaccuracies: 2
- Mistakes: 0
- Blunders: 0

#### 5.2.2 Hans Berliner (Black)

- Total moves: 21
- Accuracy: 96.1%
- Average centipawn loss: 7.9

- Best/Excellent moves: 10 / 2
- Good moves: 7
- Inaccuracies: 2
- Mistakes: 0
- Blunders: 0

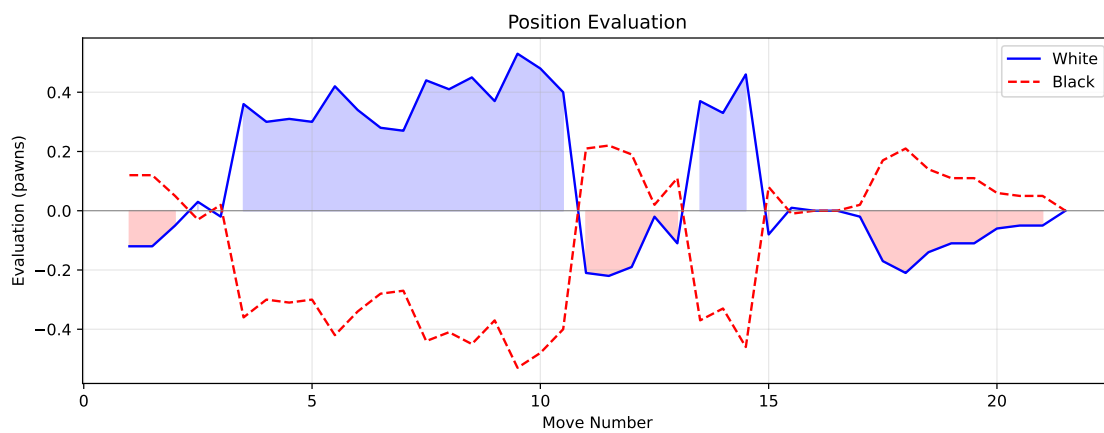
### 5.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

#### 5.3.1 Position Evaluation Over Time

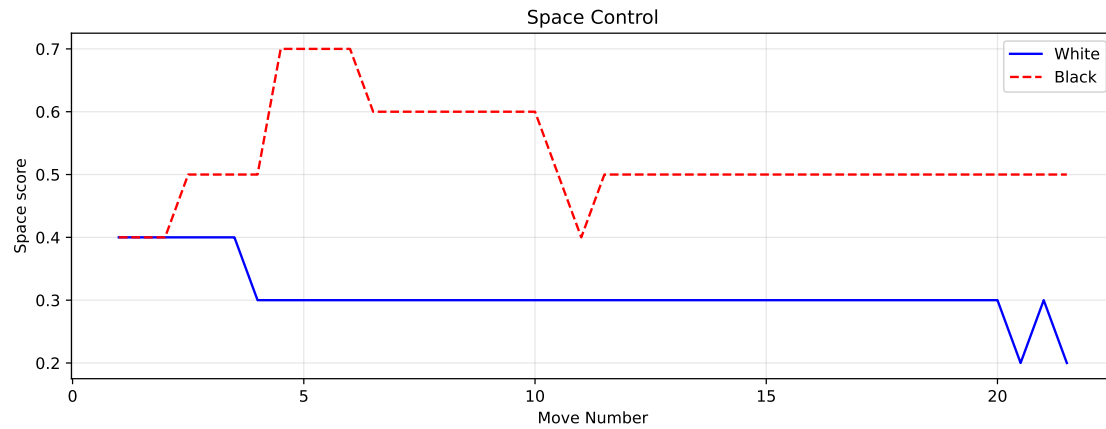
**Game Character Classification.** Evaluation range: minimum  $m_1 = -0.22$ , maximum  $m_2 = +0.53$ , spread  $d = m_2 - m_1 = 0.75$ .

This game is classified as **balanced** (one-sided). A balanced, one-sided game. White maintained the advantage throughout.



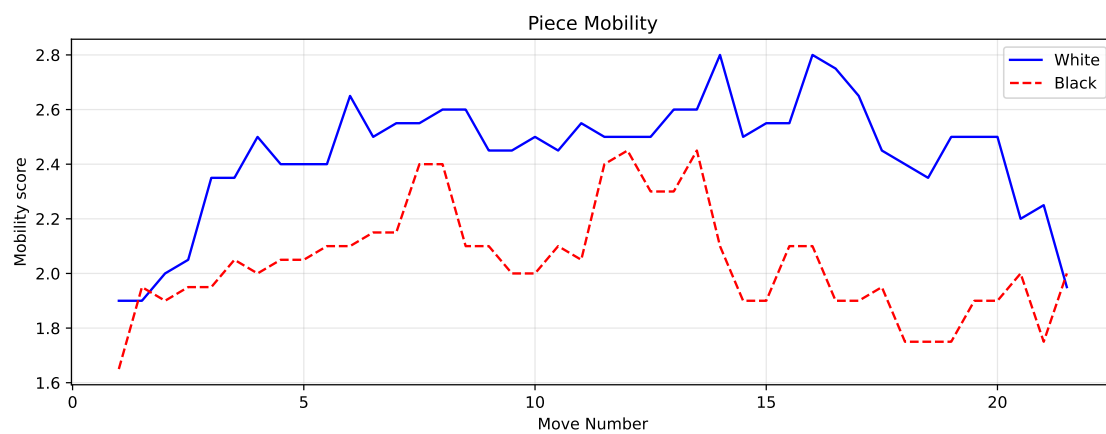
#### 5.3.2 Space Control

- White average: 0.31 (range: 0.20 to 0.40)
- Black average: 0.53 (range: 0.40 to 0.70)



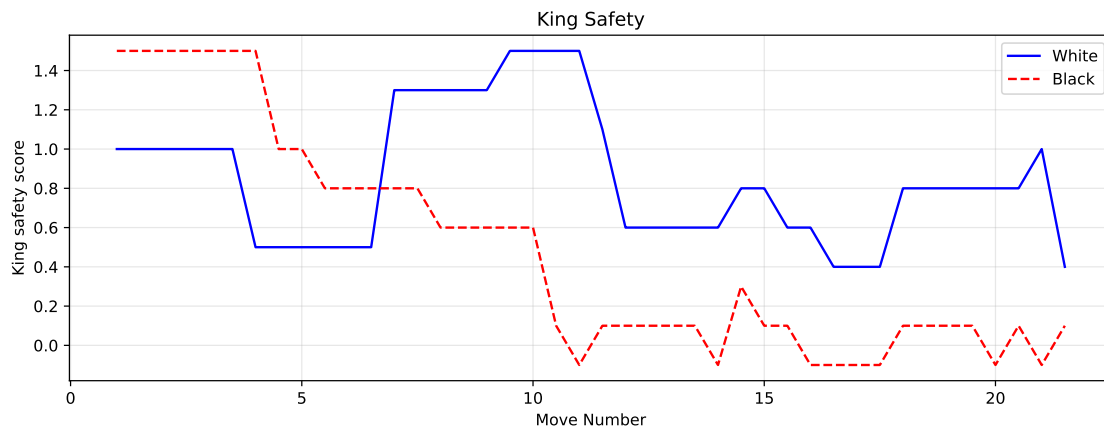
### 5.3.3 Mobility (Piece Activity)

- White average: 2.44 (range: 1.90 to 2.80)
- Black average: 2.04 (range: 1.65 to 2.45)



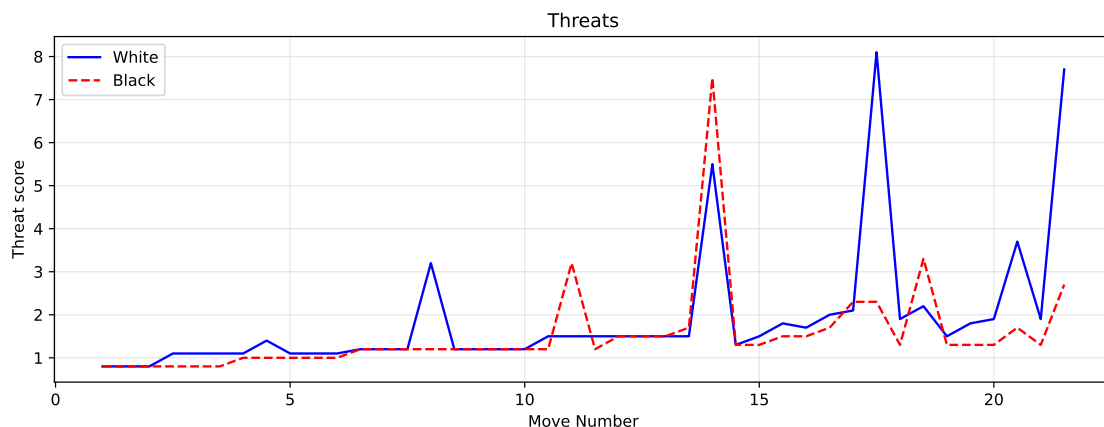
### 5.3.4 King Safety

- White average: 0.85
- Black average: 0.49



### 5.3.5 Threats

- White average: 1.91 (range: 0.80 to 8.10)
- Black average: 1.54 (range: 0.80 to 7.50)
- Average advantage: 0.37 (White maintained more threats)



## 5.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 Bg7?! 4. e4 d6 5. Be2 O-O 6. Nf3 e5 7. O-O! Nc6 8. d5 Ne7! 9. Ne1 Nd7 10. Nd3 f5! 11. exf5?! Nxf5 12. f3! Nf6 13. Nf2 Bh6?! 14. Bxh6 Nxe4?! 15. Nfe4?! Nf5 16. Qd2! Nd4! 17. Nb5 Nxe2+! 18. Qxe2 a6! 19. Nbc3! Bf5 20. Qe3 Bxe4 21. Nxe4! Nxe4!

1/2-1/2

## 5.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

- **Instead of [move]:** The move(s) that should have been played instead.

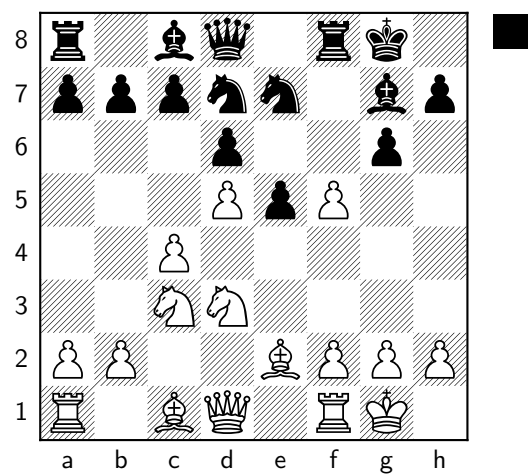
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 11. exf5 ★

*Biggest evaluation swing(s) (61cp)*

Evaluation: -0.21



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.40  |
| Mobility (MG) | 5.10  | 4.10  |
| King Safety   | 1.50  | -0.10 |
| Threats       | 1.50  | 3.20  |

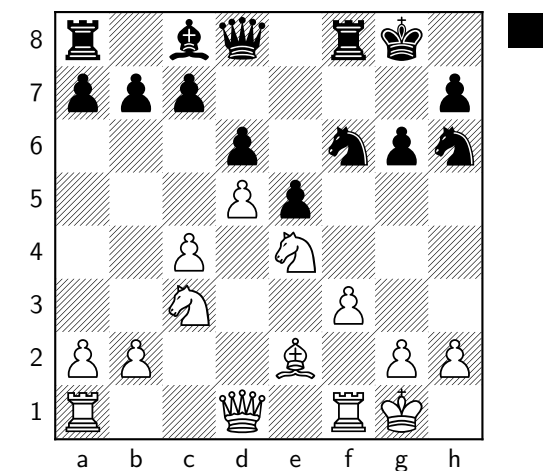
Instead of exf5: f3, Kh1, b4

Best continuation: gxf5 f4 Ng6

Position 2: After 15. Nfe4 ★

*Biggest evaluation swing(s) (54cp)*

Evaluation: -0.08



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.50  |
| Mobility (MG) | 5.10  | 3.80  |
| King Safety   | 0.80  | 0.10  |
| Threats       | 1.50  | 1.30  |

Instead of Nfe4:    g4, Qd2, Bd3                      Best continuation: Nxe4 Nxe4 Bf5

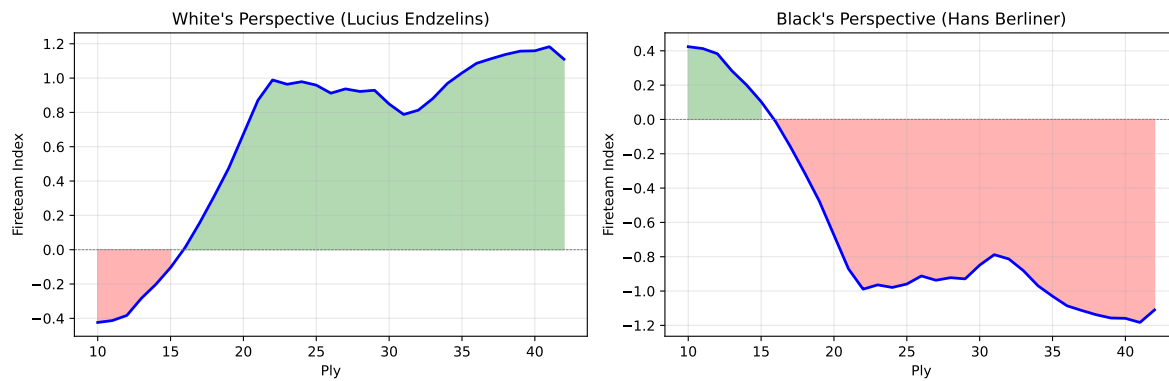
5.6 Fireteam Index Prediction

*This game ended in a draw. The Fireteam Index is analyzed from both players’ perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.*

| Standard FTI (1.0, 1.0, 1.0, 0.1)  | Player           | Algorithm | Prediction | Max Streak |
|------------------------------------|------------------|-----------|------------|------------|
|                                    | Lucius Endzelins | Per-ply   | WIN        | 26 plies   |
|                                    | Lucius Endzelins | Windowed  | WIN        | 26 plies   |
|                                    | Hans Berliner    | Per-ply   | DRAW       | 0 plies    |
|                                    | Hans Berliner    | Windowed  | DRAW       | 0 plies    |
|                                    |                  |           |            |            |
| Truncated FTI (1.2, 0.0, 0.0, 0.6) | Player           | Algorithm | Prediction | Max Streak |
|                                    | Lucius Endzelins | Per-ply   | DRAW       | 5 plies    |
|                                    | Lucius Endzelins | Windowed  | DRAW       | 9 plies    |
|                                    | Hans Berliner    | Per-ply   | WIN        | 17 plies   |
|                                    | Hans Berliner    | Windowed  | WIN        | 17 plies   |
|                                    |                  |           |            |            |

**Interpretation.** White (Lucius Endzelins) achieved sustained positional dominance, with the FT Index predicting a win. However, the game ended in a draw, suggesting Black successfully defended or White failed to convert the advantage.





## 5.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 307.1 seconds



## Chapter 6

# Arne Ericson vs Hans Berliner

### 6.1 Game Information

|                |                                       |
|----------------|---------------------------------------|
| <b>Event</b>   | 5th Correspondence World Championship |
| <b>Site</b>    | ICCF                                  |
| <b>Date</b>    | 1965.07.01                            |
| <b>Round</b>   | 14                                    |
| <b>White</b>   | Arne Ericson                          |
| <b>Black</b>   | Hans Berliner                         |
| <b>Result</b>  | 0-1                                   |
| <b>Opening</b> | B04 – Unknown                         |

### 6.2 Player Statistics

#### 6.2.1 Arne Ericson (White)

- Total moves: 52
- Accuracy: 89.3%
- Average centipawn loss: 22.5
- Best/Excellent moves: 15 / 8
- Good moves: 13
- Inaccuracies: 10
- Mistakes: 5
- Blunders: 1

#### 6.2.2 Hans Berliner (Black)

- Total moves: 52
- Accuracy: 95.5%
- Average centipawn loss: 9.1

- Best/Excellent moves: 33 / 5
- Good moves: 9
- Inaccuracies: 3
- Mistakes: 2
- Blunders: 0

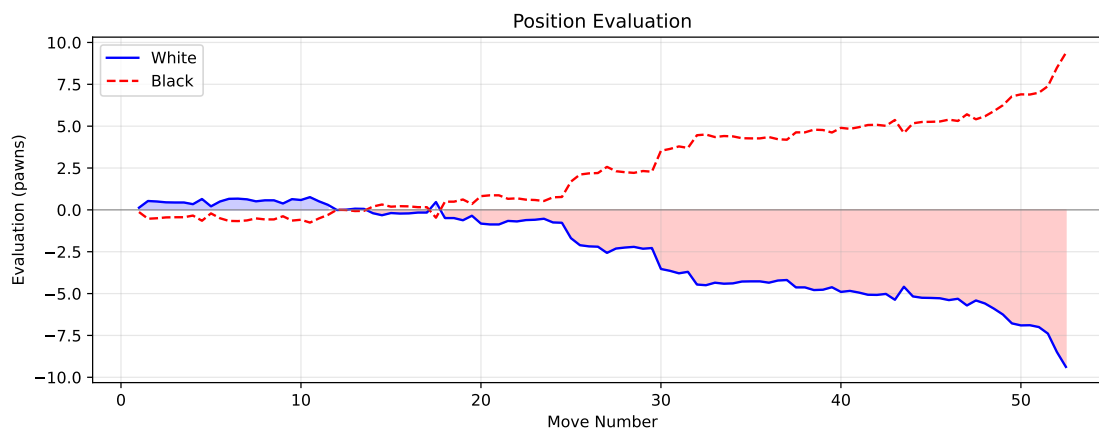
## 6.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 6.3.1 Position Evaluation Over Time

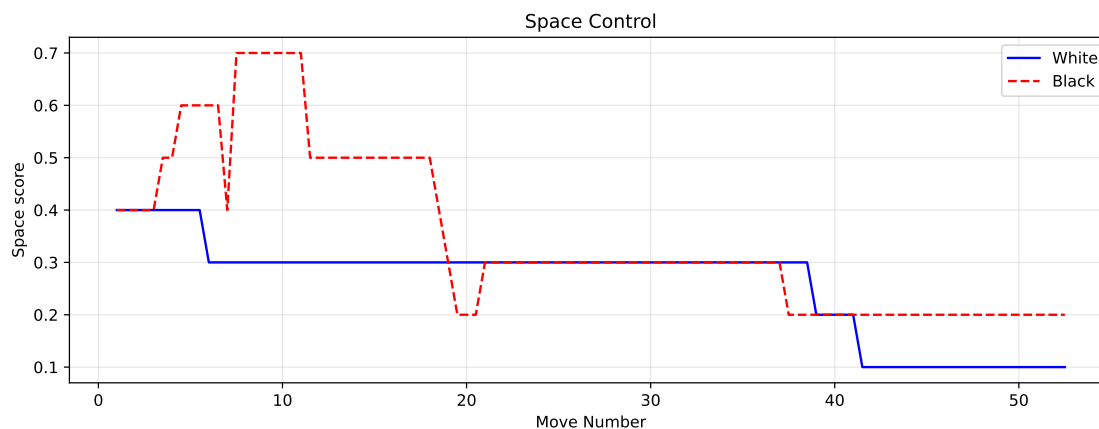
**Game Character Classification.** Evaluation range: minimum  $m_1 = -9.38$ , maximum  $m_2 = +0.76$ , spread  $d = m_2 - m_1 = 10.14$ .

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.



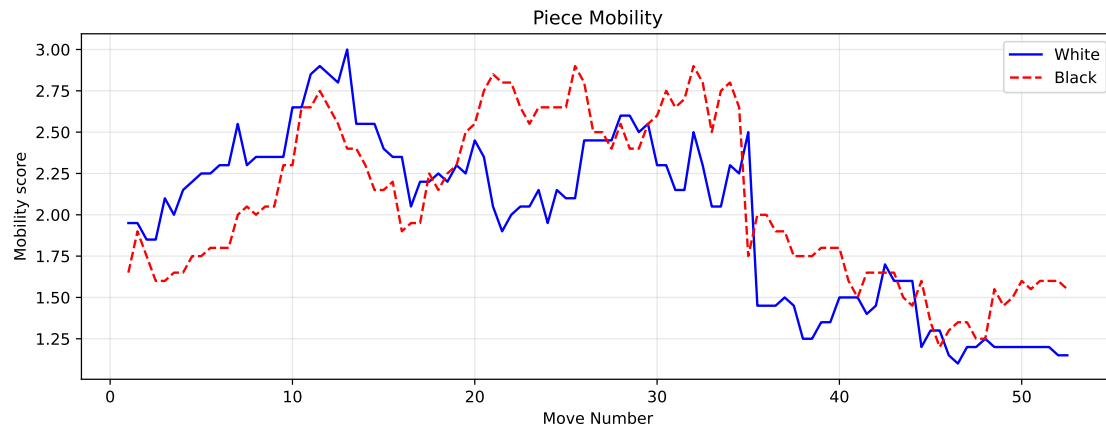
### 6.3.2 Space Control

- White average: 0.26 (range: 0.10 to 0.40)
- Black average: 0.35 (range: 0.20 to 0.70)



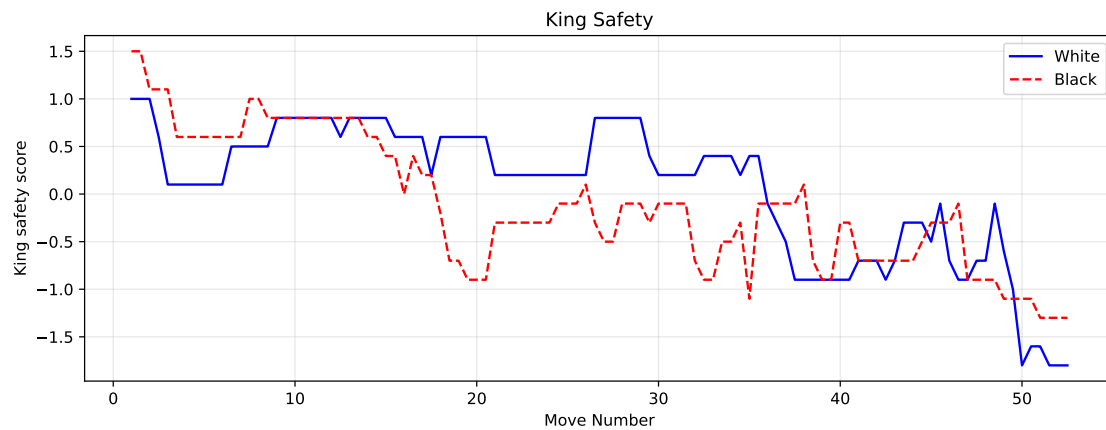
### 6.3.3 Mobility (Piece Activity)

- White average: 1.98 (range: 1.10 to 3.00)
- Black average: 2.08 (range: 1.20 to 2.90)



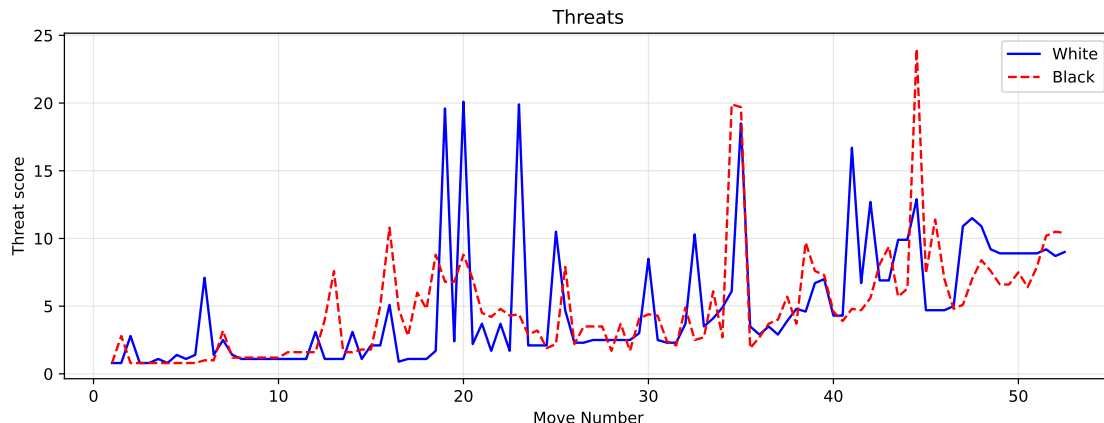
### 6.3.4 King Safety

- White average: 0.06
- Black average: -0.12



### 6.3.5 Threats

- White average: 4.87 (range: 0.80 to 20.10)
- Black average: 4.84 (range: 0.80 to 24.00)
- Average advantage: 0.02 (threats were balanced)



## 6.4 Annotated Game

1. e4 Nf6?! 2. e5 Nd5 3. d4 d6 4. Nf3 g6 5. Be2?! Bg7?! 6. c4 Nb6 7. exd6! cxd6!  
8. h3! O-O 9. O-O Nc6 10. Nc3 Bf5! 11. Be3 d5! 12. c5?! Nc4! 13. Bxc4 dxc4! 14.  
d5?! Nb4! 15. Bd4! Bd3 16. Bxg7! Kxg7! 17. Re1! Qc7? *Black loses 88cp. Consider  
instead: Nc2, Rc8, Nxd5.*

18. Re5? *White loses 86cp. Consider instead: Ne5, Qd2, Rc1.*

18...f6! 19. d6?! exd6! 20. cxd6?! Qxd6! 21. Rb5! b6 22. a3 Na6! 23. Rd5! Qe7! 24.  
Ne1 Bf5! 25. Qa4? *White loses 101cp. Consider instead: Nc2, Nf3.*

25...Nc5! 26. Qxc4 Be6! 27. Qe2 Rfe8 28. Rdd1! Nb3! 29. Rab1! Bf5! 30. Qb5??  
*White loses 126cp. Consider instead: Qxe7+.*

30...Qf7! 31. Nd3 Rad8 32. Nf4? *White loses 62cp. Consider instead: h4, Nb4, f4.*

32...Bxb1! 33. Rxb1! Re5! 34. Qc6 Qd7 35. Qxd7+! Rxd7! 36. g3 Rd2 37. Kg2  
g5! 38. Nh5+! Kg6! 39. g4 Nc5 40. Ng3 Nd3 41. Nge4? *White loses 60cp. Consider  
instead: Nce4.*

41...Rxb2 42. Rd1 Nf4+! 43. Kf1?! Ne2? *Black loses 76cp. Consider instead: Rb3,  
Re6, Rc2.*

44. a4?! Nxc3! 45. Nxc3 Rb3! 46. Ne2 Rxh3 47. Rd7?! Ra3?! 48. Ng3 Rxa4! 49. f3?!  
Ra1+! 50. Kg2?! Ra2+! 51. Kh3 Re3! 52. Rc7? *White loses 110cp. Consider instead:  
Rd8, Rd5, Rg7+.*

52...Rb3!

0-1

## 6.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

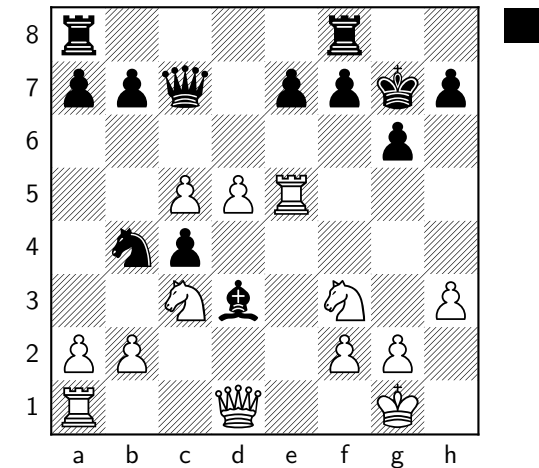
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 18. Re5 ★

Biggest evaluation swing(s) (96cp)

Evaluation: -0.49



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.50  |
| Mobility (MG) | 4.50  | 4.30  |
| King Safety   | 0.60  | -0.20 |
| Threats       | 1.10  | 4.80  |

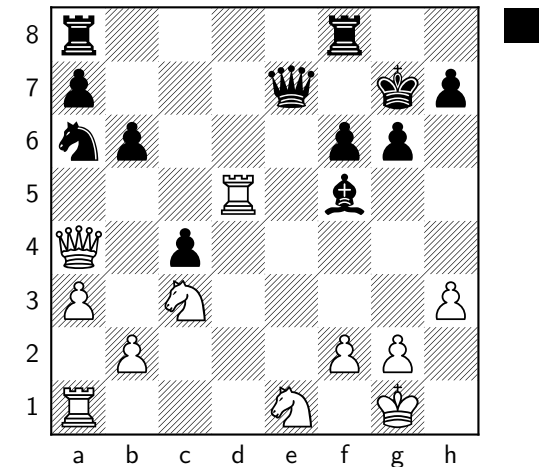
Instead of Re5: Ne5, Qd2, Rc1

Best continuation: f6 a3 fxe5

Position 2: After 25. Qa4 ★

Biggest evaluation swing(s) (92cp)

Evaluation: -1.69



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.30  |
| Mobility (MG) | 4.20  | 5.30  |
| King Safety   | 0.20  | -0.10 |
| Threats       | 10.50 | 2.20  |

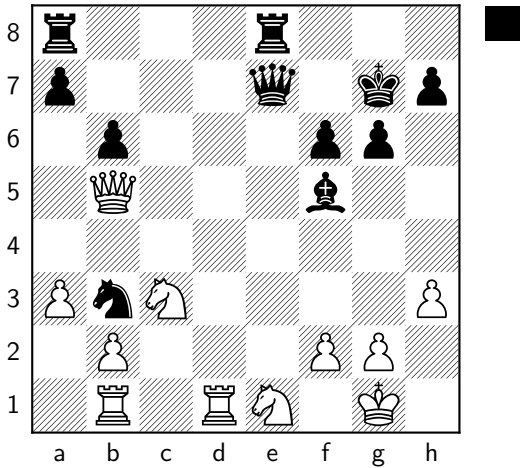
Instead of Qa4:    Nc2, Nf3

Best continuation: Nc5 Qxc4 Be6

Position 3: After 30. Qb5

White blunders (-125cp)

Evaluation: -3.53



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.30  |
| Mobility (MG) | 4.60  | 5.20  |
| King Safety   | 0.20  | -0.10 |
| Threats       | 8.50  | 4.40  |

Instead of Qb5:    Qxe7+

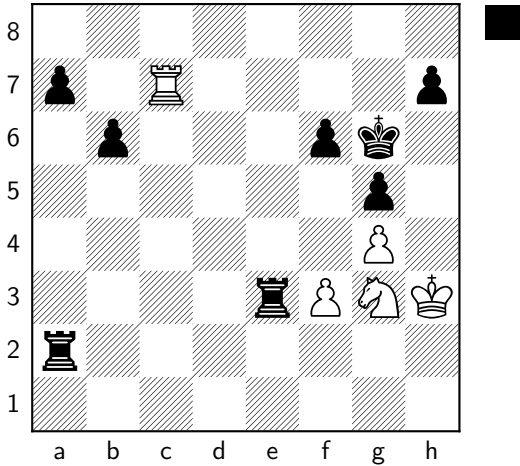
Best continuation: Qf7 Qa4 Bxb1

Position 4: After 52. Rc7

Critical moment

Evaluation: -8.49





| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.10  | 0.20  |
| Mobility (MG) | 2.30  | 3.20  |
| King Safety   | -1.80 | -1.30 |
| Threats       | 8.70  | 10.50 |

Instead of Rc7:    Rd8, Rd5, Rg7+

Best continuation: Rb3 Re7 Rbb2

6.6 Fireteam Index Prediction

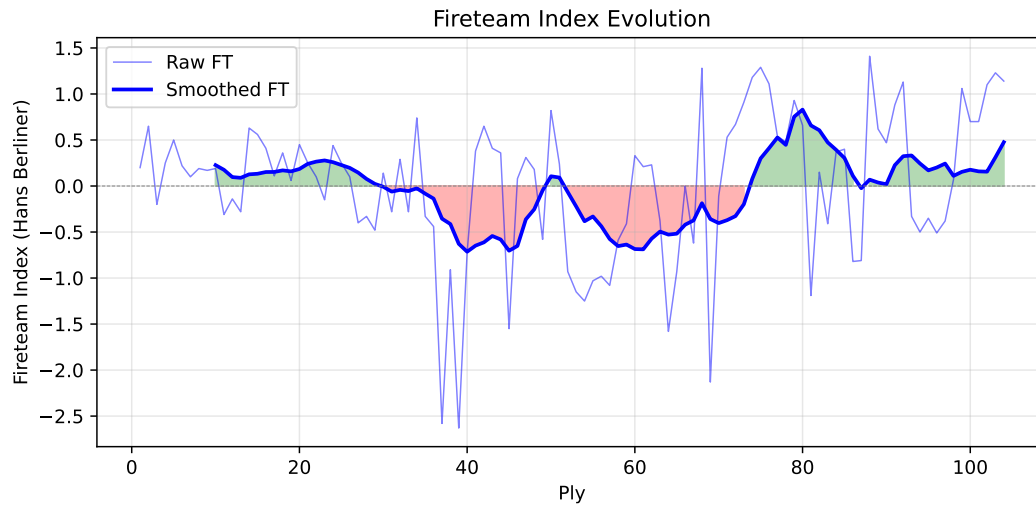
Tracking: **Hans Berliner** (Black)

| FTI Variant | Algorithm | Prediction | Max Streak |
|-------------|-----------|------------|------------|
| Standard    | Per-ply   | WIN        | 10 plies   |
| Standard    | Windowed  | WIN        | 17 plies   |
| Truncated   | Per-ply   | DRAW       | 9 plies    |
| Truncated   | Windowed  | WIN        | 26 plies   |

Actual result: 0-1 (Hans Berliner’s outcome: WIN)

Standard: Per-ply ✓    Windowed ✓

Truncated: Per-ply ✗    Windowed ✓



## 6.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 855.7 seconds

## Chapter 7

# Yakov Estrin vs Hans Berliner

### 7.1 Game Information

|                |                                       |
|----------------|---------------------------------------|
| <b>Event</b>   | 5th Correspondence World Championship |
| <b>Site</b>    | ICCF                                  |
| <b>Date</b>    | 1965.07.01                            |
| <b>Round</b>   | 12                                    |
| <b>White</b>   | Yakov Estrin                          |
| <b>Black</b>   | Hans Berliner                         |
| <b>Result</b>  | 0-1                                   |
| <b>Opening</b> | C57 – Unknown                         |

### 7.2 Player Statistics

#### 7.2.1 Yakov Estrin (White)

- Total moves: 42
- Accuracy: 88.2%
- Average centipawn loss: 25.0
- Best/Excellent moves: 14 / 8
- Good moves: 10
- Inaccuracies: 7
- Mistakes: 1
- Blunders: 2

#### 7.2.2 Hans Berliner (Black)

- Total moves: 42
- Accuracy: 93.6%
- Average centipawn loss: 13.2

- Best/Excellent moves: 25 / 3
- Good moves: 7
- Inaccuracies: 5
- Mistakes: 2
- Blunders: 0

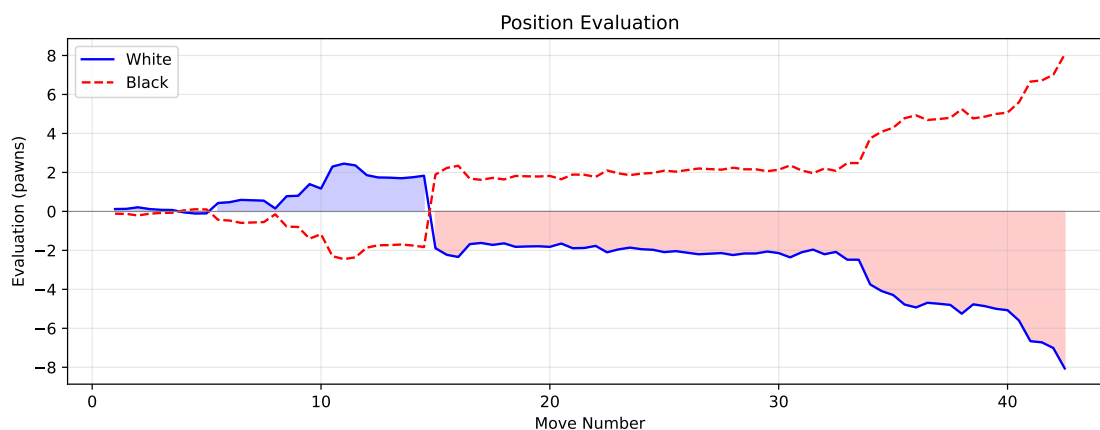
## 7.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 7.3.1 Position Evaluation Over Time

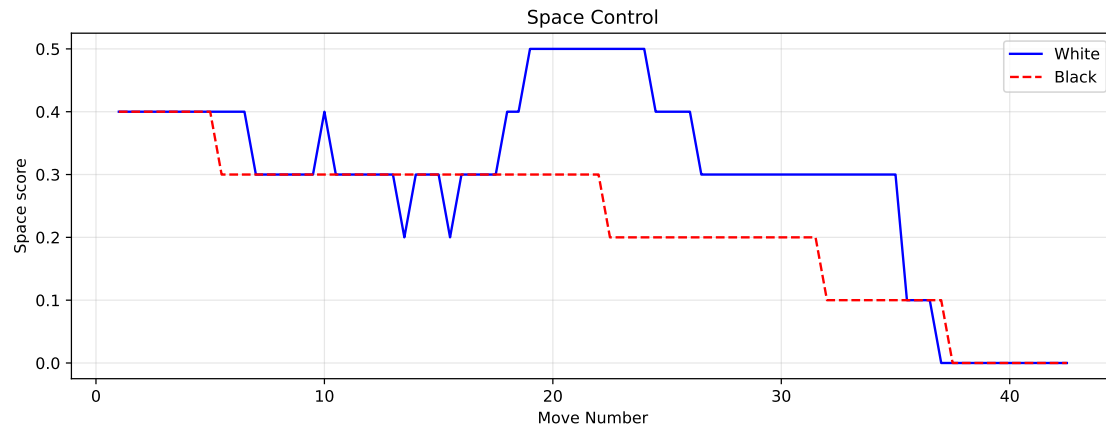
**Game Character Classification.** Evaluation range: minimum  $m_1 = -8.06$ , maximum  $m_2 = +2.45$ , spread  $d = m_2 - m_1 = 10.51$ .

This game is classified as **chaotic** (seesaw). A chaotic seesaw battle with the advantage changing hands.



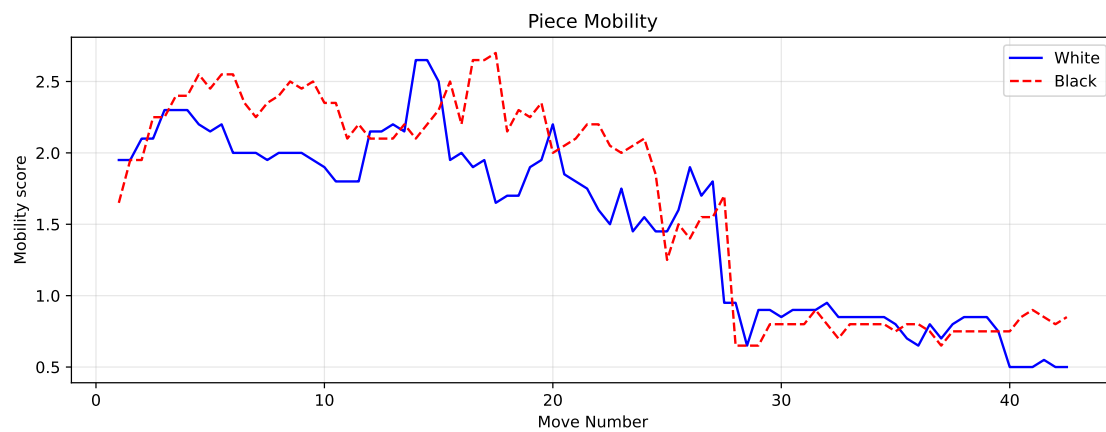
### 7.3.2 Space Control

- White average: 0.30 (range: 0.00 to 0.50)
- Black average: 0.22 (range: 0.00 to 0.40)



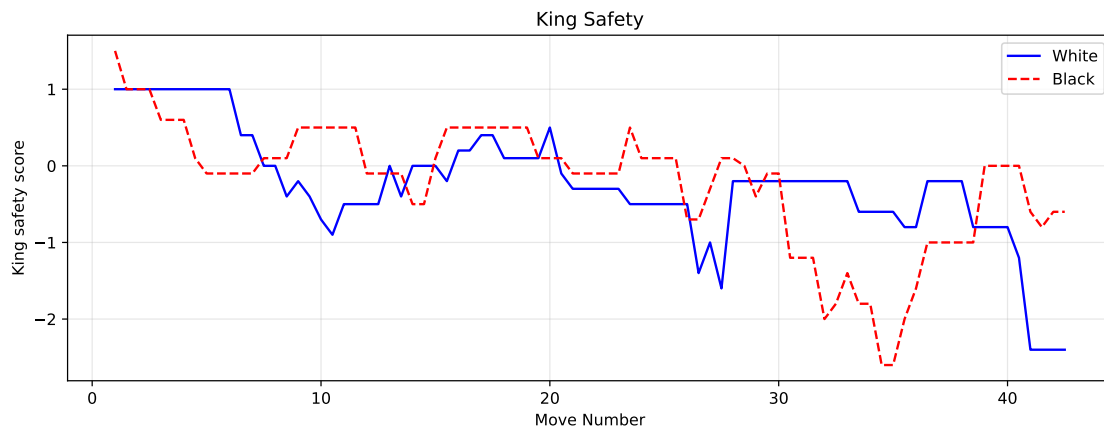
### 7.3.3 Mobility (Piece Activity)

- White average: 1.51 (range: 0.50 to 2.65)
- Black average: 1.67 (range: 0.65 to 2.70)



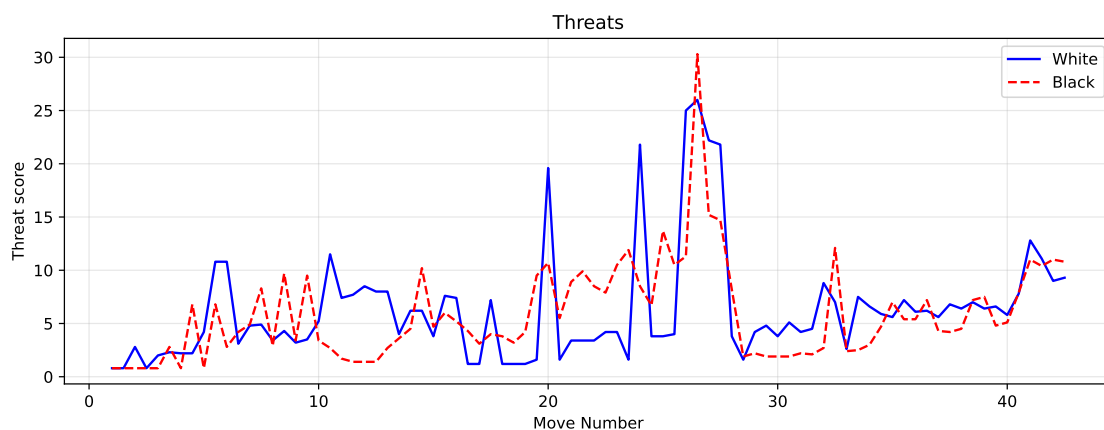
### 7.3.4 King Safety

- White average: -0.25
- Black average: -0.22



### 7.3.5 Threats

- White average: 6.35 (range: 0.80 to 26.00)
- Black average: 5.91 (range: 0.80 to 30.30)
- Average advantage: 0.43 (White maintained more threats)



## 7.4 Annotated Game

1. e4 e5 2. Nf3 Nc6 3. Bc4 Nf6 4. Ng5 d5 5. exd5 b5?! 6. Bf1 Nd4 7. c3! Nxd5! 8. Ne4?! Qh4? *Black loses 68cp. Consider instead: Ne6.*

9. Ng3 Bg4?! 10. f3 e4? *Black loses 115cp. Consider instead: Bd7, Be6.*

11. cxd4! Bd6 12. Bxb5+?! Kd8! 13. O-O! exf3! 14. Rxf3 Rb8! 15. Be2?? *White loses 357cp. Consider instead: Bf1, a4.*

15...Bxf3! 16. Bxf3! Qxd4+ 17. Kh1! Bxg3! 18. hxc3! Rb6! 19. d3! Ne3! 20. Bxe3! Qxe3 21. Bg4 h5! 22. Bh3! g5! 23. Nd2 g4! 24. Nc4 Qxg3! 25. Nxb6 gxh3! 26. Qf3 hxc3! 27. Qxg2 Qxg2+! 28. Kxg2 cxb6! 29. Rf1 Ke7 30. Re1+! Kd6! 31. Rf1! Rc8?! 32. Rxf7?! Rc7! 33. Rf2?! Ke5 34. a4?? *White loses 121cp. Consider instead: Rf8.*

34...Kd4! 35. a5?! Kxd3! 36. Rf3+?! Kc2 37. b4! b5 38. a6?! Rc4?! 39. Rf7 Rxb4!  
40. Rb7 Rg4+! 41. Kf3? *White loses 99cp.*  
41...b4! 42. Rxa7! b3?!  
0-1

7.5 Critical Positions

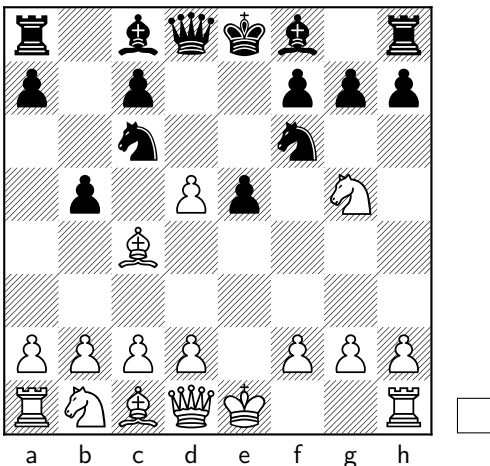
This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

Position 1: After 5...b5 ★

*Biggest evaluation swing(s) (53cp)*  
Evaluation: +0.43



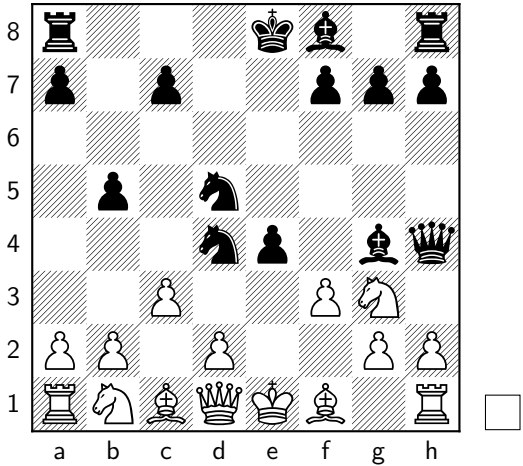
| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.40  | 0.30  |
| Mobility (MG) | 4.40  | 5.10  |
| King Safety   | 1.00  | -0.10 |
| Threats       | 10.80 | 6.80  |

Instead of b5: Na5

Best continuation: Bf1 Nd4 c3

Position 2: After 10...e4

*Critical moment*  
Evaluation: +2.30



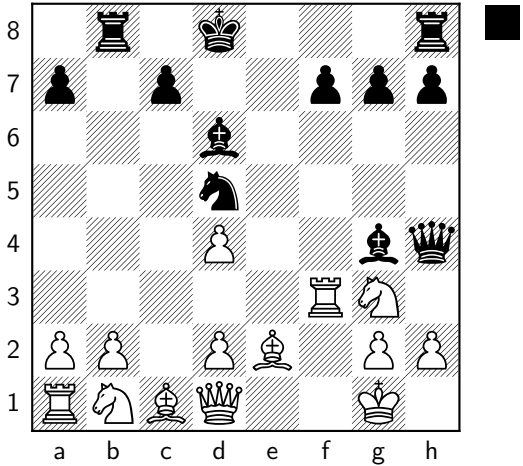
| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.30  |
| Mobility (MG) | 3.60  | 4.70  |
| King Safety   | -0.90 | 0.50  |
| Threats       | 11.50 | 2.70  |

Instead of e4:   Bd7, Be6

Best continuation: cxd4 Bd6 Qe2

Position 3: After 15. Be2

White blunders (-372cp)  
Evaluation: -1.89



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.30  |
| Mobility (MG) | 5.00  | 4.60  |
| King Safety   | 0.00  | 0.10  |
| Threats       | 3.80  | 4.70  |

Instead of Be2:   Bf1, a4

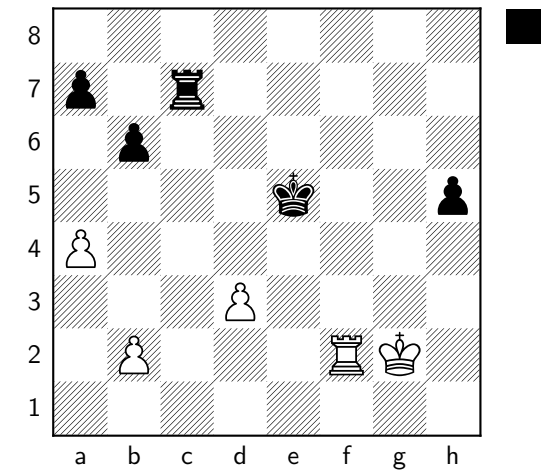
Best continuation: Bxf3 Bxf3 Re8



Position 4: After 34. a4

White blunders (-127cp)

Evaluation: -3.75



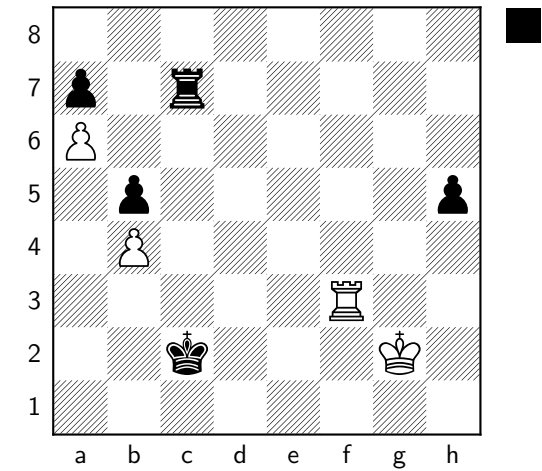
| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.10  |
| Mobility (MG) | 1.70  | 1.60  |
| King Safety   | -0.60 | -1.80 |
| Threats       | 6.60  | 3.00  |

Instead of a4: Rf8      Best continuation: Kd4 a5 bxa5

Position 5: After 38. a6 ★

Biggest evaluation swing(s) (45cp)

Evaluation: -5.25

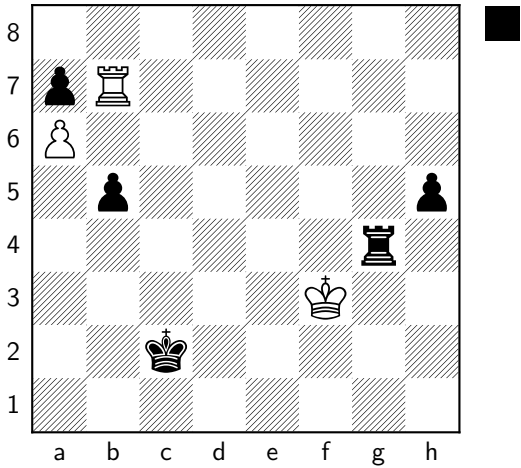


| Metric         | White                          | Black |
|----------------|--------------------------------|-------|
| Space          | 0.00                           | 0.00  |
| Mobility (MG)  | 1.70                           | 1.50  |
| King Safety    | -0.20                          | -1.00 |
| Threats        | 6.40                           | 4.50  |
| Instead of a6: | Rf5, Rh3, Kf1                  |       |
|                | Best continuation: Rc6 Rf5 Rb6 |       |

Position 6: After 41. Kf3

Critical moment

Evaluation: -6.66



| Metric          | White | Black                          |
|-----------------|-------|--------------------------------|
| Space           | 0.00  | 0.00                           |
| Mobility (MG)   | 1.00  | 1.80                           |
| King Safety     | -2.40 | -0.60                          |
| Threats         | 12.80 | 11.00                          |
| Instead of Kf3: | Kh3   | Best continuation: b4 Rc7+ Kb2 |
|                 |       |                                |

7.6 Fireteam Index Prediction

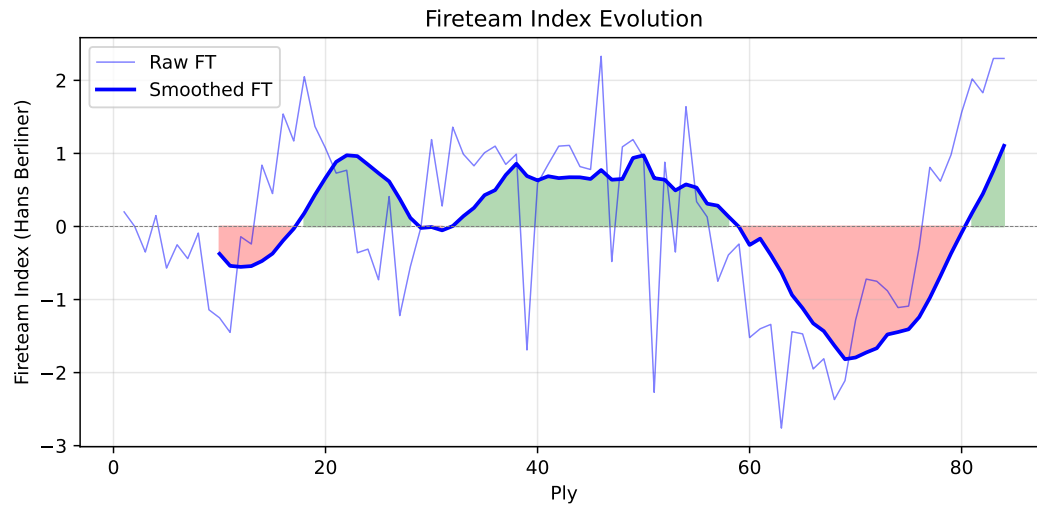
Tracking: Hans Berliner (Black)

| FTI Variant | Algorithm | Prediction | Max Streak |
|-------------|-----------|------------|------------|
| Standard    | Per-ply   | DRAW       | 9 plies    |
| Standard    | Windowed  | WIN        | 27 plies   |
| Truncated   | Per-ply   | DRAW       | 7 plies    |
| Truncated   | Windowed  | WIN        | 19 plies   |

Actual result: 0-1 (Hans Berliner’s outcome: WIN)

Standard: Per-ply ✗ Windowed ✓

Truncated: Per-ply ✗ Windowed ✓



## 7.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 700.4 seconds



## Chapter 8

# Berliner Hans J (USA) vs Husak Karel (CZE)

### 8.1 Game Information

|                |                       |
|----------------|-----------------------|
| <b>Event</b>   | Ch World (corr.)      |
| <b>Site</b>    | corr.                 |
| <b>Date</b>    | 1965.??.??            |
| <b>Round</b>   | 2                     |
| <b>White</b>   | Berliner Hans J (USA) |
| <b>Black</b>   | Husak Karel (CZE)     |
| <b>Result</b>  | 1/2-1/2               |
| <b>Opening</b> | D87 – Unknown         |

### 8.2 Player Statistics

#### 8.2.1 Berliner Hans J (USA) (White)

- Total moves: 33
- Accuracy: 96.0%
- Average centipawn loss: 8.2
- Best/Excellent moves: 17 / 7
- Good moves: 7
- Inaccuracies: 1
- Mistakes: 1
- Blunders: 0

#### 8.2.2 Husak Karel (CZE) (Black)

- Total moves: 33
- Accuracy: 96.3%

- Average centipawn loss: 7.6
- Best/Excellent moves: 19 / 4
- Good moves: 7
- Inaccuracies: 2
- Mistakes: 1
- Blunders: 0

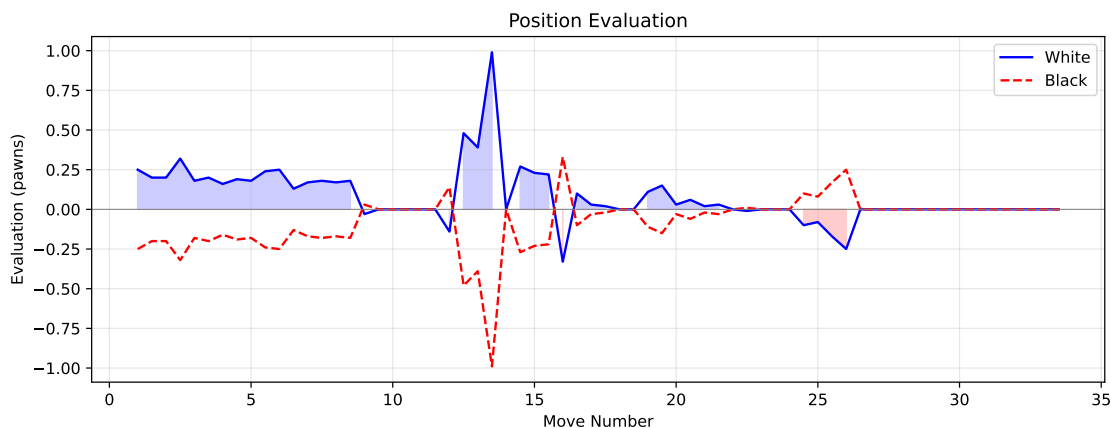
### 8.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

#### 8.3.1 Position Evaluation Over Time

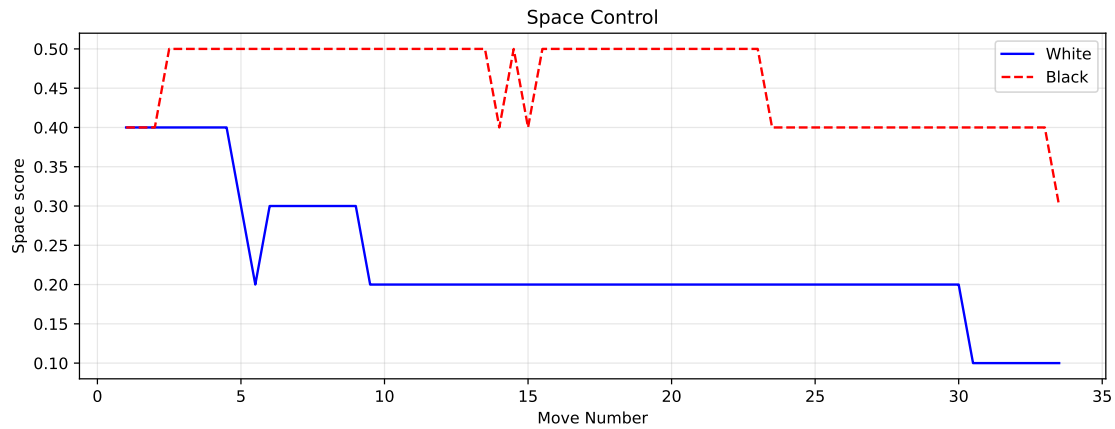
**Game Character Classification.** Evaluation range: minimum  $m_1 = -0.33$ , maximum  $m_2 = +0.99$ , spread  $d = m_2 - m_1 = 1.32$ .

This game is classified as **tense** (one-sided). A tense, one-sided game. White maintained the advantage throughout.



#### 8.3.2 Space Control

- White average: 0.23 (range: 0.10 to 0.40)
- Black average: 0.46 (range: 0.30 to 0.50)



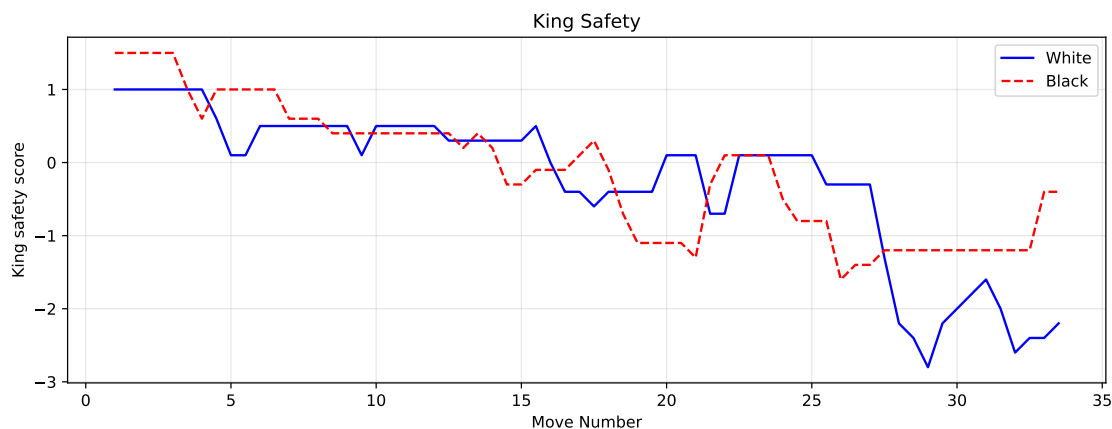
### 8.3.3 Mobility (Piece Activity)

- White average: 1.96 (range: 1.10 to 2.70)
- Black average: 1.85 (range: 1.15 to 2.30)



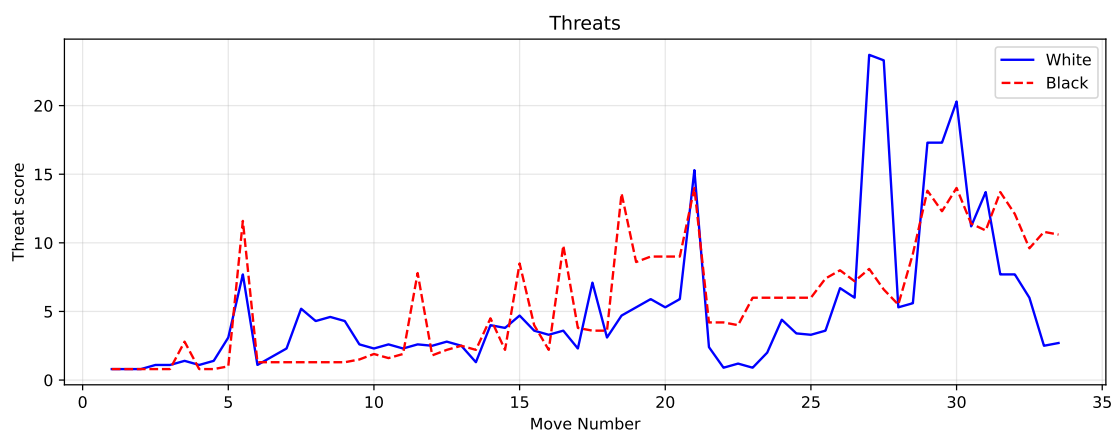
### 8.3.4 King Safety

- White average: -0.25
- Black average: -0.12



### 8.3.5 Threats

- White average: 5.17 (range: 0.80 to 23.70)
- Black average: 5.64 (range: 0.80 to 14.00)
- Average advantage: -0.47 (Black maintained more threats)



## 8.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Bc4 c5! 8. Ne2 O-O! 9. h4 cxd4! 10. cxd4! Nc6! 11. Be3! Na5! 12. Bd5 Qd6? *Black loses 61cp. Consider instead: e6, h5, Bd7.*

13. h5! Be6?! 14. hxg6? *White loses 97cp. Consider instead: Bxe6, h6, Bf4.*

14...hxg6 15. Bxe6! Qxe6 16. f3?! Nc4?! 17. Bh6! Bxh6 18. Rxh6 Kg7! 19. Qc1! Qa6 20. Kf2 Rh8! 21. Rxh8! Rxh8! 22. Qc3 Rc8! 23. Rc1! b5 24. d5+ f6! 25. Qd4 Qa3! 26. Rh1 Ne5 27. Qe3! Qxe3+! 28. Kxe3! Rc2! 29. Nd4! Nc4+ 30. Kd3! Rxg2! 31. Nxb5! Ne5+! 32. Ke3! Rxa2! 33. Rc1! f5

1/2-1/2



## 8.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

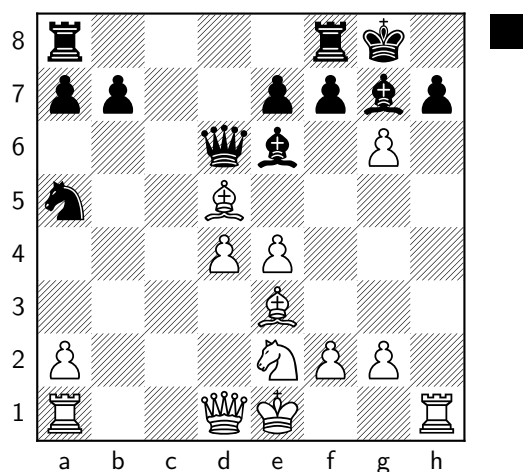
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 14. hxg6 ★

*Biggest evaluation swing(s) (99cp)*

Evaluation: +0.00



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.20  | 0.40  |
| Mobility (MG) | 5.30  | 4.40  |
| King Safety   | 0.30  | 0.20  |
| Threats       | 4.00  | 4.50  |

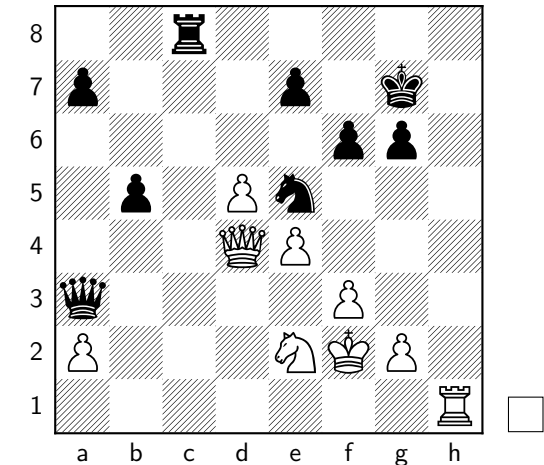
Instead of hxg6: Bxe6, h6, Bf4

Best continuation: hxg6 Bf4 Qa6

### Position 2: After 26...Ne5 ★

*Biggest evaluation swing(s) (25cp)*

Evaluation: +0.00



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.20  | 0.40  |
| Mobility (MG) | 4.00  | 4.00  |
| King Safety   | -0.30 | -1.40 |
| Threats       | 6.00  | 7.20  |

Instead of Ne5: a6, Rh8, Qa6

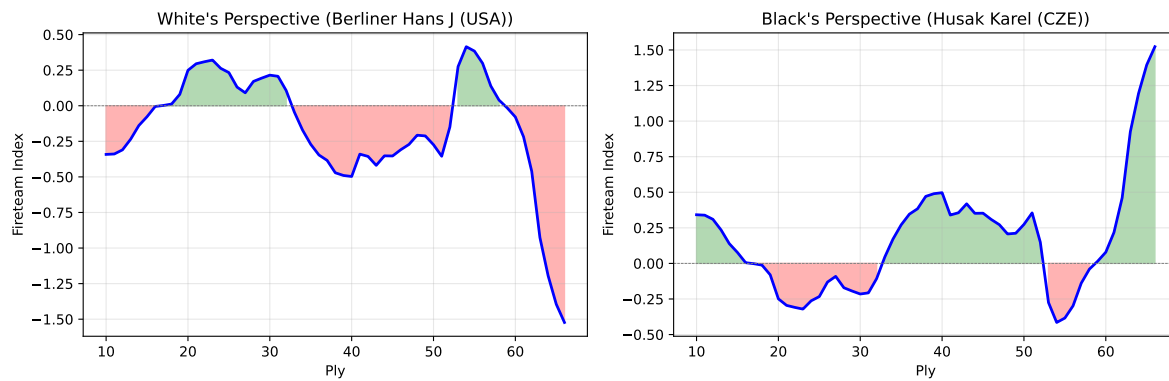
Best continuation: Qe3 Qxe3+ Kxe3

### 8.6 Fireteam Index Prediction

*This game ended in a draw. The Fireteam Index is analyzed from both players’ perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.*

|                                    | Player                | Algorithm | Prediction | Max Streak |
|------------------------------------|-----------------------|-----------|------------|------------|
| Standard FTI (1.0, 1.0, 1.0, 0.1)  | Berliner Hans J (USA) | Per-ply   | DRAW       | 5 plies    |
|                                    | Berliner Hans J (USA) | Windowed  | WIN        | 16 plies   |
|                                    | Husak Karel (CZE)     | Per-ply   | WIN        | 12 plies   |
|                                    | Husak Karel (CZE)     | Windowed  | WIN        | 20 plies   |
|                                    | Player                | Algorithm | Prediction | Max Streak |
| Truncated FTI (1.2, 0.0, 0.0, 0.6) | Berliner Hans J (USA) | Per-ply   | DRAW       | 3 plies    |
|                                    | Berliner Hans J (USA) | Windowed  | WIN        | 10 plies   |
|                                    | Husak Karel (CZE)     | Per-ply   | WIN        | 11 plies   |
|                                    | Husak Karel (CZE)     | Windowed  | WIN        | 29 plies   |

**Interpretation.** Both players achieved sustained positional dominance at different points in the game. The FT Index predicted wins for both sides, reflecting a hard-fought battle where momentum shifted. The draw is a reasonable outcome given the mutual winning chances.



## 8.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 343.4 seconds



## Chapter 9

# Berliner Hans J (USA) vs Hybl Jaroslav (CZE)

### 9.1 Game Information

|                |                            |
|----------------|----------------------------|
| <b>Event</b>   | Ch World (corr.)           |
| <b>Site</b>    | corr.                      |
| <b>Date</b>    | 1965.??.??                 |
| <b>Round</b>   | 1                          |
| <b>White</b>   | Berliner Hans J (USA)      |
| <b>Black</b>   | Hybl Jaroslav (CZE) (2172) |
| <b>Result</b>  | 1/2-1/2                    |
| <b>Opening</b> | D86 – Unknown              |

### 9.2 Player Statistics

#### 9.2.1 Berliner Hans J (USA) (White)

- Total moves: 24
- Accuracy: 96.4%
- Average centipawn loss: 7.3
- Best/Excellent moves: 12 / 3
- Good moves: 8
- Inaccuracies: 0
- Mistakes: 1
- Blunders: 0

#### 9.2.2 Hybl Jaroslav (CZE) (Black)

- Total moves: 24
- Accuracy: 97.5%

- Average centipawn loss: 5.0
- Best/Excellent moves: 14 / 1
- Good moves: 8
- Inaccuracies: 1
- Mistakes: 0
- Blunders: 0

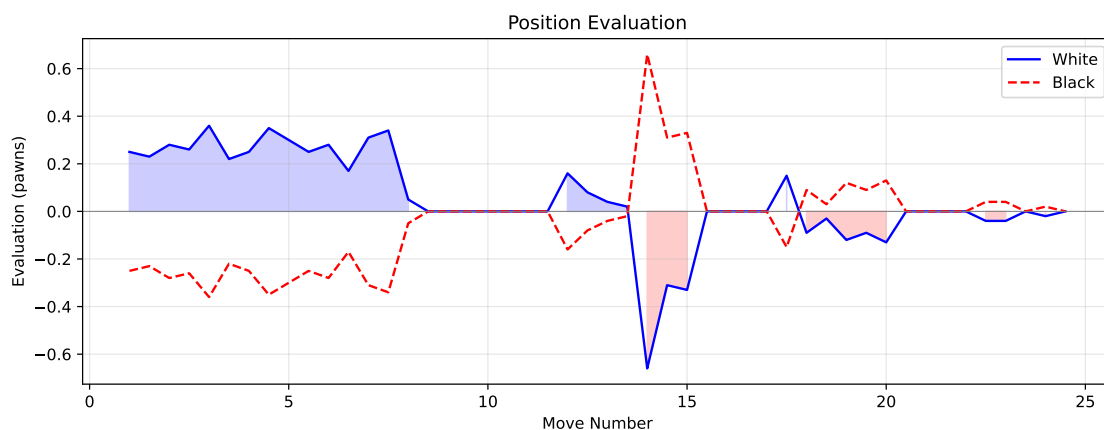
## 9.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 9.3.1 Position Evaluation Over Time

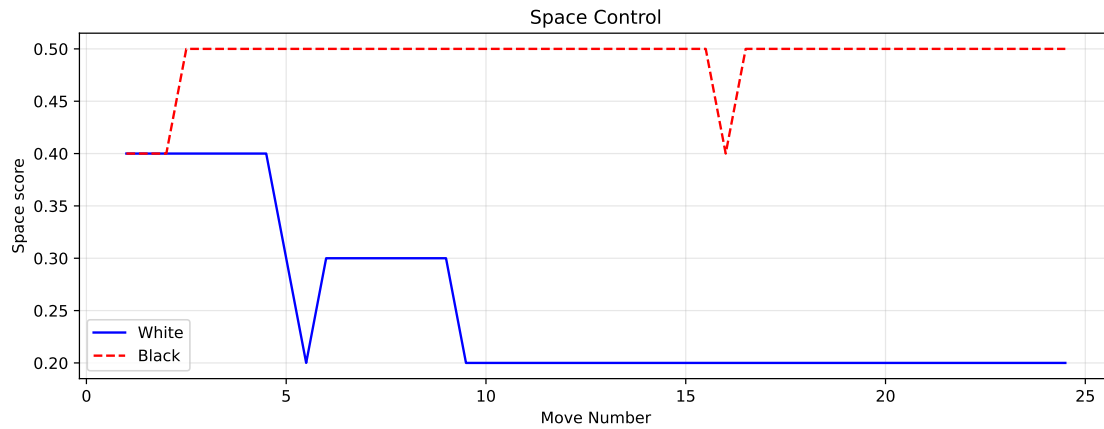
**Game Character Classification.** Evaluation range: minimum  $m_1 = -0.66$ , maximum  $m_2 = +0.36$ , spread  $d = m_2 - m_1 = 1.02$ .

This game is classified as **tense** (one-sided). A tense, one-sided game. Black maintained the advantage throughout.



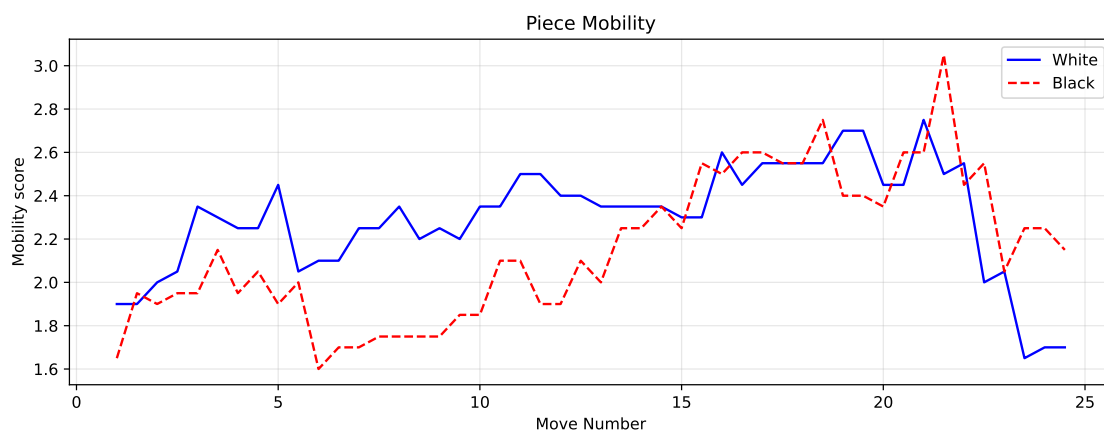
### 9.3.2 Space Control

- White average: 0.25 (range: 0.20 to 0.40)
- Black average: 0.49 (range: 0.40 to 0.50)



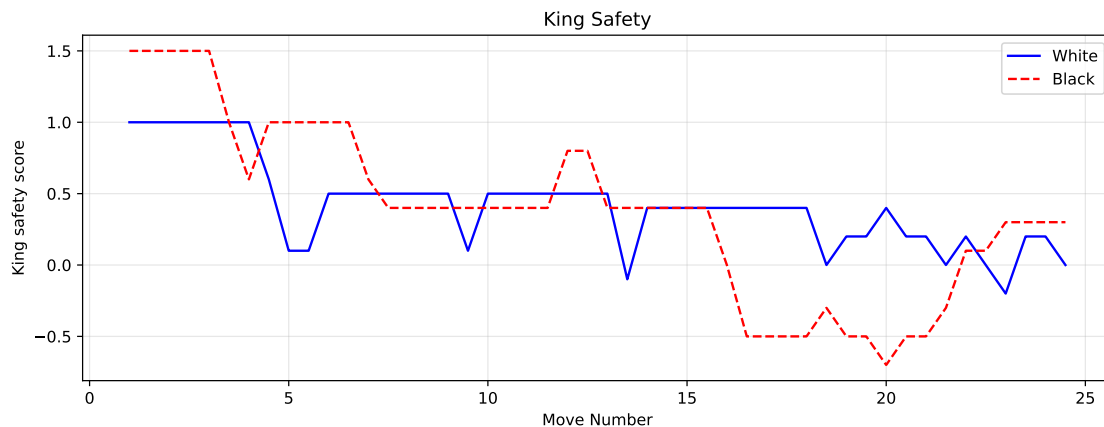
### 9.3.3 Mobility (Piece Activity)

- White average: 2.29 (range: 1.65 to 2.75)
- Black average: 2.16 (range: 1.60 to 3.05)



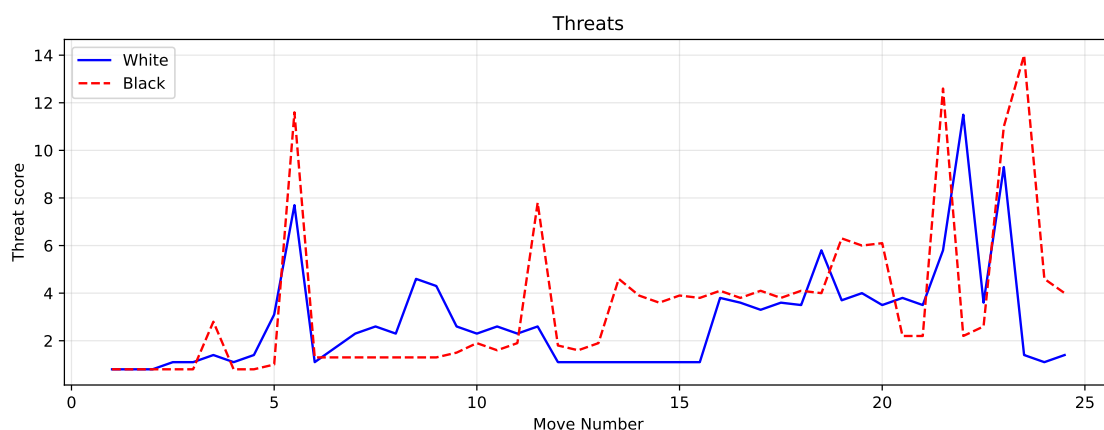
### 9.3.4 King Safety

- White average: 0.42
- Black average: 0.38



### 9.3.5 Threats

- White average: 2.82 (range: 0.80 to 11.50)
- Black average: 3.49 (range: 0.80 to 14.00)
- Average advantage: -0.67 (Black maintained more threats)



## 9.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Bc4 O-O! 8. h4 c5! 9. Ne2! cxd4 10. cxd4! Nc6! 11. Be3! Na5! 12. Bd3! Nc6! 13. e5 Qa5+! 14. Kf1? *White loses 75cp. Consider instead: Qd2.*

14...Rd8! 15. h5 Be6?! 16. hxg6! hxg6! 17. Rh4! Rd7 18. Qb1 Nxe5 19. dxe5! Rad8! 20. Nd4 Qxe5! 21. Qc2! Rxd4 22. Rxd4 Rxd4! 23. Bxd4! Qxd4! 24. Rd1! Qh4

1/2-1/2

## 9.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.



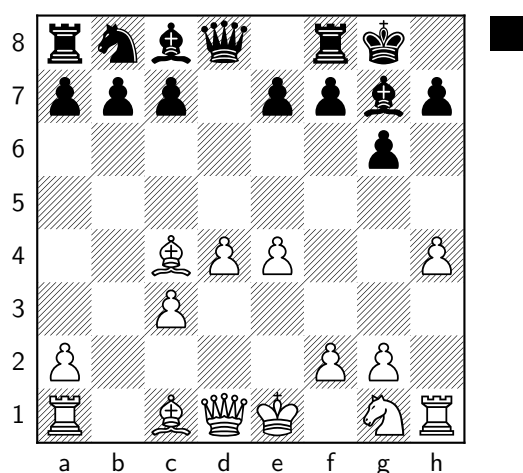
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 8. h4 ★

*Biggest evaluation swing(s) (29cp)*

Evaluation: +0.05



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.50  |
| Mobility (MG) | 4.70  | 3.50  |
| King Safety   | 0.50  | 0.40  |
| Threats       | 2.30  | 1.30  |

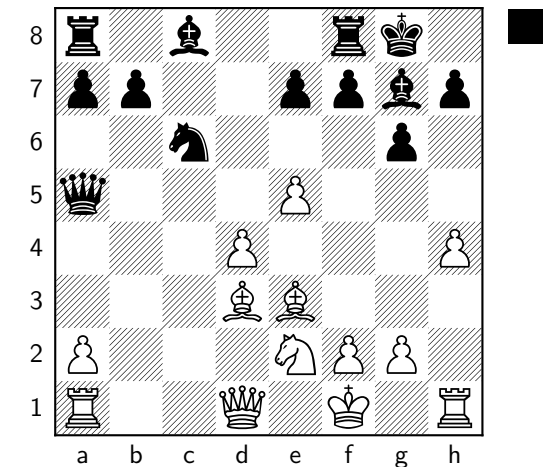
Instead of h4: Be3, Ne2, Nf3

Best continuation: c5 Ne2 cxd4

### Position 2: After 14. Kf1 ★

*Biggest evaluation swing(s) (68cp)*

Evaluation: -0.66



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.20  | 0.50  |
| Mobility (MG) | 4.70  | 4.50  |
| King Safety   | 0.40  | 0.40  |
| Threats       | 1.10  | 3.90  |

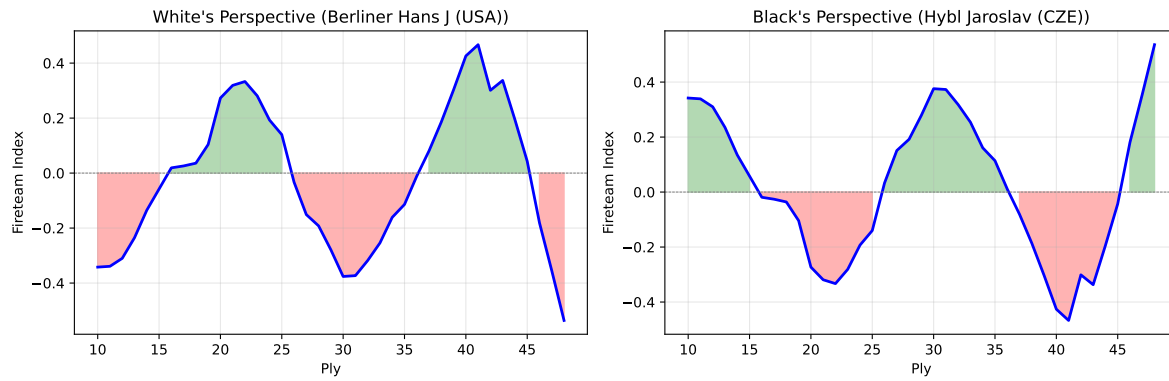
Instead of Kf1: Qd2      Best continuation: Rd8 h5 Nxe5

### 9.6 Fireteam Index Prediction

*This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.*

| Standard FTI (1.0, 1.0, 1.0, 0.1)  | Player                | Algorithm | Prediction | Max Streak |
|------------------------------------|-----------------------|-----------|------------|------------|
|                                    | Berliner Hans J (USA) | Per-ply   | DRAW       | 5 plies    |
|                                    | Berliner Hans J (USA) | Windowed  | DRAW       | 9 plies    |
|                                    | Hybl Jaroslav (CZE)   | Per-ply   | DRAW       | 5 plies    |
|                                    | Hybl Jaroslav (CZE)   | Windowed  | WIN        | 11 plies   |
| Truncated FTI (1.2, 0.0, 0.0, 0.6) | Player                | Algorithm | Prediction | Max Streak |
|                                    | Berliner Hans J (USA) | Per-ply   | DRAW       | 2 plies    |
|                                    | Berliner Hans J (USA) | Windowed  | DRAW       | 7 plies    |
|                                    | Hybl Jaroslav (CZE)   | Per-ply   | WIN        | 15 plies   |
|                                    | Hybl Jaroslav (CZE)   | Windowed  | WIN        | 25 plies   |

**Interpretation.** Black (Hybl Jaroslav (CZE)) achieved sustained positional dominance, with the FT Index predicting a win. However, the game ended in a draw, suggesting White successfully defended or Black failed to convert the advantage.



## 9.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 266.7 seconds



## Chapter 10

# Hans Berliner vs Kenneth Charles Messere

### 10.1 Game Information

|                |                                       |
|----------------|---------------------------------------|
| <b>Event</b>   | 5th Correspondence World Championship |
| <b>Site</b>    | ICCF                                  |
| <b>Date</b>    | 1965.07.01                            |
| <b>Round</b>   | 13                                    |
| <b>White</b>   | Hans Berliner                         |
| <b>Black</b>   | Kenneth Charles Messere               |
| <b>Result</b>  | 1-0                                   |
| <b>Opening</b> | D86 – Unknown                         |

### 10.2 Player Statistics

#### 10.2.1 Hans Berliner (White)

- Total moves: 22
- Accuracy: 92.1%
- Average centipawn loss: 16.4
- Best/Excellent moves: 9 / 2
- Good moves: 8
- Inaccuracies: 1
- Mistakes: 1
- Blunders: 1

#### 10.2.2 Kenneth Charles Messere (Black)

- Total moves: 21
- Accuracy: 77.6%

- Average centipawn loss: 50.6
- Best/Excellent moves: 3 / 1
- Good moves: 7
- Inaccuracies: 2
- Mistakes: 4
- Blunders: 4

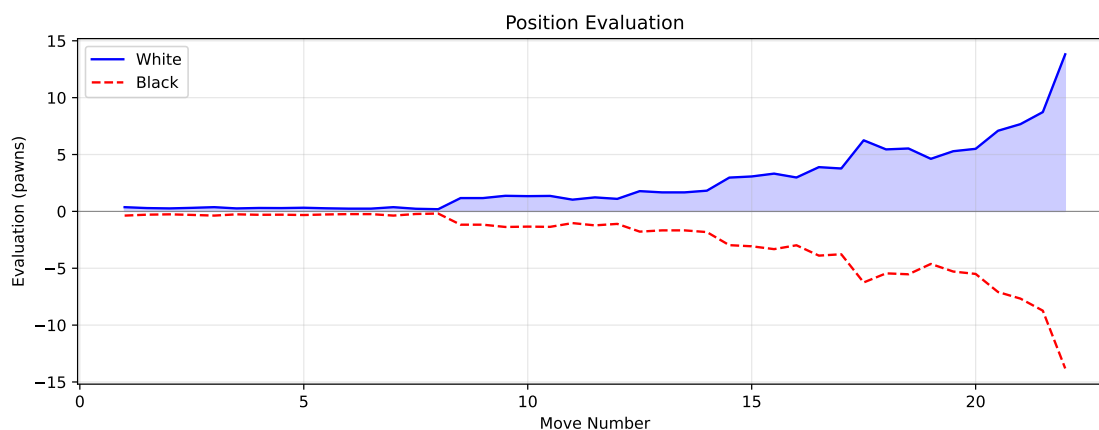
## 10.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 10.3.1 Position Evaluation Over Time

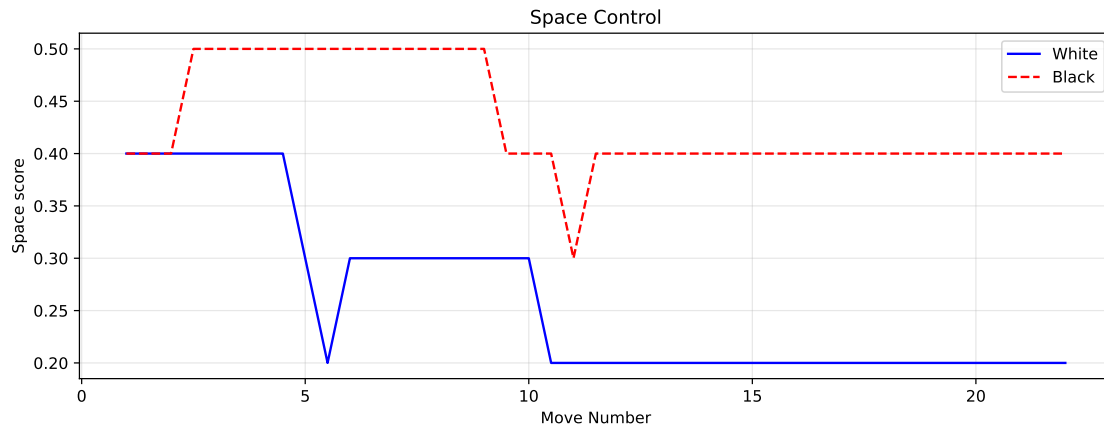
**Game Character Classification.** Evaluation range: minimum  $m_1 = +0.19$ , maximum  $m_2 = +13.81$ , spread  $d = m_2 - m_1 = 13.62$ .

This game is classified as **chaotic** (one-sided). A chaotic, one-sided game. White maintained the advantage throughout.



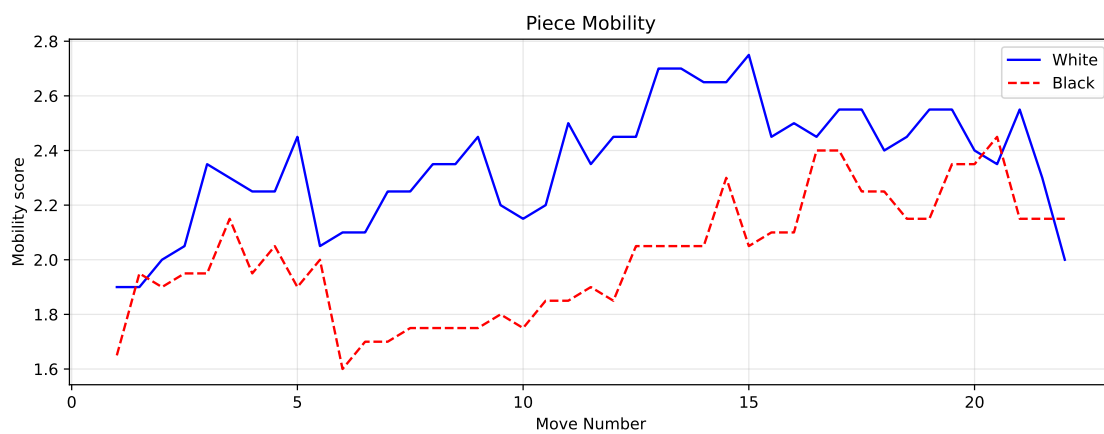
### 10.3.2 Space Control

- White average: 0.26 (range: 0.20 to 0.40)
- Black average: 0.43 (range: 0.30 to 0.50)



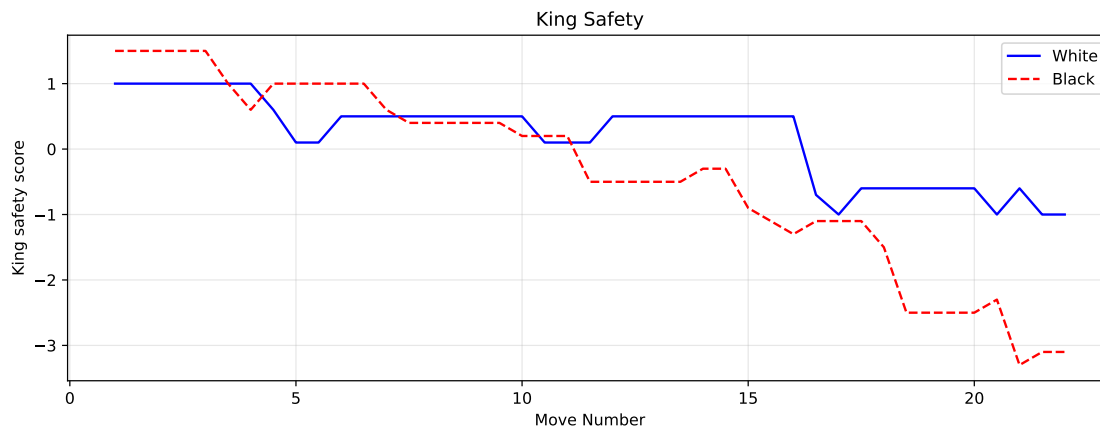
### 10.3.3 Mobility (Piece Activity)

- White average: 2.35 (range: 1.90 to 2.75)
- Black average: 2.01 (range: 1.60 to 2.45)



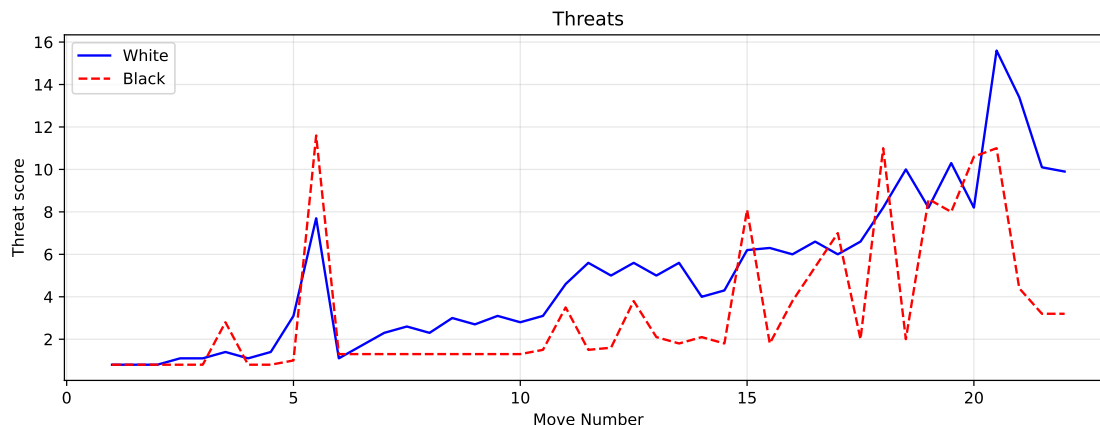
### 10.3.4 King Safety

- White average: 0.19
- Black average: -0.37



### 10.3.5 Threats

- White average: 5.01 (range: 0.80 to 15.60)
- Black average: 3.31 (range: 0.80 to 11.60)
- Average advantage: 1.69 (White maintained more threats)



## 10.4 Annotated Game

1. d4 Nf6 2. c4 g6 3. Nc3 d5 4. cxd5 Nxd5 5. e4 Nxc3 6. bxc3 Bg7 7. Bc4 O-O 8. Ne2! b6? *Black loses 92cp. Consider instead: c5, Nc6, Re8.*
9. h4 e5?! 10. h5! exd4! 11. hxg6 hxg6 12. cxd4 Qe7? *Black loses 68cp. Consider instead: b5, Qe7.*
13. Qd3! Re8! 14. Bh6! Nc6?? *Black loses 140cp. Consider instead: Bxh6, Be6, Ba6.*
15. Bxg7! Kxg7 16. Qd2?! Qxe4? *Black loses 64cp. Consider instead: Qb4.*
17. O-O-O Qe7?? *Black loses 244cp. Consider instead: Qf5, Nb4, Bf5.*
18. Qh6+? *White loses 89cp.*
- 18...Kf6? *Black loses 62cp.*
19. Rd3?? *White loses 144cp. Consider instead: Qh4+.*



19...Bg4?! 20. Rhh3 Nxd4?? *Black loses 142cp. Consider instead: Qb4, Qe4, Qd7.*

21. Rxd4 Bxe2?? *Black loses 125cp. Consider instead: Qe5.*

22. Rf4+

1-0

## 10.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

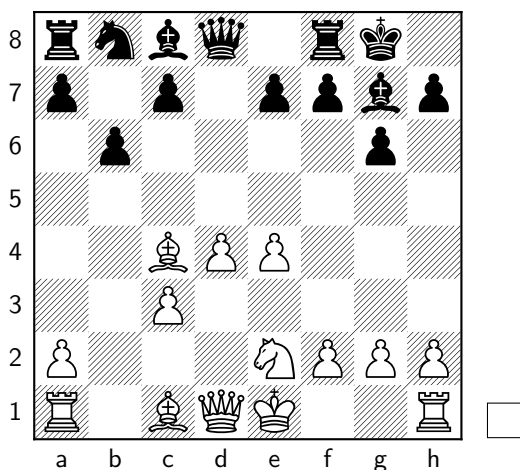
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 8...b6 ★

*Biggest evaluation swing(s) (98cp)*

Evaluation: +1.17



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.50  |
| Mobility (MG) | 4.70  | 3.50  |
| King Safety   | 0.50  | 0.40  |
| Threats       | 3.00  | 1.30  |

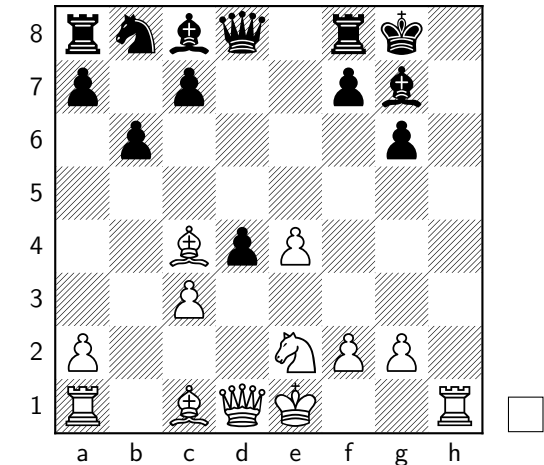
Instead of b6: c5, Nc6, Re8

Best continuation: h4 Nc6 e5

### Position 2: After 11...hxc6 ★

*Biggest evaluation swing(s) (20cp)*

Evaluation: +1.23



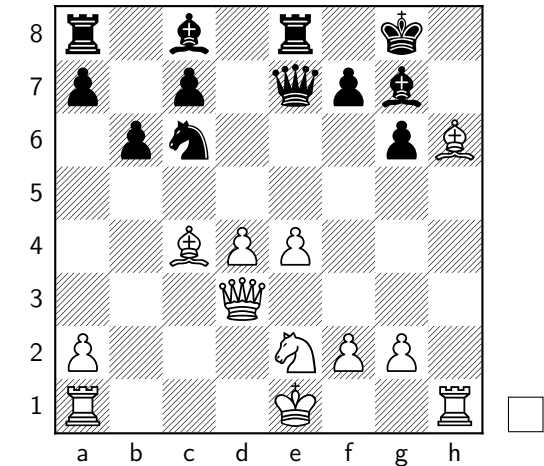
| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.20  | 0.40  |
| Mobility (MG) | 4.70  | 3.80  |
| King Safety   | 0.10  | -0.50 |
| Threats       | 5.60  | 1.50  |

Best continuation: Qd3 Re8 cxd4

Position 3: After 14...Nc6

Black blunders (+115cp)

Evaluation: +2.97



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.20  | 0.40  |
| Mobility (MG) | 5.30  | 4.60  |
| King Safety   | 0.50  | -0.30 |
| Threats       | 4.30  | 1.80  |

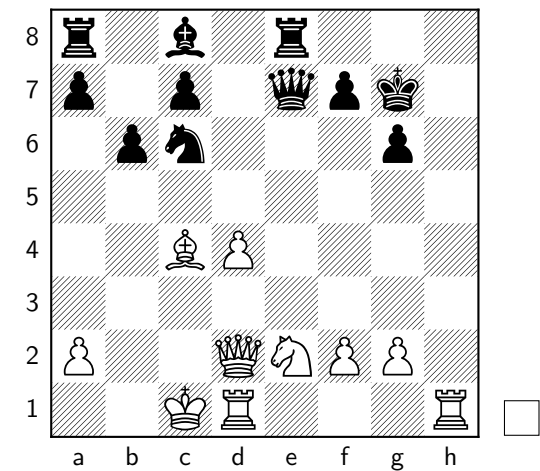
Instead of Nc6: Bxh6, Be6, Ba6

Best continuation: Bxg7 Kxg7 Qd2

Position 4: After 17...Qe7

Black blunders (+248cp)

Evaluation: +6.25



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.20  | 0.40  |
| Mobility (MG) | 5.10  | 4.50  |
| King Safety   | -0.60 | -1.10 |
| Threats       | 6.60  | 2.00  |

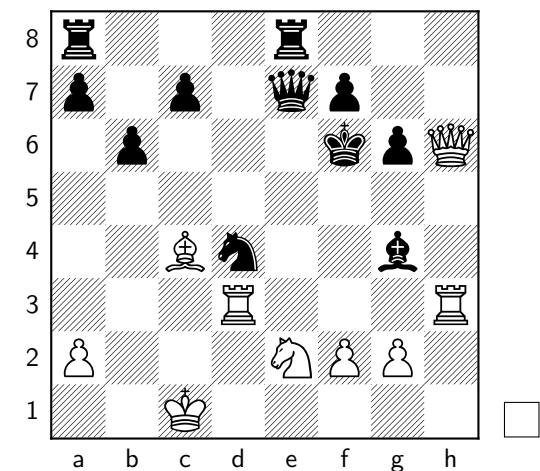
Instead of Qe7: Qf5, Nb4, Bf5

Best continuation: Qh6+ Kf6 Qh4+

Position 5: After 20...Nxd4

Black blunders (+159cp)

Evaluation: +7.09



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.20  | 0.40  |
| Mobility (MG) | 4.70  | 4.90  |
| King Safety   | -1.00 | -2.30 |
| Threats       | 15.60 | 11.00 |

Instead of Nxd4: Qb4, Qe4, Qd7

Best continuation: Qh4+ Kg7 Qh6+

## 10.6 Fireteam Index Prediction

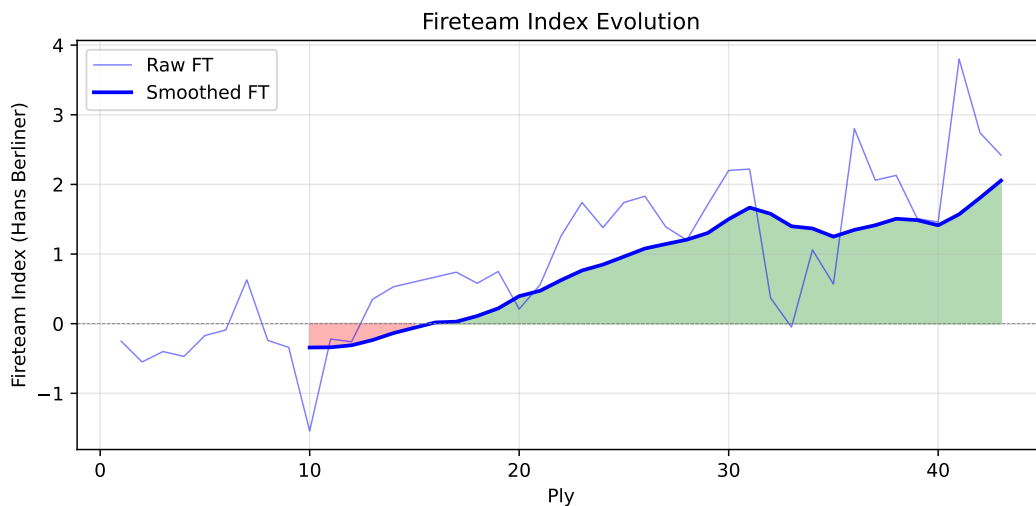
Tracking: **Hans Berliner** (White)

| FTI Variant | Algorithm | Prediction | Max Streak |
|-------------|-----------|------------|------------|
| Standard    | Per-ply   | WIN        | 16 plies   |
| Standard    | Windowed  | WIN        | 27 plies   |
| Truncated   | Per-ply   | WIN        | 12 plies   |
| Truncated   | Windowed  | WIN        | 27 plies   |

Actual result: 1-0 (Hans Berliner's outcome: WIN)

Standard: Per-ply ✓ Windowed ✓

Truncated: Per-ply ✓ Windowed ✓



## 10.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 1645.7 seconds

# Chapter 11

## Hans Berliner vs Julius Nielsen

### 11.1 Game Information

|                |                                       |
|----------------|---------------------------------------|
| <b>Event</b>   | 5th Correspondence World Championship |
| <b>Site</b>    | ICCF                                  |
| <b>Date</b>    | 1965.07.01                            |
| <b>Round</b>   | 7                                     |
| <b>White</b>   | Hans Berliner                         |
| <b>Black</b>   | Julius Nielsen                        |
| <b>Result</b>  | 1-0                                   |
| <b>Opening</b> | D35 – Unknown                         |

### 11.2 Player Statistics

#### 11.2.1 Hans Berliner (White)

- Total moves: 48
- Accuracy: 95.3%
- Average centipawn loss: 9.6
- Best/Excellent moves: 27 / 6
- Good moves: 10
- Inaccuracies: 3
- Mistakes: 2
- Blunders: 0

#### 11.2.2 Julius Nielsen (Black)

- Total moves: 47
- Accuracy: 91.5%
- Average centipawn loss: 17.7

- Best/Excellent moves: 17 / 7
- Good moves: 13
- Inaccuracies: 7
- Mistakes: 2
- Blunders: 1

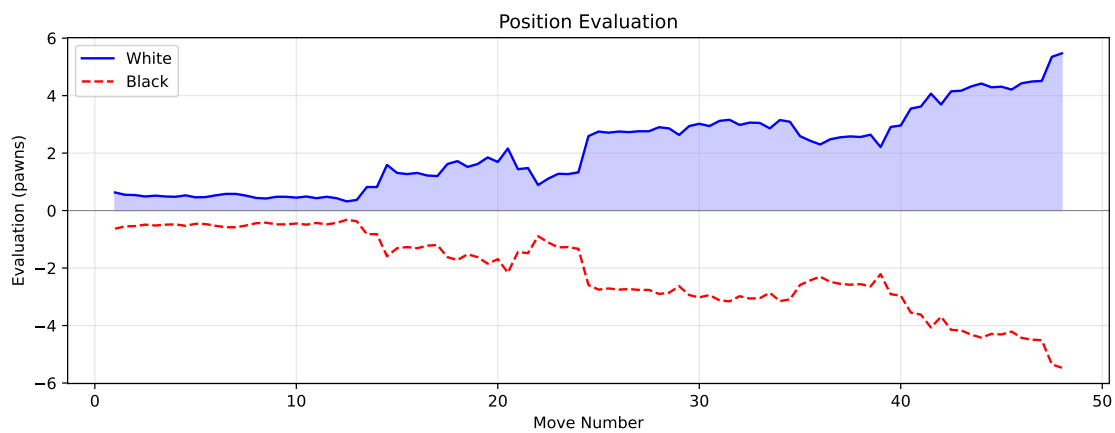
## 11.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 11.3.1 Position Evaluation Over Time

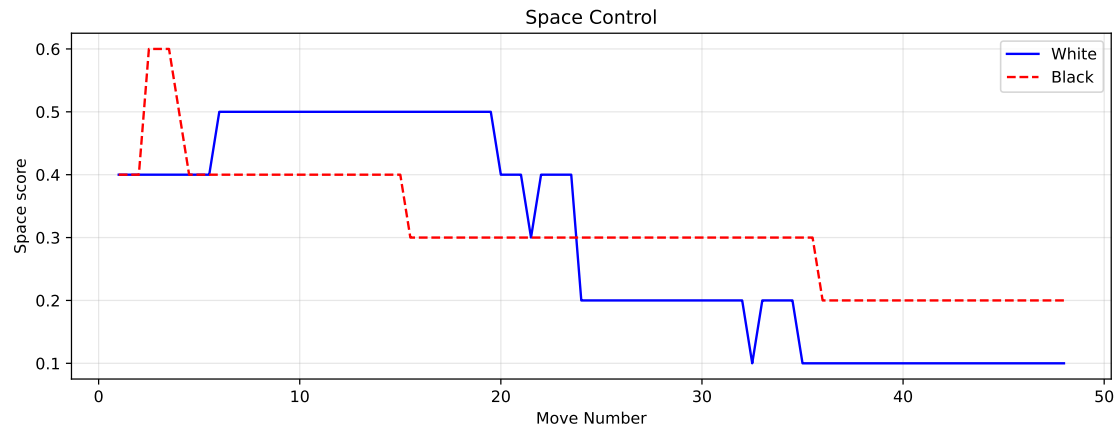
**Game Character Classification.** Evaluation range: minimum  $m_1 = +0.32$ , maximum  $m_2 = +5.47$ , spread  $d = m_2 - m_1 = 5.15$ .

This game is classified as **tactical** (one-sided). A tactical, one-sided game. White maintained the advantage throughout.



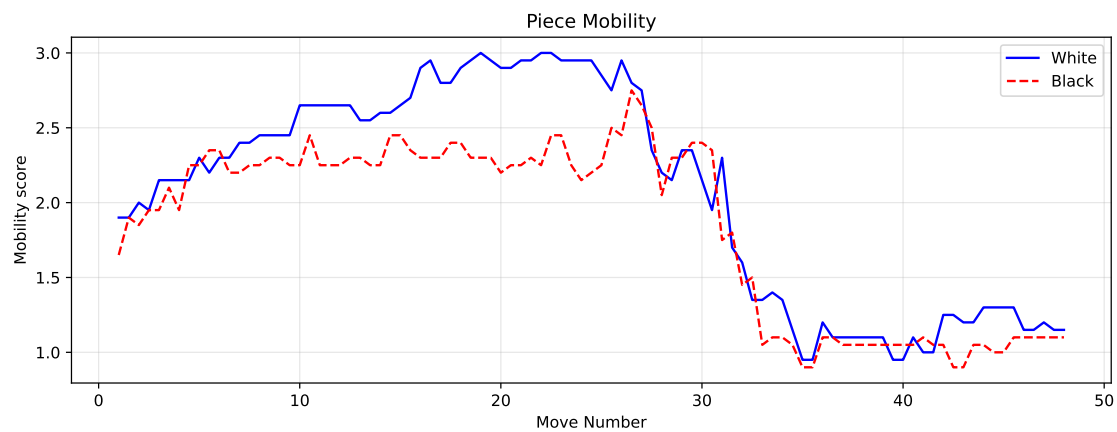
### 11.3.2 Space Control

- White average: 0.30 (range: 0.10 to 0.50)
- Black average: 0.31 (range: 0.20 to 0.60)



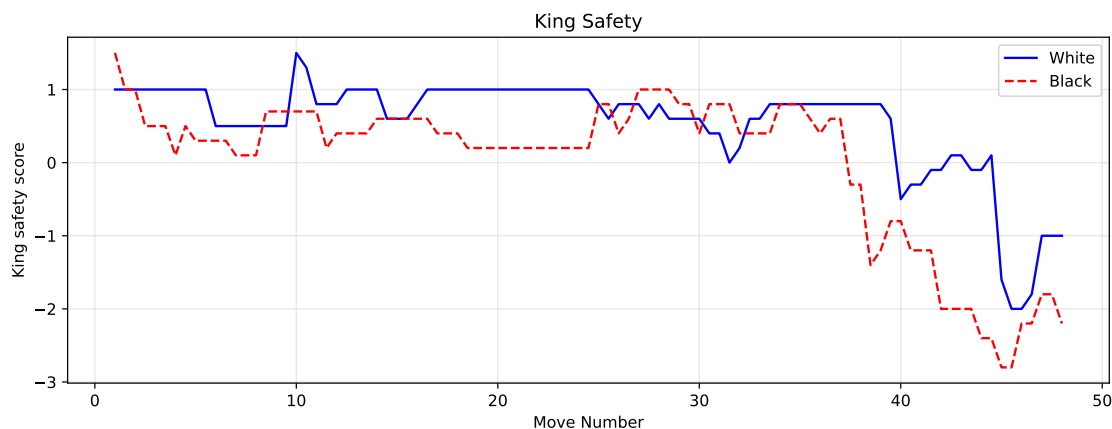
### 11.3.3 Mobility (Piece Activity)

- White average: 2.07 (range: 0.95 to 3.00)
- Black average: 1.84 (range: 0.90 to 2.75)



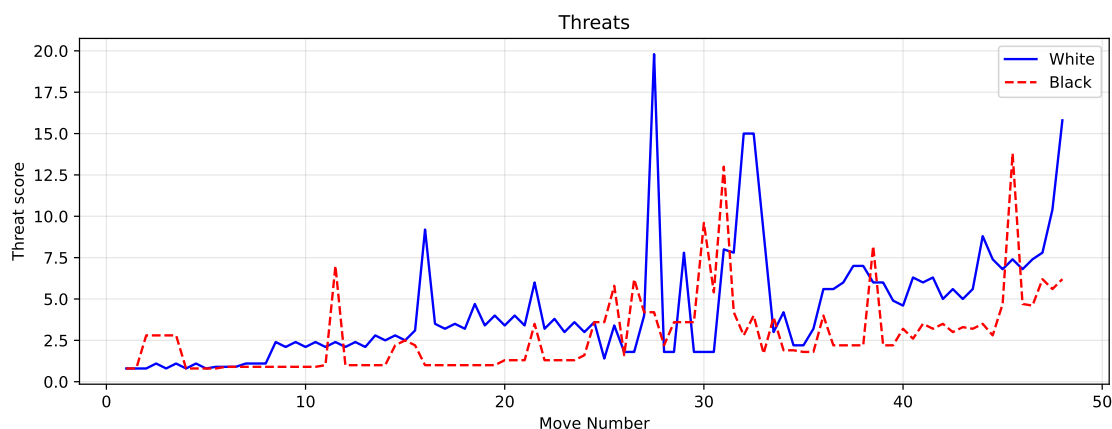
### 11.3.4 King Safety

- White average: 0.54
- Black average: 0.01



### 11.3.5 Threats

- White average: 4.33 (range: 0.80 to 19.80)
- Black average: 2.78 (range: 0.80 to 13.80)
- Average advantage: 1.55 (White maintained more threats)



## 11.4 Annotated Game

1. d4 d5 2. c4 e6 3. Nc3 Nf6 4. cxd5 exd5 5. Bg5 c6 6. e3 Nbd7 7. Bd3! Be7! 8. Nge2 O-O 9. Qc2! Re8 10. O-O Nf8 11. f3 h6 12. Bh4! Ne6! 13. Rad1 b6?! 14. Bf2 Qc7? *Black loses 87cp. Consider instead: b5, Bb7, Nh5.*

15. Kh1?! c5 16. Bg3! Qb7! 17. Be5! Rd8 18. Ng3! Ne8 19. Nf5 Bf6?! 20. f4 Nf8?! 21. Be2? *White loses 65cp. Consider instead: g4.*

21...cxd4 22. Bxd4? *White loses 64cp. Consider instead: exd4.*

22...Be6?! 23. Bf3! Qc8 24. e4 Ng6?? *Black loses 122cp. Consider instead: Kh8, Bxf5.*

25. Ne3 dxe4 26. Qxe4! Rxd4! 27. Qxa8! Qxa8 28. Bxa8! Nxf4! 29. Bc6 Nc7 30. Ng4! Bxg4! 31. Rxd4! Bxd4! 32. Rxf4 Bxc3! 33. bxc3! Be6! 34. Ra4! a5! 35. c4?! Kf8! 36. c5! bxc5! 37. Rxa5! Ke7! 38. Kg1! Kd6 39. Bf3?! Nd5?! 40. Kf2! Nb4?! 41. a4!



Bb3?! 42. Ra8 c4 43. a5! c3 44. Rc8! c2 45. Ke3! Be6 46. Rc3! Bf5 47. Kd2! h5?  
*Black loses 87cp. Consider instead: Bd3, Be6.*

48. Rc4!

1-0

## 11.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

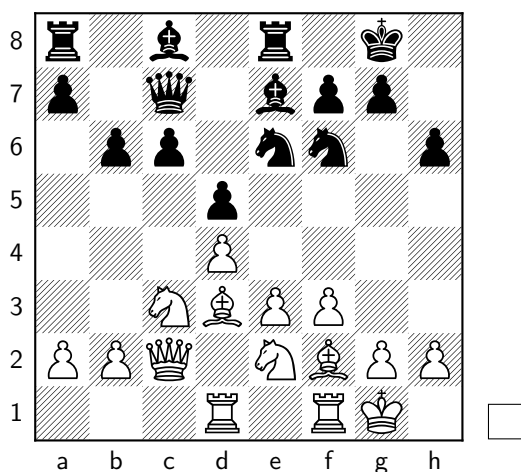
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 14...Qc7 ★

*Biggest evaluation swing(s) (77cp)*

Evaluation: +1.59



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.50  | 0.40  |
| Mobility (MG) | 5.20  | 4.90  |
| King Safety   | 0.60  | 0.60  |
| Threats       | 2.80  | 2.20  |

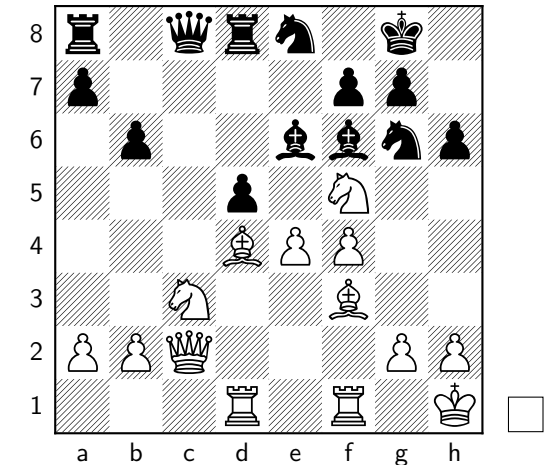
Instead of Qc7: b5, Bb7, Nh5

Best continuation: e4 dxe4 fxe4

### Position 2: After 24...Ng6

*Black blunders (+126cp)*

Evaluation: +2.59



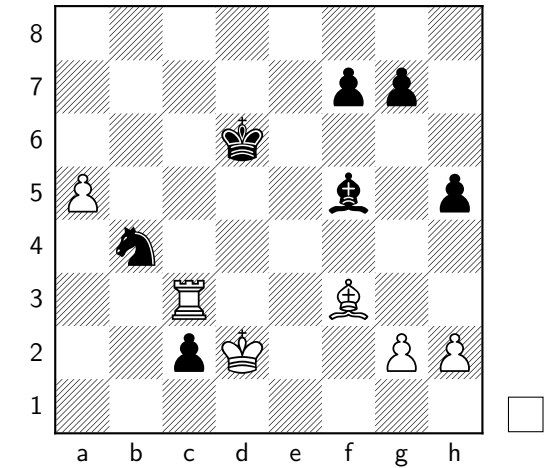
| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.20  | 0.30  |
| Mobility (MG) | 5.90  | 4.40  |
| King Safety   | 1.00  | 0.20  |
| Threats       | 3.60  | 3.60  |

Instead of Ng6:    Kh8, Bxf5                      Best continuation: Nxh6+ gxh6 f5

Position 3: After 47...h5 ★

*Biggest evaluation swing(s) (84cp)*

Evaluation: +5.35



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.10  | 0.20  |
| Mobility (MG) | 2.30  | 2.20  |
| King Safety   | -1.00 | -1.80 |
| Threats       | 10.40 | 5.60  |

Instead of h5:    Bd3, Be6                      Best continuation: Rc4 c1=Q+ Rxc1

## 11.6 Fireteam Index Prediction

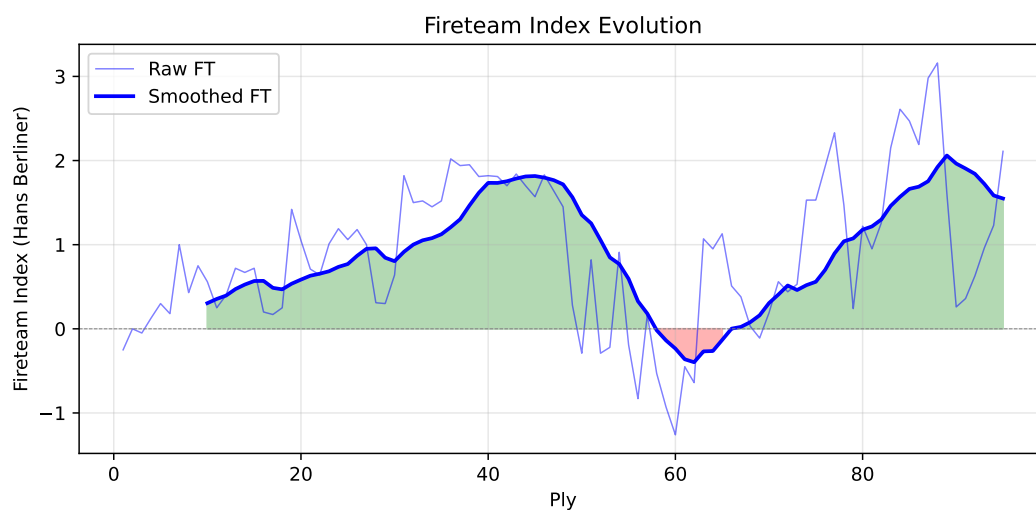
Tracking: **Hans Berliner** (White)

| FTI Variant | Algorithm | Prediction | Max Streak |
|-------------|-----------|------------|------------|
| Standard    | Per-ply   | WIN        | 33 plies   |
| Standard    | Windowed  | WIN        | 41 plies   |
| Truncated   | Per-ply   | WIN        | 25 plies   |
| Truncated   | Windowed  | WIN        | 36 plies   |

Actual result: 1-0 (Hans Berliner's outcome: WIN)

Standard: Per-ply ✓ Windowed ✓

Truncated: Per-ply ✓ Windowed ✓



## 11.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 820.9 seconds



## Chapter 12

# Hans Berliner vs Sture Valentin Nyman

### 12.1 Game Information

|                |                                       |
|----------------|---------------------------------------|
| <b>Event</b>   | 5th Correspondence World Championship |
| <b>Site</b>    | ICCF                                  |
| <b>Date</b>    | 1965.07.01                            |
| <b>Round</b>   | 15                                    |
| <b>White</b>   | Hans Berliner                         |
| <b>Black</b>   | Sture Valentin Nyman                  |
| <b>Result</b>  | 1-0                                   |
| <b>Opening</b> | A56 – Unknown                         |

### 12.2 Player Statistics

#### 12.2.1 Hans Berliner (White)

- Total moves: 46
- Accuracy: 95.2%
- Average centipawn loss: 9.8
- Best/Excellent moves: 20 / 8
- Good moves: 12
- Inaccuracies: 6
- Mistakes: 0
- Blunders: 0

#### 12.2.2 Sture Valentin Nyman (Black)

- Total moves: 45
- Accuracy: 90.9%

- Average centipawn loss: 19.1
- Best/Excellent moves: 12 / 9
- Good moves: 13
- Inaccuracies: 8
- Mistakes: 3
- Blunders: 0

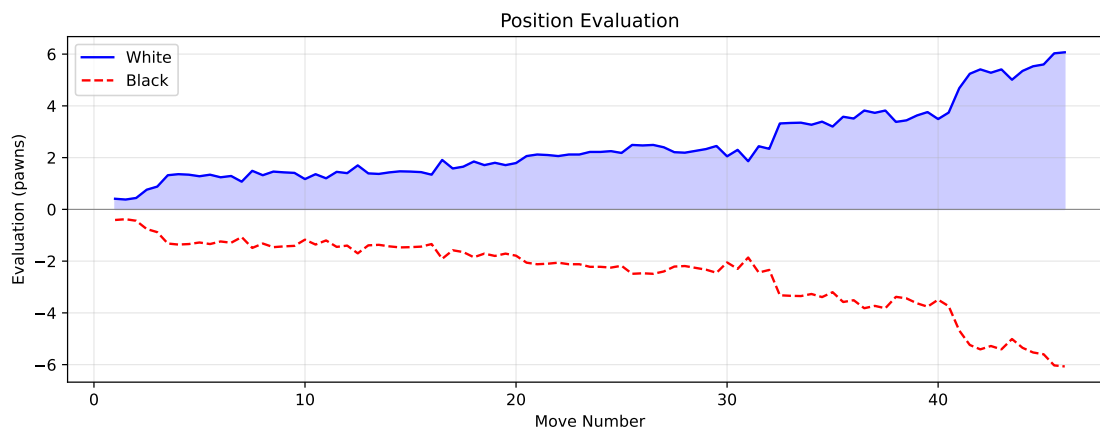
## 12.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 12.3.1 Position Evaluation Over Time

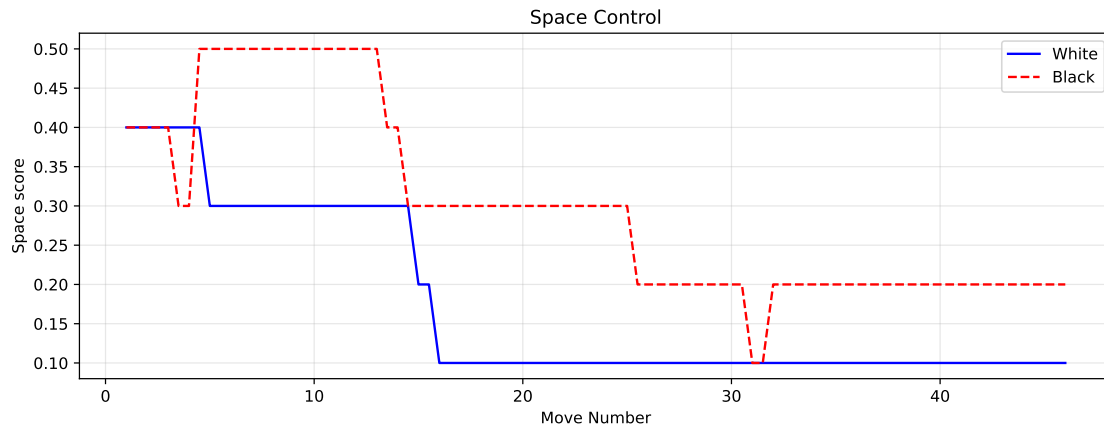
**Game Character Classification.** Evaluation range: minimum  $m_1 = +0.38$ , maximum  $m_2 = +6.07$ , spread  $d = m_2 - m_1 = 5.69$ .

This game is classified as **tactical** (one-sided). A tactical, one-sided game. White maintained the advantage throughout.



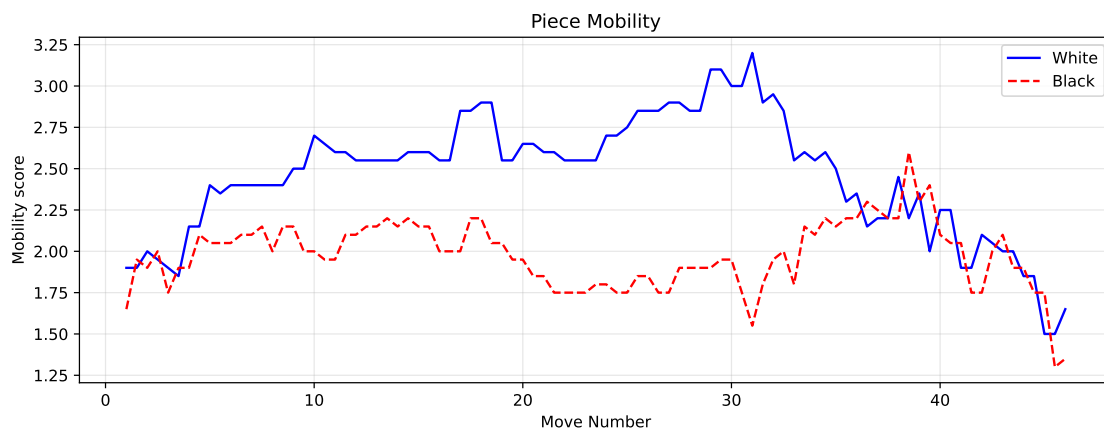
### 12.3.2 Space Control

- White average: 0.17 (range: 0.10 to 0.40)
- Black average: 0.30 (range: 0.10 to 0.50)



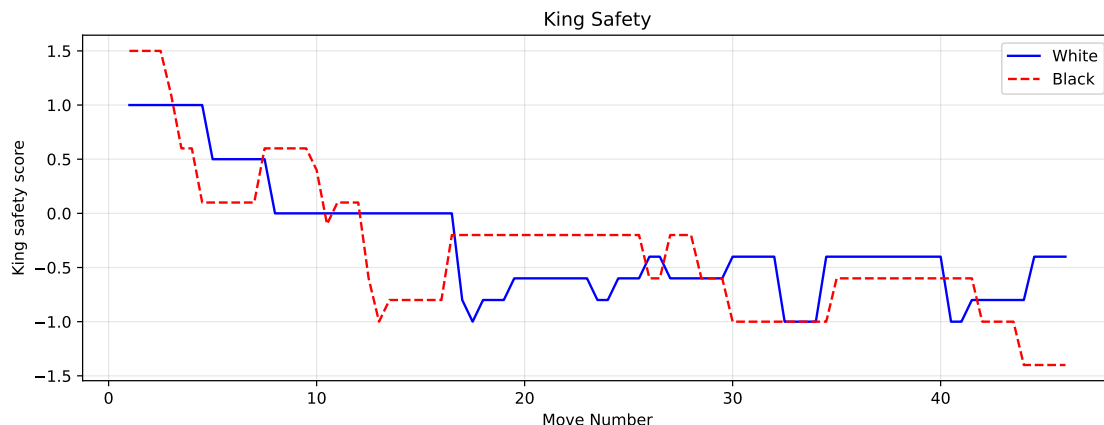
### 12.3.3 Mobility (Piece Activity)

- White average: 2.46 (range: 1.50 to 3.20)
- Black average: 1.98 (range: 1.30 to 2.60)



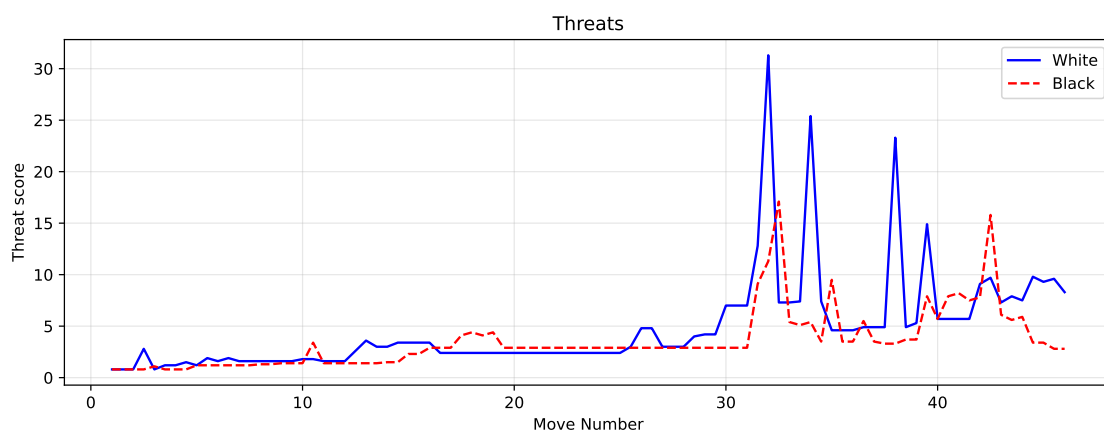
### 12.3.4 King Safety

- White average: -0.27
- Black average: -0.34



### 12.3.5 Threats

- White average: 4.79 (range: 0.80 to 31.30)
- Black average: 3.55 (range: 0.80 to 17.10)
- Average advantage: 1.24 (White maintained more threats)



## 12.4 Annotated Game

1. d4 Nf6 2. c4 c5?! 3. d5 e5?! 4. Nc3 d6 5. e4 g6 6. Bd3 Bg7 7. Nge2 O-O?! 8. f3 Na6 9. h4! Nc7 10. Bg5 h6 11. Be3 Bd7 12. Qd2 Kh7 13. h5?! g5 14. a3 b6 15. g4 Ng8 16. b4 Kh8? *Black loses 72cp. Consider instead: a5, Qf6.*

17. Kf2 Qf6! 18. Kg2! Rfb8! 19. Rhb1 Qd8! 20. Ra2 Qc8?! 21. Bc2 Bf8 22. Rab2 Ne8 23. Qd3! Ngf6 24. Ng3! Ng8 25. Bf2 f6?! 26. Nf5 Ne7 27. Ne3 Qc7 28. Qd1 Kh7 29. Ba4! Nc8?! 30. Nf5?! Ng7 31. bxc5?! Bxa4? *Black loses 61cp. Consider instead: dxc5.*

32. cxb6 Qxc4? *Black loses 90cp. Consider instead: Nxb6, axb6.*

33. Nxa4 axb6 34. Rb4! Qc7 35. Nxb5! Bxb5 36. Qb3! b5 37. Nc3! Na7! 38. Nxb5?! Nxb5! 39. Rxb5! Rxb5 40. Qxb5?! Qc2! 41. Rb4! Qa2 42. Qd7! Qxa3! 43. Rb7 Rg8! 44. Qe7?! Qa8?! 45. Rd7! Qf8?! 46. Qe6!



1-0

## 12.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

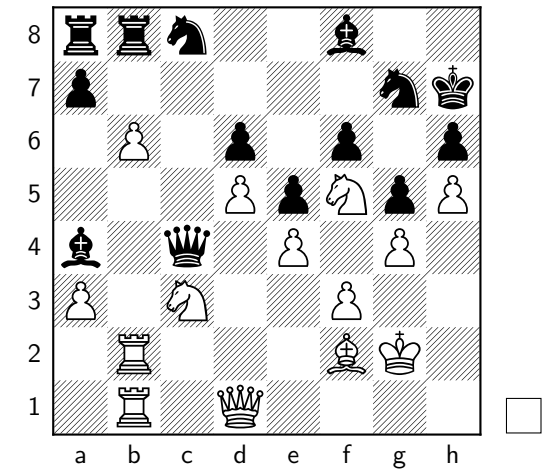
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 32...Qxc4 ★

*Biggest evaluation swing(s) (98cp)*

Evaluation: +3.32



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.10  | 0.20  |
| Mobility (MG) | 5.70  | 4.00  |
| King Safety   | -1.00 | -1.00 |
| Threats       | 7.30  | 17.10 |

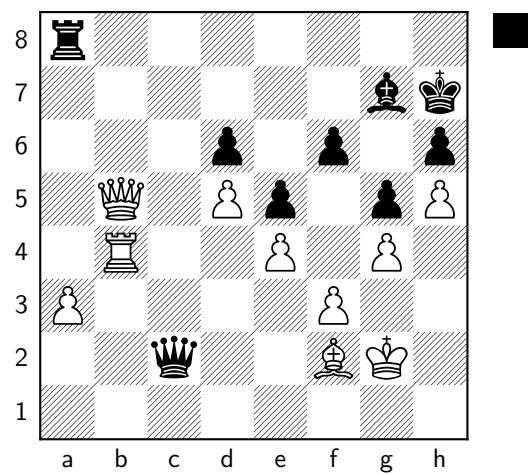
Instead of Qxc4: Nxb6, axb6

Best continuation: Nxa4 axb6 Rb4

### Position 2: After 41. Rb4 ★

*Biggest evaluation swing(s) (94cp)*

Evaluation: +4.68



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.10  | 0.20  |
| Mobility (MG) | 3.80  | 4.10  |
| King Safety   | -1.00 | -0.60 |
| Threats       | 5.70  | 8.20  |

Instead of Rb4:    Rb2                      Best continuation: Qc1 Qd7 Qc8

## 12.6 Fireteam Index Prediction

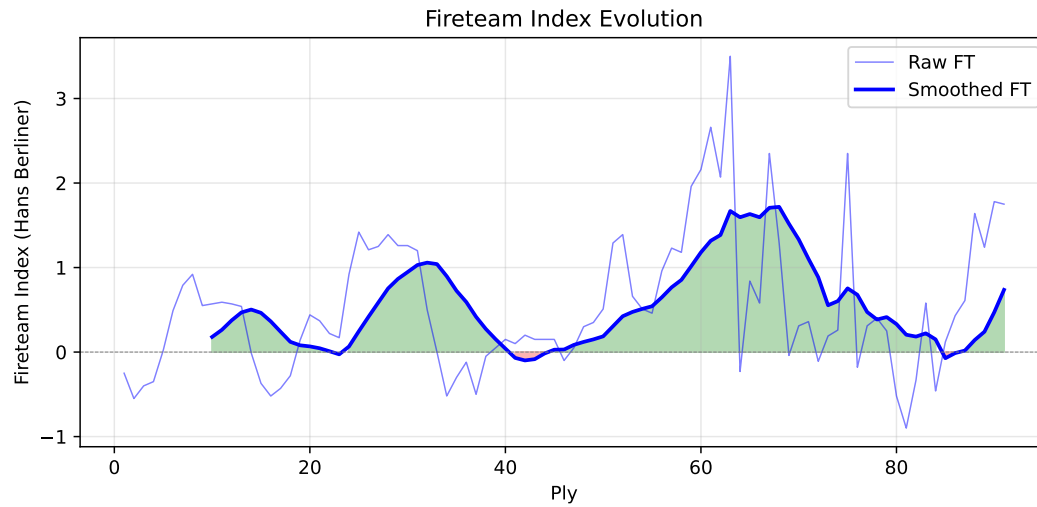
Tracking: **Hans Berliner** (White)

| FTI Variant | Algorithm | Prediction | Max Streak |
|-------------|-----------|------------|------------|
| Standard    | Per-ply   | WIN        | 17 plies   |
| Standard    | Windowed  | WIN        | 40 plies   |
| Truncated   | Per-ply   | DRAW       | 8 plies    |
| Truncated   | Windowed  | WIN        | 29 plies   |

Actual result: 1-0 (Hans Berliner's outcome: WIN)

Standard: Per-ply ✓    Windowed ✓

Truncated: Per-ply ✗    Windowed ✓



## 12.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 724.5 seconds



# Chapter 13

## Josef Richter vs Hans Berliner

### 13.1 Game Information

|                |                                       |
|----------------|---------------------------------------|
| <b>Event</b>   | 5th Correspondence World Championship |
| <b>Site</b>    | ICCF                                  |
| <b>Date</b>    | 1965.07.01                            |
| <b>Round</b>   | 8                                     |
| <b>White</b>   | Josef Richter                         |
| <b>Black</b>   | Hans Berliner                         |
| <b>Result</b>  | 0-1                                   |
| <b>Opening</b> | B03 – Unknown                         |

### 13.2 Player Statistics

#### 13.2.1 Josef Richter (White)

- Total moves: 39
- Accuracy: 90.8%
- Average centipawn loss: 19.2
- Best/Excellent moves: 8 / 12
- Good moves: 12
- Inaccuracies: 3
- Mistakes: 4
- Blunders: 0

#### 13.2.2 Hans Berliner (Black)

- Total moves: 39
- Accuracy: 96.3%
- Average centipawn loss: 7.6

- Best/Excellent moves: 24 / 6
- Good moves: 6
- Inaccuracies: 1
- Mistakes: 2
- Blunders: 0

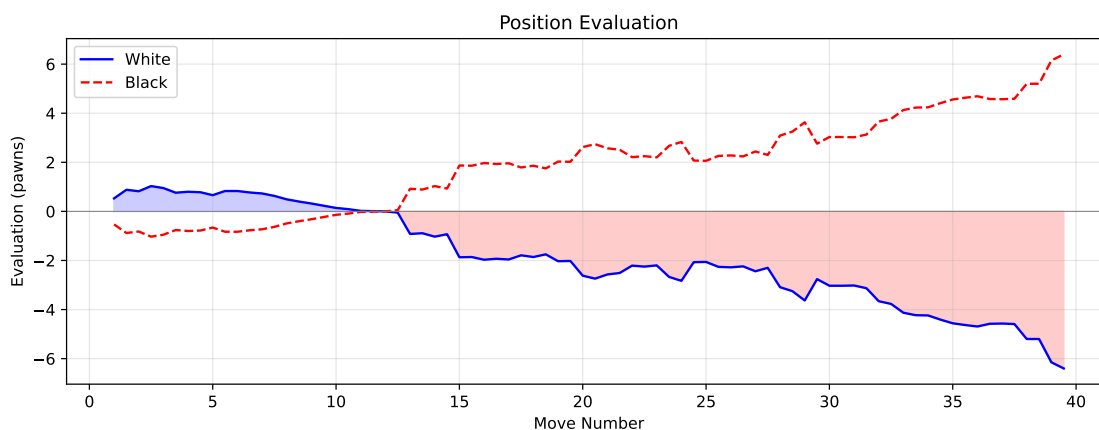
### 13.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

#### 13.3.1 Position Evaluation Over Time

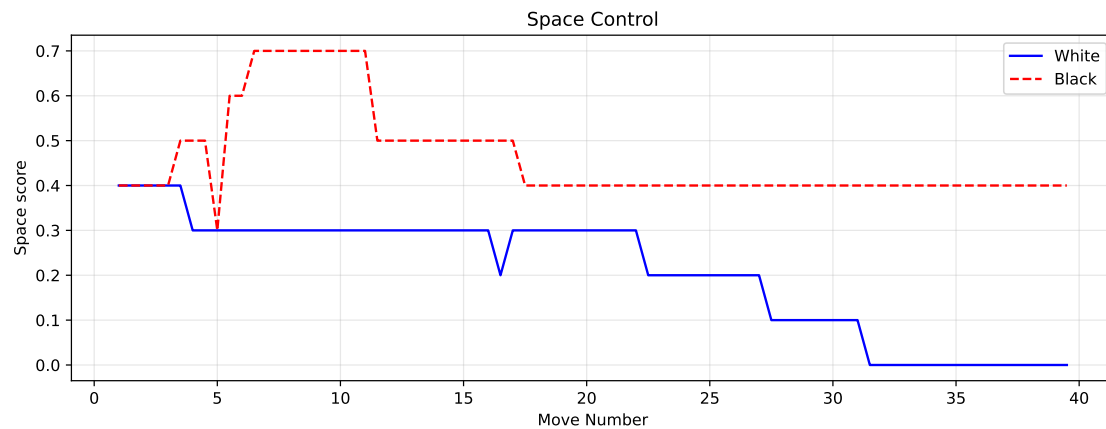
**Game Character Classification.** Evaluation range: minimum  $m_1 = -6.40$ , maximum  $m_2 = +1.03$ , spread  $d = m_2 - m_1 = 7.43$ .

This game is classified as **chaotic** (seesaw). A chaotic seesaw battle with the advantage changing hands.



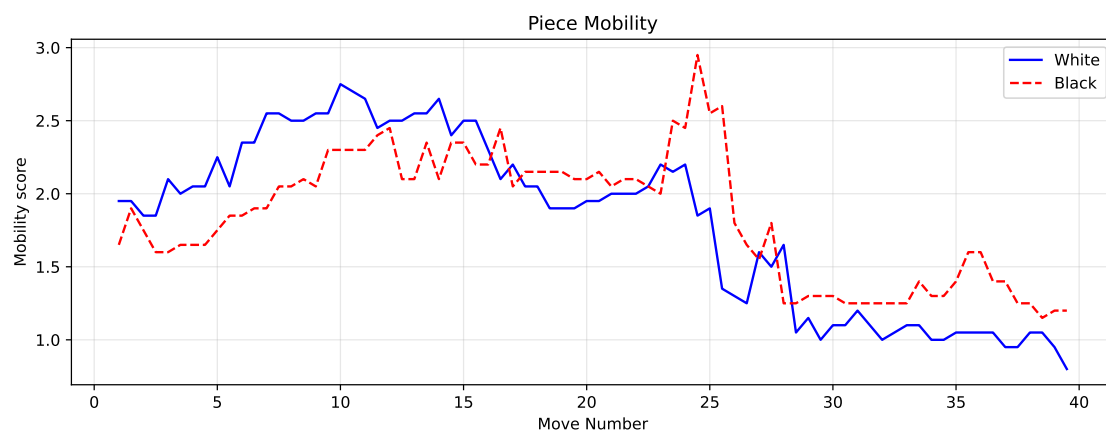
#### 13.3.2 Space Control

- White average: 0.21 (range: 0.00 to 0.40)
- Black average: 0.46 (range: 0.30 to 0.70)



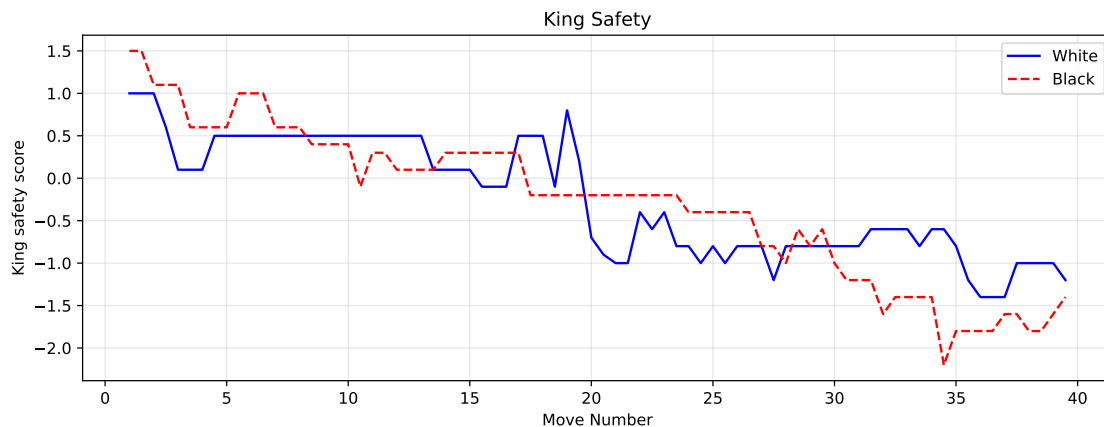
### 13.3.3 Mobility (Piece Activity)

- White average: 1.82 (range: 0.80 to 2.75)
- Black average: 1.84 (range: 1.15 to 2.95)



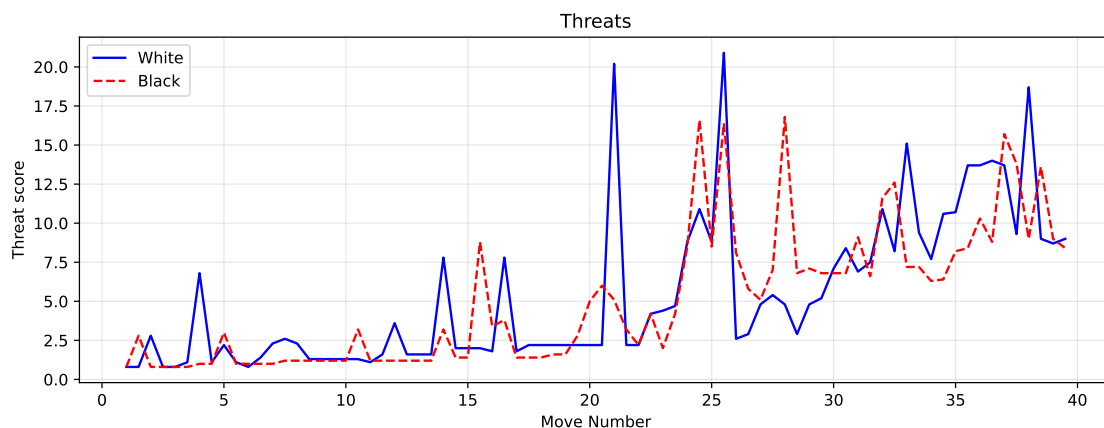
### 13.3.4 King Safety

- White average: -0.24
- Black average: -0.29



### 13.3.5 Threats

- White average: 5.37 (range: 0.80 to 20.90)
- Black average: 5.09 (range: 0.80 to 16.80)
- Average advantage: 0.28 (White maintained more threats)



## 13.4 Annotated Game

1. e4 Nf6?! 2. e5 Nd5 3. d4 d6 4. c4 Nb6 5. exd6 cxd6 6. Nc3 g6 7. Bg5 Bg7! 8. Rc1 O-O! 9. h3 Nc6! 10. Nf3 h6! 11. Be3 e5 12. d5 Ne7! 13. Qb3? *White loses 92cp. Consider instead: Bd3, g4, Be2.*

13...Nf5 14. Bxb6 axb6! 15. Be2? *White loses 98cp. Consider instead: Bd3.*

15...Nd4 16. Qd1 Nxf3+! 17. Bxf3! f5 18. Be2! e4 19. O-O Qh4 20. Kh2?! Be5+! 21. g3 Qg5 22. Kh1! f4! 23. Bg4! fxg3! 24. Nxe4! Bxg4? *Black loses 68cp. Consider instead: g2+.*

25. Qxg4 Qxg4! 26. hxc4 Rxa2 27. fxg3 Rxb2! 28. Rxf8+?! Kxf8! 29. g5 hxc5? *Black loses 83cp. Consider instead: Kg7.*



30. Rf1+ Kg7! 31. Nxg5 Bxg3! 32. Rf7+?! Kh6! 33. Ne4 Be5! 34. Nf2 Kg5! 35. Nd3 Rh2+! 36. Kg1 Bg3! 37. Ne1! Rb2! 38. Nd3? *White loses 78cp. Consider instead: Nf3+.*

38...Rd2 39. Nc1? *White loses 60cp. Consider instead: Rf3, Nb4.*

39...Bf4!

0-1

## 13.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

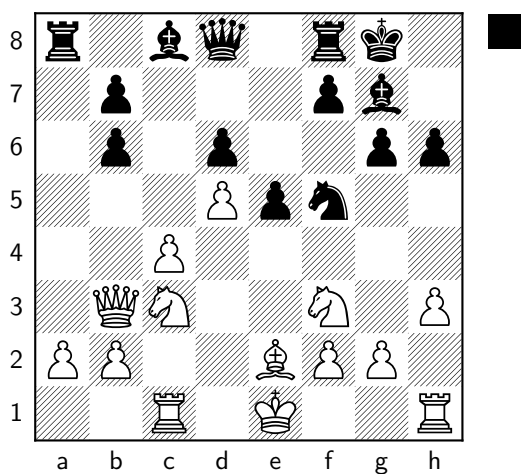
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 15. Be2 ★

*Biggest evaluation swing(s) (94cp)*

Evaluation: -1.87



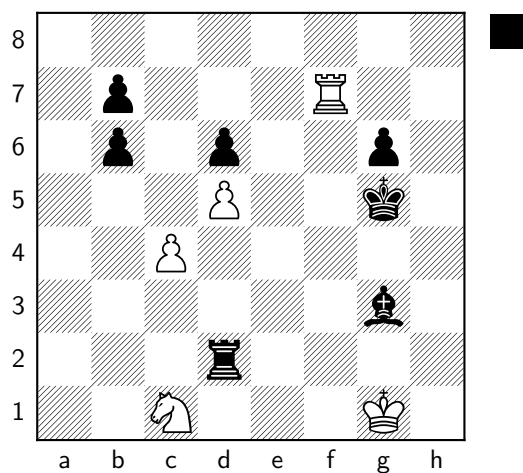
| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.50  |
| Mobility (MG) | 5.00  | 4.70  |
| King Safety   | 0.10  | 0.30  |
| Threats       | 2.00  | 1.40  |

Instead of Be2: Bd3

Best continuation: Re8 Qc2 Nd4

**Position 2: After 39. Nc1 ★***Biggest evaluation swing(s) (95cp)*

Evaluation: -6.15



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.00  | 0.40  |
| Mobility (MG) | 1.90  | 2.40  |
| King Safety   | -1.00 | -1.60 |
| Threats       | 8.70  | 9.00  |

Instead of Nc1: Rf3, Nb4

Best continuation: Bf4 Kf1 Rd1+

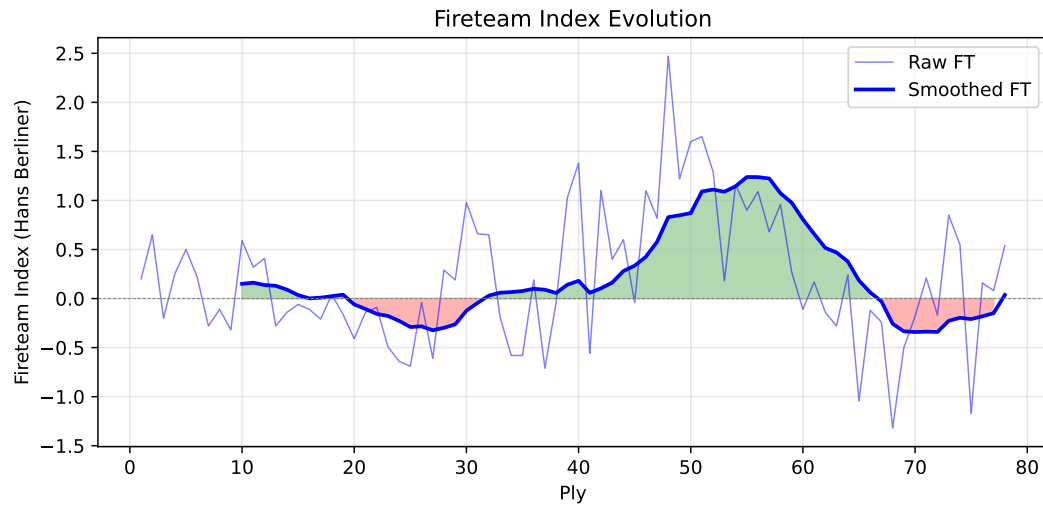
**13.6 Fireteam Index Prediction**Tracking: **Hans Berliner** (Black)

| FTI Variant | Algorithm | Prediction | Max Streak |
|-------------|-----------|------------|------------|
| Standard    | Per-ply   | WIN        | 14 plies   |
| Standard    | Windowed  | WIN        | 35 plies   |
| Truncated   | Per-ply   | DRAW       | 9 plies    |
| Truncated   | Windowed  | WIN        | 17 plies   |

Actual result: 0-1 (Hans Berliner's outcome: WIN)

Standard: Per-ply ✓ Windowed ✓

Truncated: Per-ply ✗ Windowed ✓



## 13.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 879.6 seconds



## Chapter 14

# Berliner Hans J (USA) vs Rokhlin Yakov (RUS)

### 14.1 Game Information

|                |                       |
|----------------|-----------------------|
| <b>Event</b>   | Ch World (corr.)      |
| <b>Site</b>    | corr.                 |
| <b>Date</b>    | 1965.??.??            |
| <b>Round</b>   | 9                     |
| <b>White</b>   | Berliner Hans J (USA) |
| <b>Black</b>   | Rokhlin Yakov (RUS)   |
| <b>Result</b>  | 1-0                   |
| <b>Opening</b> | E31 – Unknown         |

### 14.2 Player Statistics

#### 14.2.1 Berliner Hans J (USA) (White)

- Total moves: 46
- Accuracy: 90.2%
- Average centipawn loss: 20.5
- Best/Excellent moves: 15 / 12
- Good moves: 10
- Inaccuracies: 4
- Mistakes: 4
- Blunders: 1

#### 14.2.2 Rokhlin Yakov (RUS) (Black)

- Total moves: 46
- Accuracy: 88.0%

- Average centipawn loss: 25.6
- Best/Excellent moves: 17 / 2
- Good moves: 14
- Inaccuracies: 7
- Mistakes: 4
- Blunders: 2

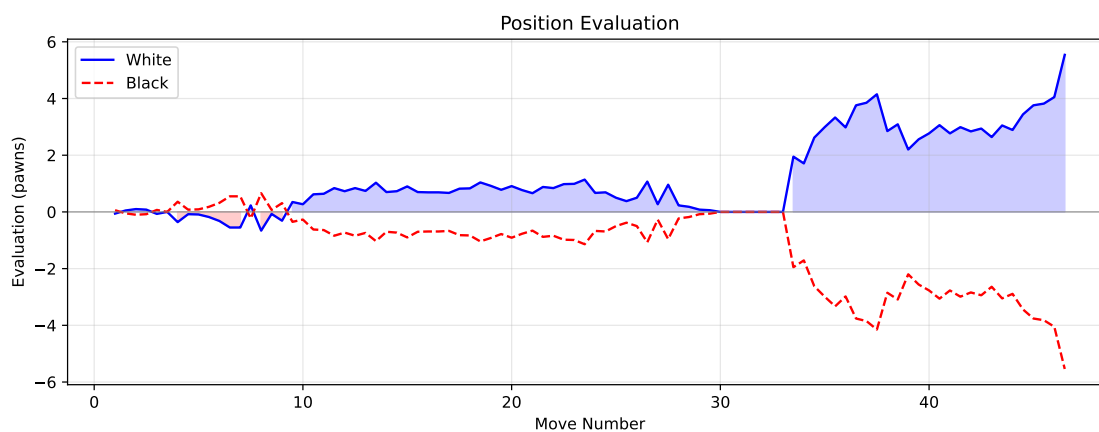
### 14.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

#### 14.3.1 Position Evaluation Over Time

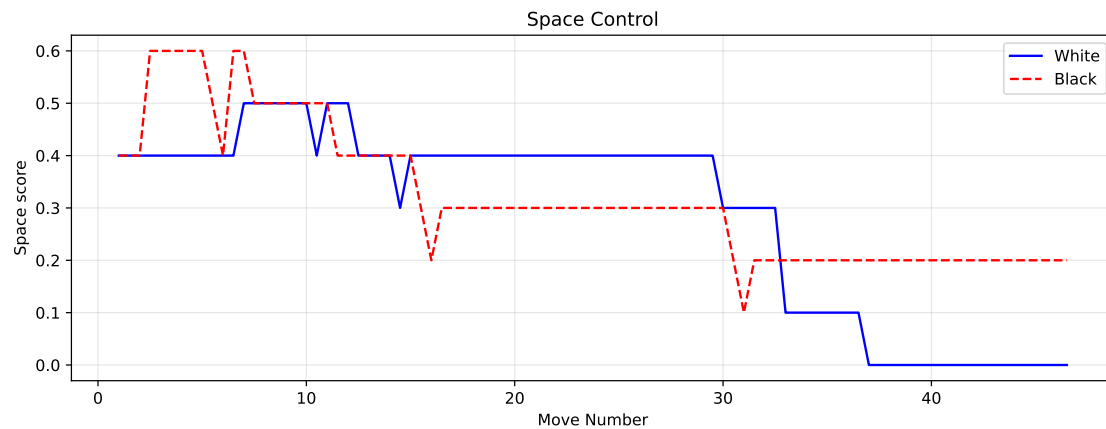
**Game Character Classification.** Evaluation range: minimum  $m_1 = -0.66$ , maximum  $m_2 = +5.54$ , spread  $d = m_2 - m_1 = 6.20$ .

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.



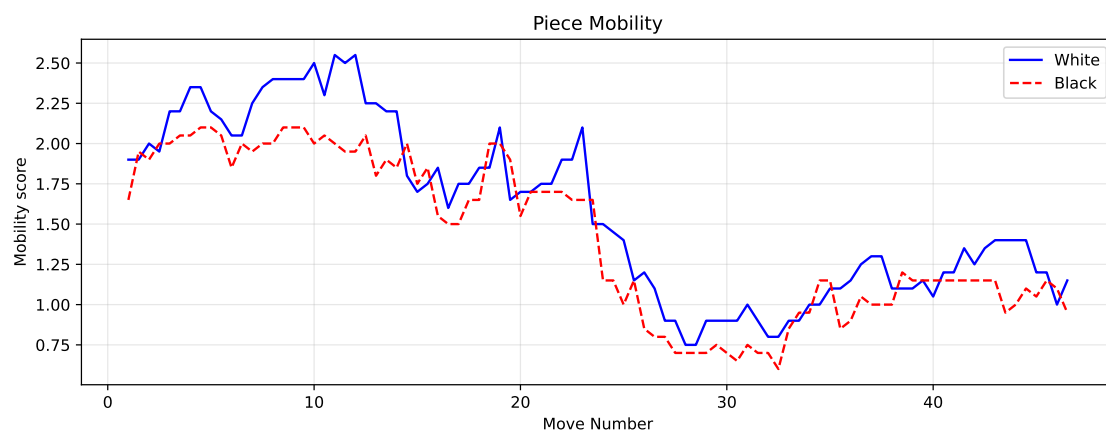
#### 14.3.2 Space Control

- White average: 0.29 (range: 0.00 to 0.50)
- Black average: 0.32 (range: 0.10 to 0.60)



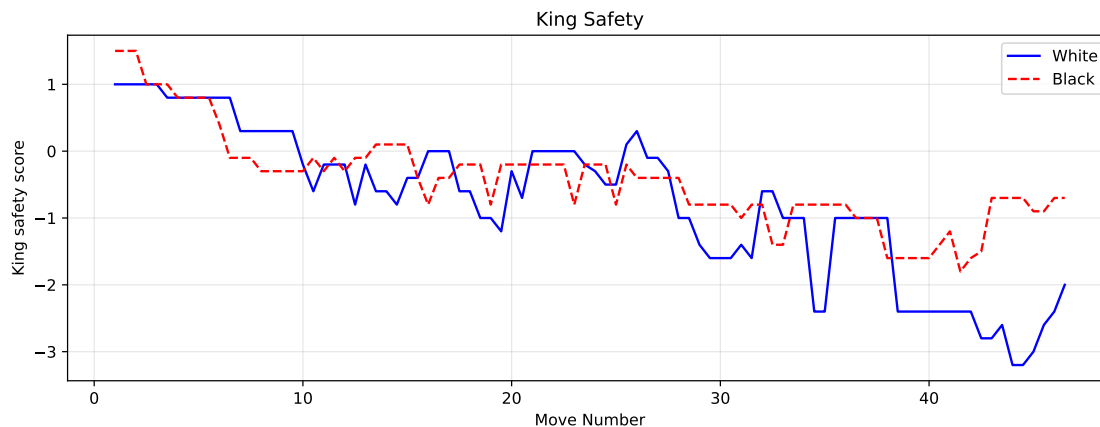
### 14.3.3 Mobility (Piece Activity)

- White average: 1.59 (range: 0.75 to 2.55)
- Black average: 1.42 (range: 0.60 to 2.10)



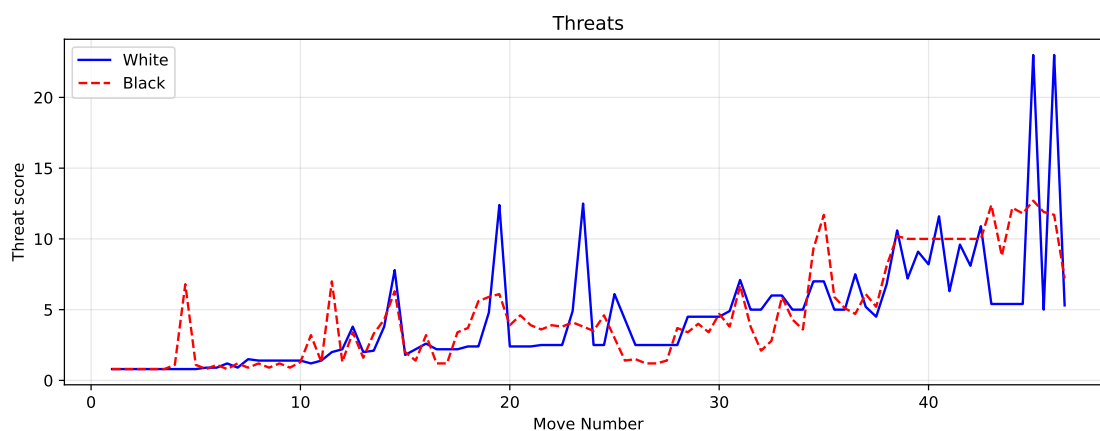
### 14.3.4 King Safety

- White average: -0.76
- Black average: -0.43



### 14.3.5 Threats

- White average: 4.46 (range: 0.80 to 23.00)
- Black average: 4.63 (range: 0.80 to 12.70)
- Average advantage: -0.17 (Black maintained more threats)



## 14.4 Annotated Game

1. d4 Nf6 2. c4 e6 3. Nc3 Bb4 4. Bg5?! h6 5. Bh4 c5 6. d5 d6 7. e3 e5? *Black loses 80cp. Consider instead: g5, Bxc3+, Qe7.*

8. Qc2? *White loses 89cp. Consider instead: Ne2.*

8...Nbd7?! 9. Bd3?! Qe7? *Black loses 62cp. Consider instead: Bxc3+, g5.*

10. f3 e4?! 11. Bxe4 g5 12. Bg3! Nxe4 13. fxe4! Ne5 14. a3 Bxc3+! 15. Qxc3! f6! 16. Bxe5 fxe5! 17. Ne2 O-O! 18. Ng3 Bg4 19. Rf1 Rxf1+ 20. Kxf1 Rf8+! 21. Kg1 h5! 22. Qc2 Qh7 23. Rf1! Rxf1+! 24. Kxf1?! h4 25. Nf5! Bxf5! 26. exf5! g4?! 27. Qf2? *White loses 106cp. Consider instead: b4, Ke2, Ke1.*

27...Qh6? *Black loses 84cp. Consider instead: Kf7, Qh5, Kg7.*

28. Ke2? *White loses 73cp. Consider instead: h3.*



28...Kf7! 29. Kd3 Qg5! 30. b4 b6 31. bxc5! bxc5! 32. Kc2! Kf6! 33. e4! Kf7?? *Black loses 195cp. Consider instead: g3.*

34. Kb3 Qc1?! 35. Qe2! Qf4? *Black loses 62cp. Consider instead: Qb1+, Qf4, Ke8.*

36. h3?! gxh3 37. gxh3! Ke7 38. Qh5?? *White loses 143cp. Consider instead: Ka4.*

38...Qc1! 39. Qxh4+? *White loses 116cp. Consider instead: Qe2.*

39...Kd7! 40. Qh7+ Kd8?! 41. Qg8+ Kc7?! 42. Qg7+! Kb6 43. Qg2 Qb1+ 44. Kc3 Qc1+?! 45. Qc2! Qe3+! 46. Qd3! Qf4?? *Black loses 156cp. Consider instead: Qe1+.*

1-0

## 14.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

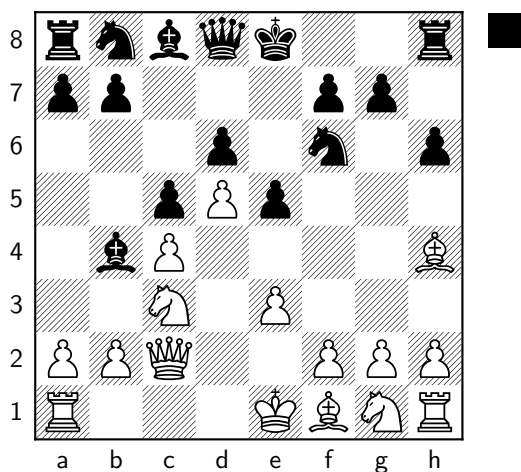
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 8. Qc2 ★

*Biggest evaluation swing(s) (89cp)*

Evaluation: -0.66



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.50  | 0.50  |
| Mobility (MG) | 4.80  | 4.00  |
| King Safety   | 0.30  | -0.30 |
| Threats       | 1.40  | 1.20  |

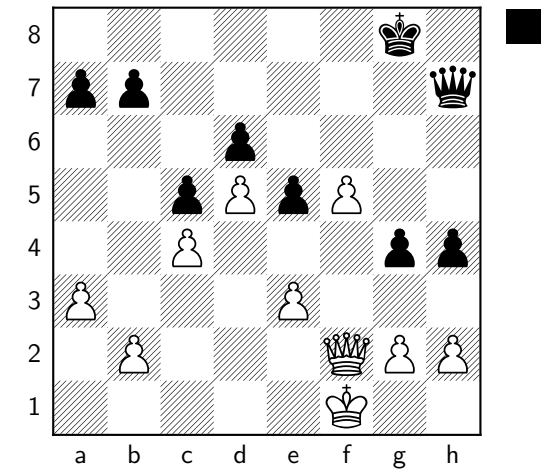
Instead of Qc2: Ne2

Best continuation: Bxc3+ bxc3 g5

Position 2: After 27. Qf2 ★

Biggest evaluation swing(s) (80cp)

Evaluation: +0.27



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.40  | 0.30  |
| Mobility (MG) | 1.80  | 1.60  |
| King Safety   | -0.10 | -0.40 |
| Threats       | 2.50  | 1.20  |

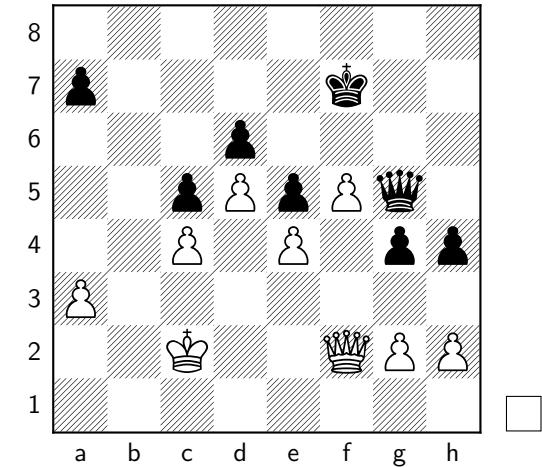
Instead of Qf2: b4, Ke2, Ke1

Best continuation: Qh5 Ke1 Qg5

Position 3: After 33...Kf7

Black blunders (+195cp)

Evaluation: +1.95



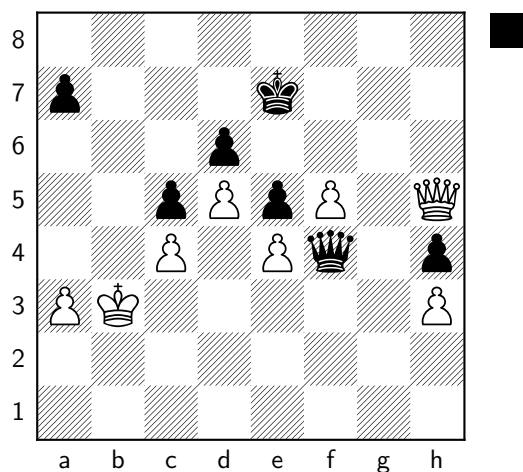
| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.10  | 0.20  |
| Mobility (MG) | 1.80  | 1.90  |
| King Safety   | -1.00 | -0.80 |
| Threats       | 5.00  | 4.30  |

Instead of Kf7: g3                      Best continuation: Kb3 Qd8 Qe2

#### Position 4: After 38. Qh5

*White blunders (-130cp)*

Evaluation: +2.85



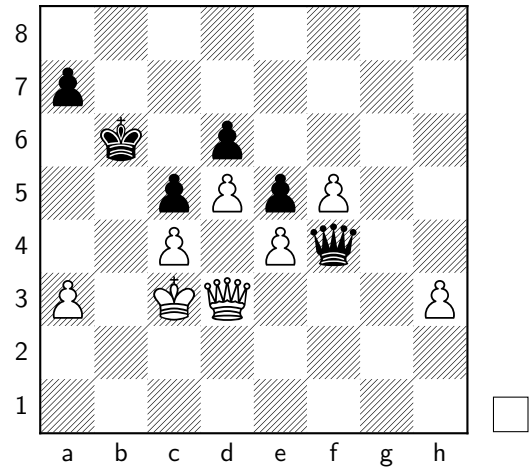
| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.00  | 0.20  |
| Mobility (MG) | 2.20  | 2.00  |
| King Safety   | -1.00 | -1.60 |
| Threats       | 6.80  | 8.10  |

Instead of Qh5: Ka4                      Best continuation: Qc1 Qe2 Qa1

#### Position 5: After 46...Qf4

*Black blunders (+149cp)*

Evaluation: +5.54



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.00  | 0.20  |
| Mobility (MG) | 2.30  | 1.90  |
| King Safety   | -2.00 | -0.70 |
| Threats       | 5.30  | 7.20  |

Instead of Qf4:    Qe1+                      Best continuation: Qb1+ Kc7 Kb3

14.6    Fireteam Index Prediction

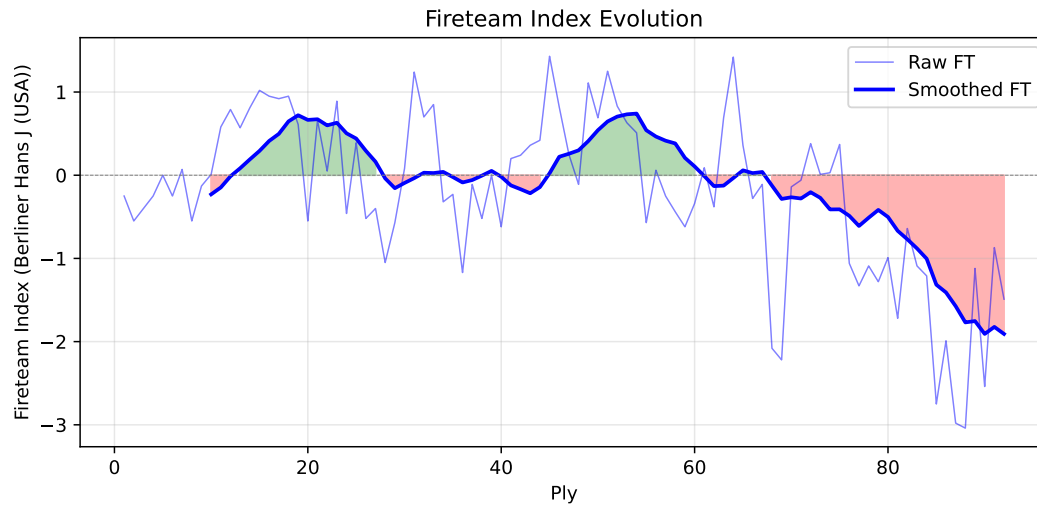
Tracking: **Berliner Hans J (USA)** (White)

| FTI Variant | Algorithm | Prediction | Max Streak |
|-------------|-----------|------------|------------|
| Standard    | Per-ply   | DRAW       | 7 plies    |
| Standard    | Windowed  | WIN        | 16 plies   |
| Truncated   | Per-ply   | DRAW       | 6 plies    |
| Truncated   | Windowed  | WIN        | 24 plies   |

Actual result: 1-0 (Berliner Hans J (USA)’s outcome: WIN)

Standard: Per-ply ✗    Windowed ✓

Truncated: Per-ply ✗    Windowed ✓



## 14.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 2997.9 seconds



# Chapter 15

## Werner Stern vs Hans Berliner

### 15.1 Game Information

|                |                                       |
|----------------|---------------------------------------|
| <b>Event</b>   | 5th Correspondence World Championship |
| <b>Site</b>    | ICCF                                  |
| <b>Date</b>    | 1965.07.01                            |
| <b>Round</b>   | 11                                    |
| <b>White</b>   | Werner Stern                          |
| <b>Black</b>   | Hans Berliner                         |
| <b>Result</b>  | 0-1                                   |
| <b>Opening</b> | B04 – Unknown                         |

### 15.2 Player Statistics

#### 15.2.1 Werner Stern (White)

- Total moves: 20
- Accuracy: 79.5%
- Average centipawn loss: 45.8
- Best/Excellent moves: 2 / 5
- Good moves: 5
- Inaccuracies: 2
- Mistakes: 3
- Blunders: 3

#### 15.2.2 Hans Berliner (Black)

- Total moves: 20
- Accuracy: 94.9%
- Average centipawn loss: 10.4

- Best/Excellent moves: 9 / 3
- Good moves: 7
- Inaccuracies: 0
- Mistakes: 1
- Blunders: 0

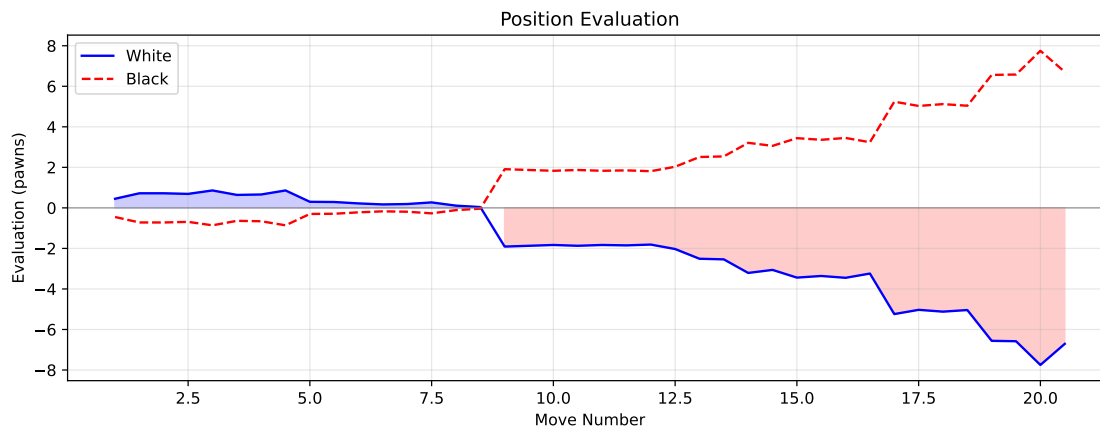
## 15.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 15.3.1 Position Evaluation Over Time

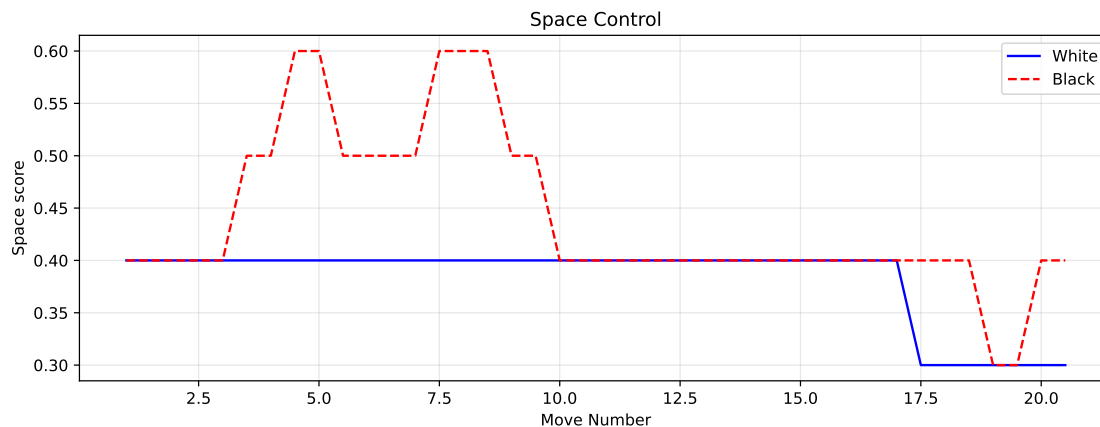
**Game Character Classification.** Evaluation range: minimum  $m_1 = -7.75$ , maximum  $m_2 = +0.86$ , spread  $d = m_2 - m_1 = 8.61$ .

This game is classified as **chaotic** (normal). A chaotic game with moderate swings.



### 15.3.2 Space Control

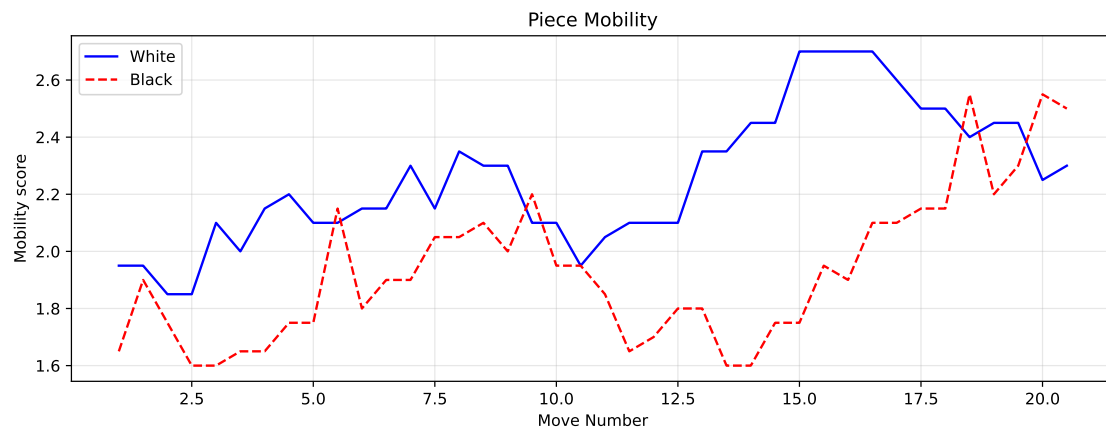
- White average: 0.38 (range: 0.30 to 0.40)
- Black average: 0.44 (range: 0.30 to 0.60)





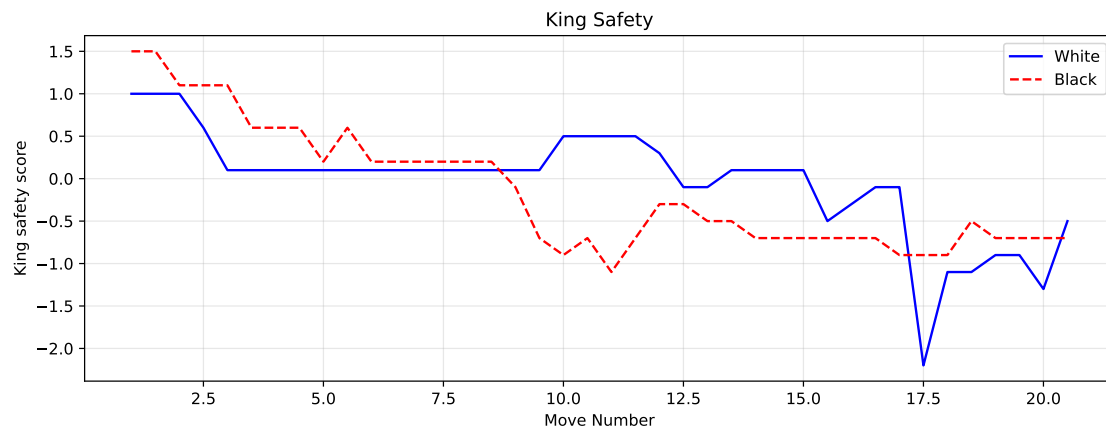
### 15.3.3 Mobility (Piece Activity)

- White average: 2.26 (range: 1.85 to 2.70)
- Black average: 1.93 (range: 1.60 to 2.55)



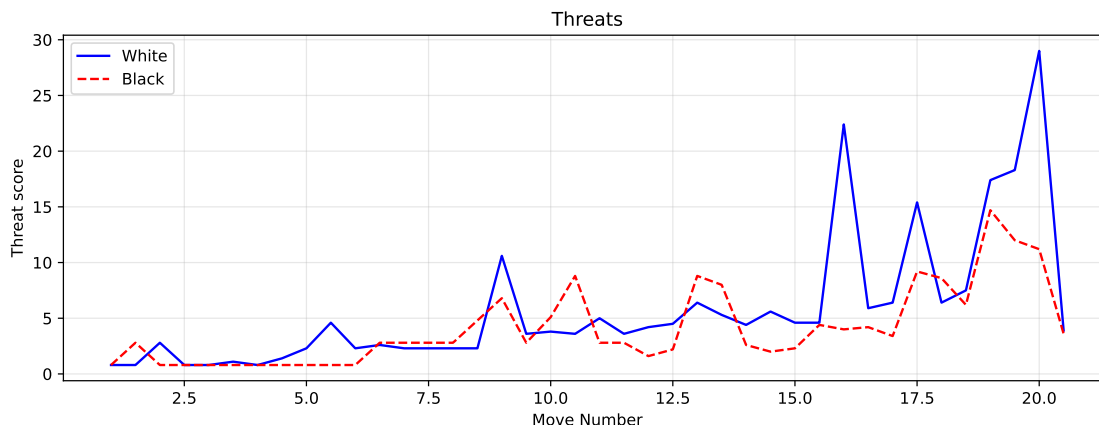
### 15.3.4 King Safety

- White average: -0.04
- Black average: -0.15



### 15.3.5 Threats

- White average: 5.82 (range: 0.80 to 29.00)
- Black average: 4.12 (range: 0.80 to 14.70)
- Average advantage: 1.69 (White maintained more threats)



## 15.4 Annotated Game

1. e4 Nf6 2. e5 Nd5 3. d4 d6 4. Nf3 g6 5. Ng5?! dxe5 6. dxe5 Bg7 7. Bc4 c6! 8. Nc3 h6! 9. Nxf7?? *White loses 199cp. Consider instead: Nge4, Nf3.*

9...Kxf7 10. Nxd5! cxd5! 11. Bxd5+! Ke8 12. e6 Rf8! 13. Qg4? *White loses 66cp. Consider instead: O-O, c4.*

13...Rf6! 14. Qc4? *White loses 61cp. Consider instead: Qa4+, Qe4, Qc4.*

14...Na6! 15. Be3?! Qa5+ 16. Bd2 Qb6 17. Bc3?? *White loses 210cp. Consider instead: O-O, O-O-O.*

17...Qxf2+ 18. Kd1 Bxe6! 19. Bxf6?? *White loses 135cp. Consider instead: Bxe6, Qb5+.*

19...Rd8! 20. Bh4? *White loses 114cp. Consider instead: Bd4, Qb5+.*

20...Qb6? *Black loses 109cp. Consider instead: Qe3.*

0-1

## 15.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

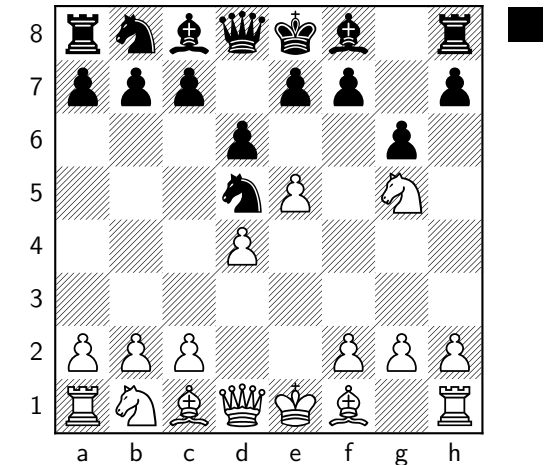
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 5. Ng5 ★

*Biggest evaluation swing(s) (56cp)*

Evaluation: +0.30



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.40  | 0.60  |
| Mobility (MG) | 4.20  | 3.50  |
| King Safety   | 0.10  | 0.20  |
| Threats       | 2.30  | 0.80  |

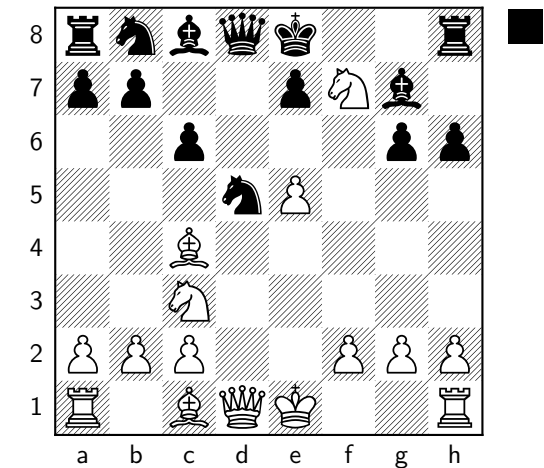
Instead of Ng5: c4, h3, Bc4

Best continuation: dxe5 Bc4 c6

Position 2: After 9. Nxf7

White blunders (-195cp)

Evaluation: -1.91



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.40  | 0.50  |
| Mobility (MG) | 4.60  | 4.00  |
| King Safety   | 0.10  | -0.10 |
| Threats       | 10.60 | 6.80  |

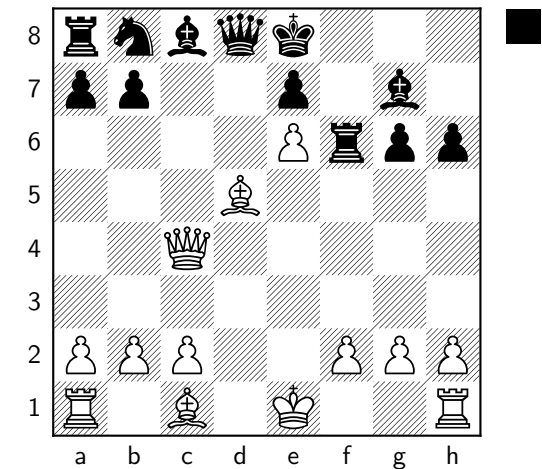
Instead of Nxf7: Nge4, Nf3

Best continuation: Kxf7 Nxd5 cxd5

Position 3: After 14. Qc4 ★

*Biggest evaluation swing(s) (67cp)*

Evaluation: -3.21



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.40  | 0.40  |
| Mobility (MG) | 4.90  | 3.20  |
| King Safety   | 0.10  | -0.70 |
| Threats       | 4.40  | 2.60  |

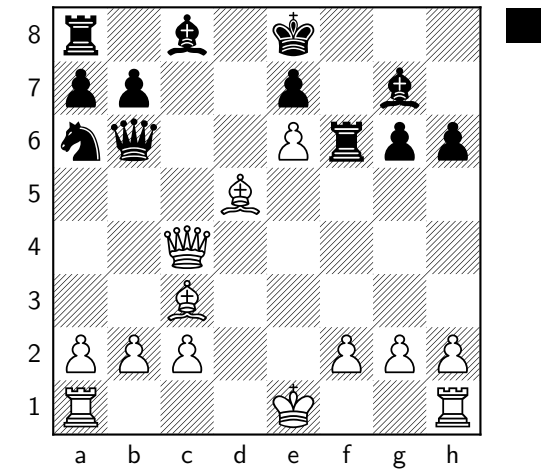
Instead of Qc4: Qa4+, Qe4

Best continuation: Na6 Qa4+ Kf8

Position 4: After 17. Bc3

*White blunders (-200cp)*

Evaluation: -5.24



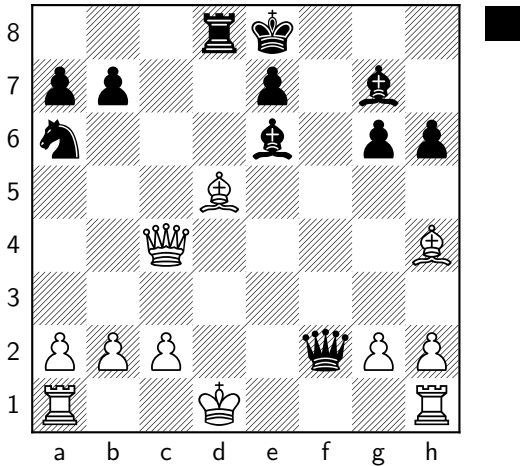
| Metric          | White      | Black |
|-----------------|------------|-------|
| Space           | 0.40       | 0.40  |
| Mobility (MG)   | 5.20       | 4.20  |
| King Safety     | -0.10      | -0.90 |
| Threats         | 6.40       | 3.40  |
| Instead of Bc3: | O-O, O-O-O |       |

Best continuation: Qxf2+ Kd1 Bxe6

Position 5: After 20. Bh4

Critical moment

Evaluation: -7.75



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.30  | 0.40  |
| Mobility (MG) | 4.50  | 5.10  |
| King Safety   | -1.30 | -0.70 |
| Threats       | 29.00 | 11.20 |

Instead of Bh4: Bd4, Qb5+

Best continuation: Qe3 Qb5+ Rd7

15.6 Fireteam Index Prediction

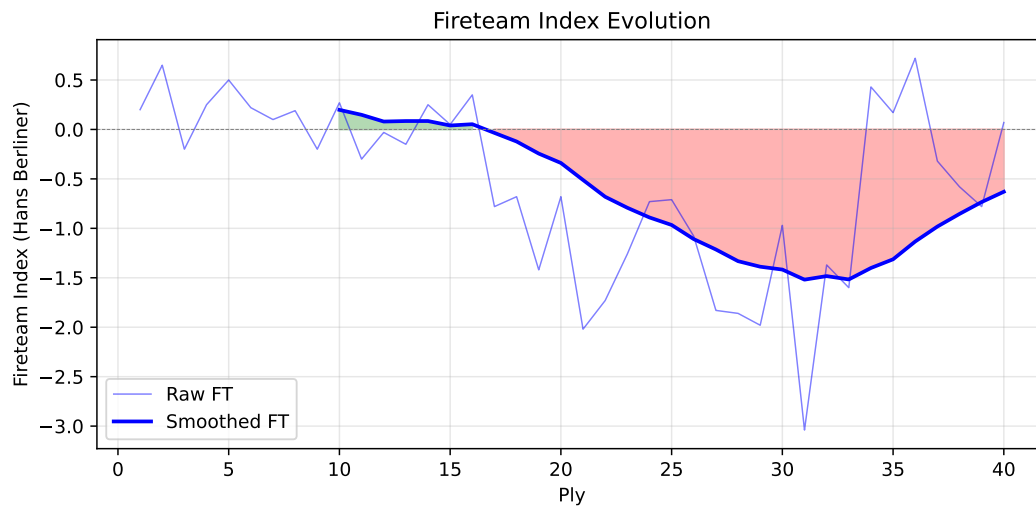
Tracking: Hans Berliner (Black)

| FTI Variant | Algorithm | Prediction | Max Streak |
|-------------|-----------|------------|------------|
| Standard    | Per-ply   | DRAW       | 3 plies    |
| Standard    | Windowed  | DRAW       | 0 plies    |
| Truncated   | Per-ply   | DRAW       | 2 plies    |
| Truncated   | Windowed  | DRAW       | 4 plies    |

Actual result: 0-1 (Hans Berliner's outcome: WIN)

Standard: Per-ply ✗ Windowed ✗

Truncated: Per-ply ✗ Windowed ✗



## 15.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 3016.7 seconds

## Chapter 16

# Hans Berliner vs Vladimir Zagorovsky

### 16.1 Game Information

|                |                                       |
|----------------|---------------------------------------|
| <b>Event</b>   | 5th Correspondence World Championship |
| <b>Site</b>    | ICCF                                  |
| <b>Date</b>    | 1965.07.01                            |
| <b>Round</b>   | 3                                     |
| <b>White</b>   | Hans Berliner                         |
| <b>Black</b>   | Vladimir Zagorovsky                   |
| <b>Result</b>  | 1/2-1/2                               |
| <b>Opening</b> | D35 – Unknown                         |

### 16.2 Player Statistics

#### 16.2.1 Hans Berliner (White)

- Total moves: 38
- Accuracy: 93.8%
- Average centipawn loss: 12.7
- Best/Excellent moves: 14 / 11
- Good moves: 8
- Inaccuracies: 4
- Mistakes: 1
- Blunders: 0

#### 16.2.2 Vladimir Zagorovsky (Black)

- Total moves: 38
- Accuracy: 94.0%
- Average centipawn loss: 12.3

- Best/Excellent moves: 19 / 5
- Good moves: 10
- Inaccuracies: 3
- Mistakes: 0
- Blunders: 1

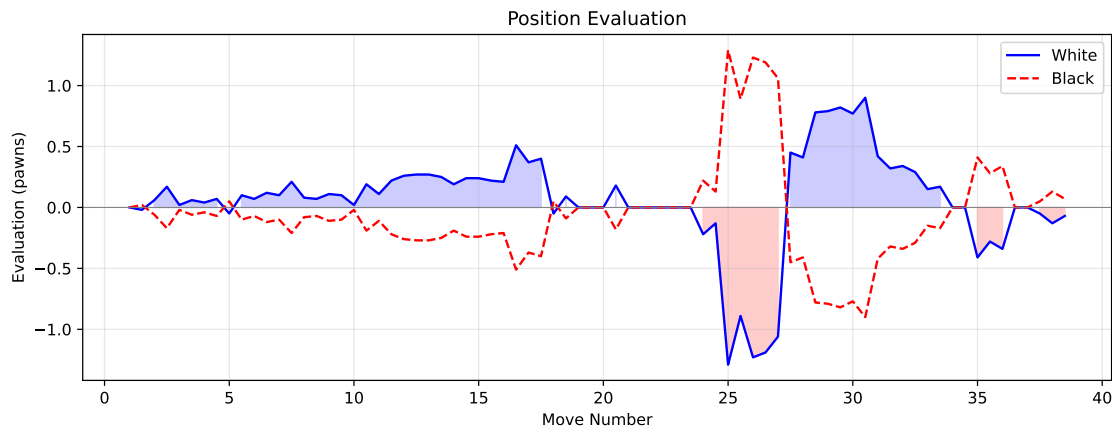
## 16.3 Positional Analysis

This section shows how key positional factors evolved throughout the game.

### 16.3.1 Position Evaluation Over Time

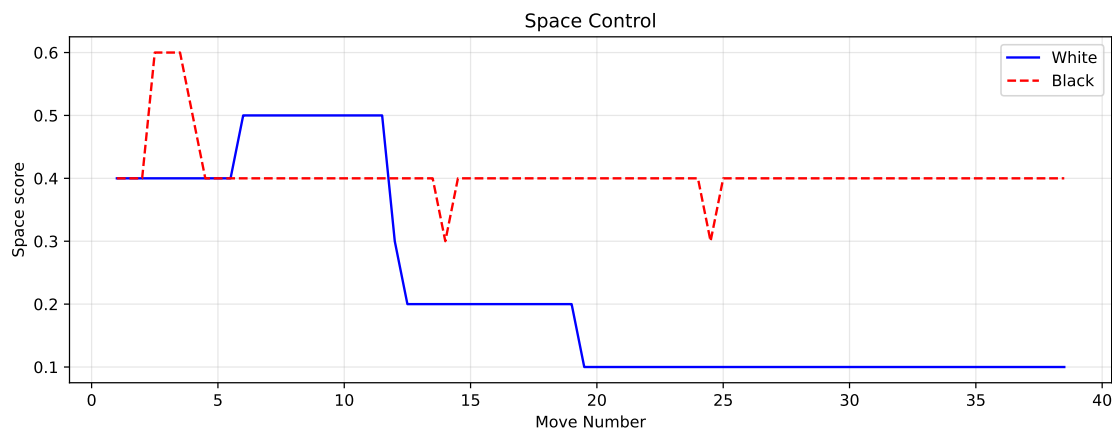
**Game Character Classification.** Evaluation range: minimum  $m_1 = -1.29$ , maximum  $m_2 = +0.90$ , spread  $d = m_2 - m_1 = 2.19$ .

This game is classified as **tense** (normal). A tense game with moderate swings.



### 16.3.2 Space Control

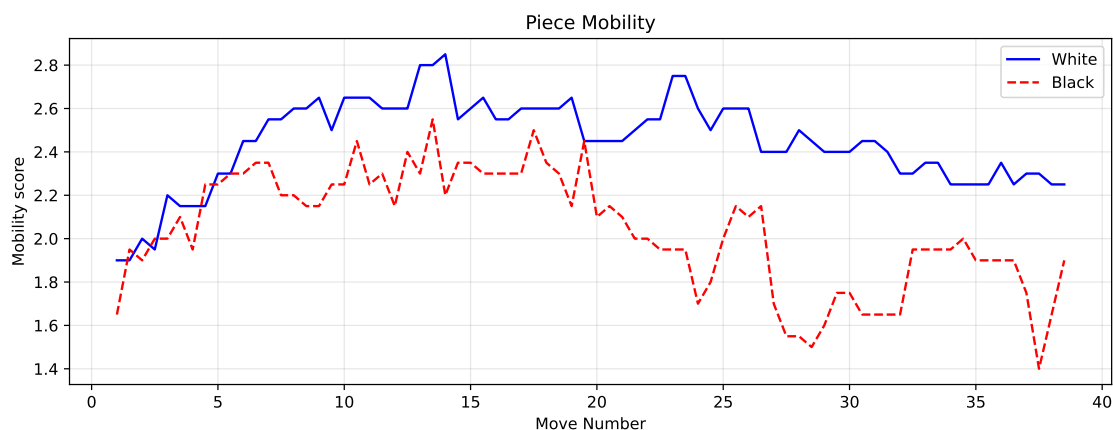
- White average: 0.22 (range: 0.10 to 0.50)
- Black average: 0.41 (range: 0.30 to 0.60)





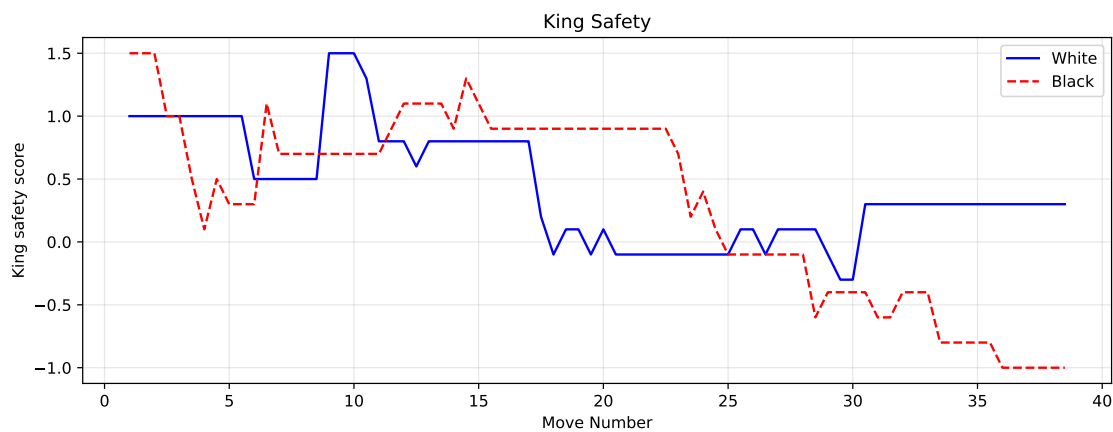
### 16.3.3 Mobility (Piece Activity)

- White average: 2.45 (range: 1.90 to 2.85)
- Black average: 2.04 (range: 1.40 to 2.55)



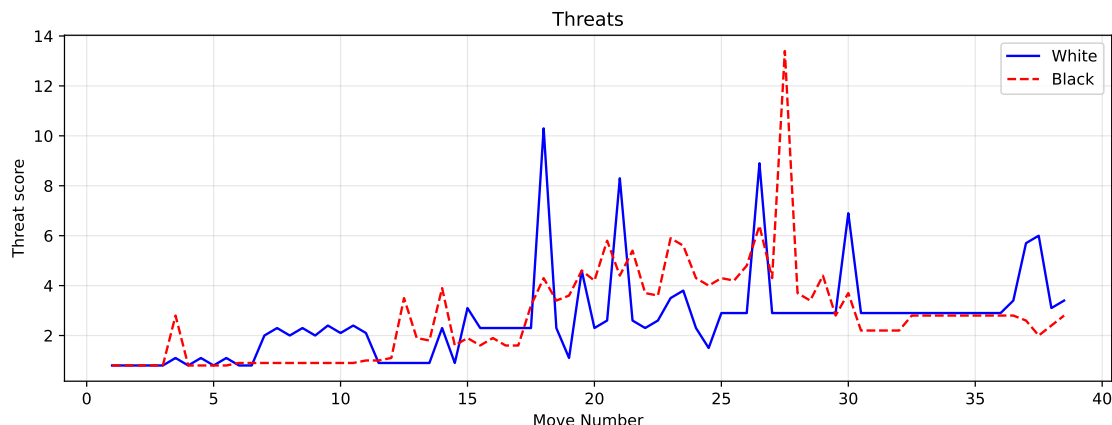
### 16.3.4 King Safety

- White average: 0.44
- Black average: 0.32



### 16.3.5 Threats

- White average: 2.64 (range: 0.80 to 10.30)
- Black average: 2.74 (range: 0.80 to 13.40)
- Average advantage: -0.10 (Black maintained more threats)



## 16.4 Annotated Game

1. d4 Nf6 2. c4 e6 3. Nc3 d5 4. cxd5 exd5 5. Bg5 Be7 6. e3 O-O 7. Bd3! Nbd7 8. Nge2 Re8 9. O-O! c6! 10. Qc2 Nf8 11. f3 Ng6 12. e4! dxe4! 13. fxe4 Be6! 14. Bxf6! Bxf6! 15. e5! Be7! 16. Rad1! Bf8?! 17. Kh1 Qh4 18. g3?! Qh6 19. Nf4 Nxf4! 20. gxf4! Bg4 21. Be2! Bh3! 22. Rf3! Rad8! 23. Rg3 Kh8! 24. f5 g6! 25. f6? *White loses 113cp. Consider instead: Bg4, fxg6.*

25...Be6?! 26. Bg4?! Bxg4! 27. Rxxg4! Qh5?? *Black loses 149cp. Consider instead: c5.*

28. Qe2 h6?! 29. Re4! Qf5 30. Rf1 Qe6! 31. Qg2?! h5! 32. Qf3 Bh6 33. Rd1 Kh7 34. Reel Rd7 35. b3?! Red8! 36. Ne4! b6 37. Nd6! Bf8 38. Ne4 Bh6!

1/2-1/2

## 16.5 Critical Positions

This section highlights critical moments where the evaluation shifted substantially—either due to errors or missed opportunities. Each diagram shows the position *after* the move was played.

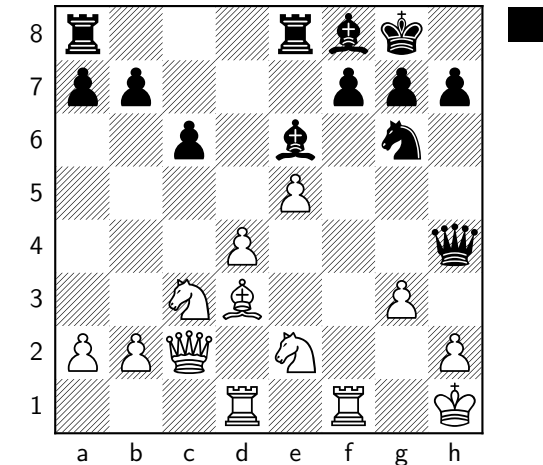
- **Instead of [move]:** The move(s) that should have been played instead.
- **Best continuation:** The optimal sequence of moves from the diagrammed position.

Positions marked with ★ represent the biggest evaluation swings in the game. The positional metrics table summarizes the spatial control, piece activity, king safety, and tactical threats for each side.

### Position 1: After 18. g3 ★

*Biggest evaluation swing(s) (45cp)*

Evaluation: -0.05



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.20  | 0.40  |
| Mobility (MG) | 5.20  | 4.70  |
| King Safety   | -0.10 | 0.90  |
| Threats       | 10.30 | 4.30  |

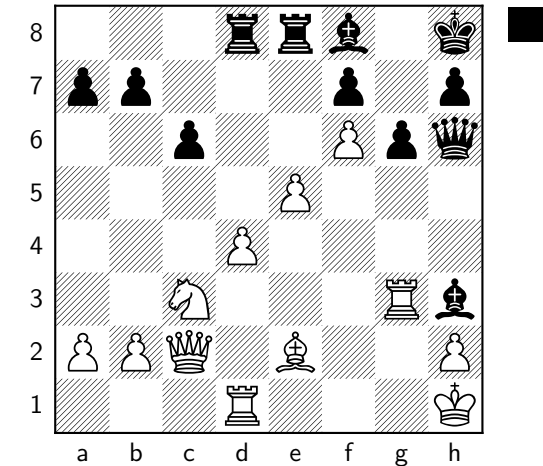
Instead of g3: Bf5, Rde1

Best continuation: Qh5 Bf5 Rad8

Position 2: After 25. f6

Critical moment

Evaluation: -1.29



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.10  | 0.40  |
| Mobility (MG) | 5.20  | 4.00  |
| King Safety   | -0.10 | -0.10 |
| Threats       | 2.90  | 4.30  |

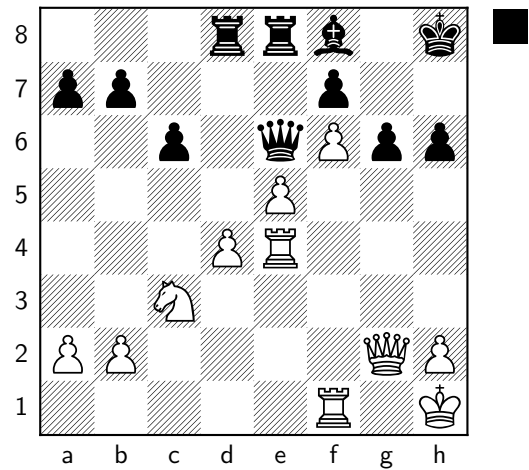
Instead of f6: Bg4, fxg6

Best continuation: Bf5 Bd3 Be6

Position 3: After 31. Qg2 ★

Biggest evaluation swing(s) (48cp)

Evaluation: +0.42



| Metric        | White | Black |
|---------------|-------|-------|
| Space         | 0.10  | 0.40  |
| Mobility (MG) | 4.90  | 3.30  |
| King Safety   | 0.30  | -0.60 |
| Threats       | 2.90  | 2.20  |

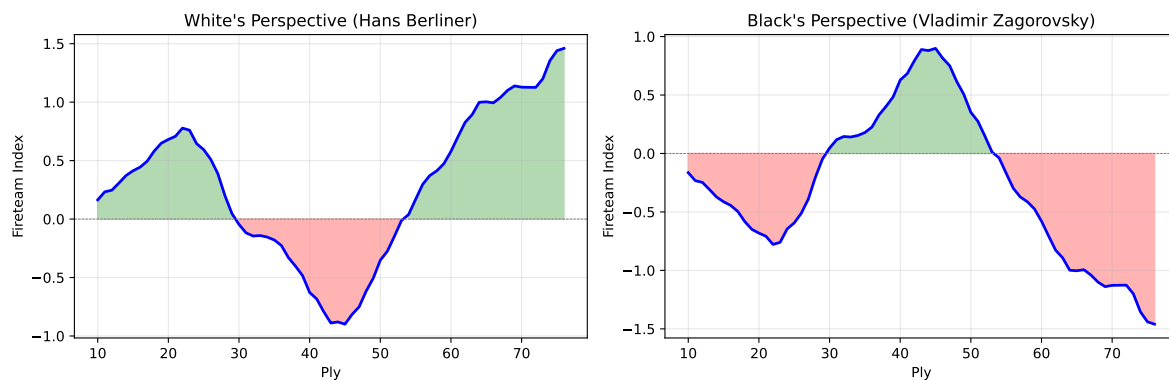
Instead of Qg2: Rf3, Qe3, Rg1      Best continuation: h5 Qf3 Bh6

16.6 Fireteam Index Prediction

*This game ended in a draw. The Fireteam Index is analyzed from both players' perspectives to determine whether either side achieved sustained positional dominance that might have predicted a win.*

| Standard FTI (1.0, 1.0, 1.0, 0.1)  | Player              | Algorithm | Prediction | Max Streak |
|------------------------------------|---------------------|-----------|------------|------------|
|                                    | Hans Berliner       | Per-ply   | WIN        | 22 plies   |
|                                    | Hans Berliner       | Windowed  | WIN        | 23 plies   |
|                                    | Vladimir Zagorovsky | Per-ply   | WIN        | 12 plies   |
|                                    | Vladimir Zagorovsky | Windowed  | WIN        | 24 plies   |
|                                    |                     |           |            |            |
| Truncated FTI (1.2, 0.0, 0.0, 0.6) | Player              | Algorithm | Prediction | Max Streak |
|                                    | Hans Berliner       | Per-ply   | DRAW       | 6 plies    |
|                                    | Hans Berliner       | Windowed  | WIN        | 10 plies   |
|                                    | Vladimir Zagorovsky | Per-ply   | WIN        | 10 plies   |
|                                    | Vladimir Zagorovsky | Windowed  | WIN        | 28 plies   |

**Interpretation.** Both players achieved sustained positional dominance at different points in the game. The FT Index predicted wins for both sides, reflecting a hard-fought battle where momentum shifted. The draw is a reasonable outcome given the mutual winning chances.



## 16.7 Analysis Information

- Engine: Stockfish 17.1
- Depth: 23
- Analysis time: 714.0 seconds



## Chapter 17

# Fireteam Index Prediction Summary

This chapter summarizes the accuracy of the Fireteam Index prediction algorithms across all games in this collection.

### 17.1 Overall Results

| Metric                      | Value |
|-----------------------------|-------|
| Total games                 | 16    |
| Decisive games              | 12    |
| Draws (both sides analyzed) | 4     |
| Games with prediction       | 16    |

#### 17.1.1 Standard FTI (Weights: 1.0, 1.0, 1.0, 0.1)

| Algorithm           | Correct | Accuracy |
|---------------------|---------|----------|
| Per-ply (raw)       | 8/16    | 50.0%    |
| Windowed (smoothed) | 11/16   | 68.8%    |

#### 17.1.2 Truncated FTI (Weights: 1.2, 0.0, 0.0, 0.6)

The truncated FTI uses only Space and Threats components, testing whether a simpler metric focusing on territorial control and tactical pressure is more predictive.

| Algorithm           | Correct | Accuracy |
|---------------------|---------|----------|
| Per-ply (raw)       | 4/16    | 25.0%    |
| Windowed (smoothed) | 11/16   | 68.8%    |

#### 17.1.3 Comparison

Both FTI variants show similar predictive accuracy (standard: 68.8%, truncated: 68.8%), suggesting that the additional Mobility and King Safety components provide marginal benefit.

## 17.2 Interpretation

These predictions tracked the eventual winner in each decisive game. A “WIN” prediction means the Fireteam Index exceeded the threshold for the required number of consecutive plies, indicating sustained positional dominance.

The per-ply algorithm uses raw positional deltas, while the windowed algorithm smooths values over a 10-ply (5 full move) rolling average to reduce noise.



## Appendix A

# Methodology: How Metrics Are Computed

This appendix explains how the positional metrics are computed.

This section explains how Stockfish’s classical evaluation computes each positional factor. Note that modern Stockfish primarily uses NNUE (neural network) evaluation, but the classical evaluation terms provide interpretable metrics for educational purposes.

### A.1 Space

Space evaluation counts squares in the center (files c-f, ranks 2-4 for White, 5-7 for Black) that are:

1. Attacked by at least one friendly pawn, AND
2. Not occupied by any enemy piece

The computation is weighted by the number of pieces behind the pawn chain—this rewards positions where you control territory AND have pieces to exploit that space. The space term is scaled by total piece count; it matters more in closed positions with many pieces.

### A.2 Mobility

Mobility counts legal moves for each piece type, with different weights:

- **Knights:** Safe squares attacked (excluding squares defended by enemy pawns)
- **Bishops:** Safe squares on diagonals (bonus for long, unobstructed diagonals)
- **Rooks:** Squares on files/ranks (bonus for open files, 7th rank penetration)
- **Queens:** Combination of bishop and rook mobility patterns

“Safe squares” exclude those defended by enemy pawns or where exchanges would lose material. Mobility in the opponent’s territory is valued higher than mobility in your own.

### A.3 King Safety

King safety combines several factors:

- **Pawn shield:** Pawns on 2nd/3rd rank directly in front of the castled king
- **King tropism:** Distance of enemy pieces to your king (closer = more dangerous)
- **Attack units:** Count of enemy pieces attacking squares adjacent to your king
- **Safe checks:** Number of safe checking squares available to the opponent

### A.4 Threats

Threat evaluation considers:

- Hanging pieces (attacked but not adequately defended)
- Safe pawn advances that create threats
- Minor pieces attacking major pieces
- Weak squares in the pawn structure

### A.5 Middlegame vs Endgame (MG/EG)

Each term has separate middlegame (MG) and endgame (EG) values. Stockfish blends these based on remaining material (the “phase”). With all pieces on the board, MG dominates; as pieces trade, EG values become more important. For example, space matters greatly in the middlegame but is nearly irrelevant in endgames.

### A.6 Game Character Classification

Games are classified based on evaluation volatility and directionality. Let  $m_2$  be the maximum evaluation for White during the game (in pawns) and  $m_1$  be the minimum. The *spread*  $d = m_2 - m_1$  measures the total evaluation range.

**Spread Classifications.** Based on the magnitude of  $d$ :

- **Balanced** ( $d < 1$ ): Minimal advantage shifts; controlled, technical play with neither side gaining a significant edge.
- **Tense** ( $1 \leq d < 3$ ): Normal competitive tension; typical of well-played games where small advantages are contested.
- **Tactical** ( $3 \leq d < 6$ ): Significant evaluation swings; tactical complications, missed opportunities, or speculative play.
- **Chaotic** ( $d \geq 6$ ): Wild swings in evaluation; likely includes major blunders, speculative sacrifices, or complex tactical melees.

**Directionality Classifications.** Based on whether the advantage changed hands:

- **One-sided:** Either  $m_1 > -\frac{1}{2}$  (White always comfortable) or  $m_2 < \frac{1}{2}$  (Black always comfortable). The leading side maintained control throughout.
- **Seesaw:** Both  $m_1 < -1$  and  $m_2 > 1$ . The advantage genuinely changed hands, with each side holding a significant advantage at some point.
- **Normal:** Neither one-sided nor seesaw; moderate swings without complete reversals.

These classifications combine to describe the overall game character. For example, a “tactical seesaw battle” indicates significant evaluation swings ( $3 \leq d < 6$ ) with the advantage changing hands multiple times.

## A.7 Fireteam Index (FTI) and Win Prediction

The Fireteam Index is a composite metric inspired by the ancient mathematical board game *Rithmo-machia*, where a “progression victory” is achieved when a coordinated group of pieces (a “fireteam”) establishes control deep in enemy territory. Similarly, in chess, sustained positional dominance across multiple factors often precedes victory.

**Standard FTI Formula.** The standard Fireteam Index combines all four positional metrics into a single value measuring overall positional advantage from a given player’s perspective:

$$FT_{\text{std}} = 1.0 \cdot \Delta\text{Space} + 1.0 \cdot \Delta\text{Mobility} + 1.0 \cdot \Delta\text{King Safety} + 0.1 \cdot \Delta\text{Threats} \quad (\text{A.1})$$

where each  $\Delta$  represents (Player’s value – Opponent’s value). Positive FT indicates the player has overall positional superiority; negative indicates the opponent is better.

The Threats component is weighted at 0.1 because its raw values are typically an order of magnitude larger than the other metrics, and empirical testing showed this weighting provides better predictive accuracy.

**Truncated FTI Formula.** The truncated Fireteam Index uses only Space and Threats, testing whether a simpler metric focusing on territorial control and tactical pressure is more predictive for decisive games:

$$FT_{\text{trunc}} = 1.2 \cdot \Delta\text{Space} + 0.6 \cdot \Delta\text{Threats} \quad (\text{A.2})$$

This variant hypothesizes that Mobility and King Safety may introduce noise that obscures the core signal of who controls territory and applies pressure. The increased weight on Space (1.2 vs 1.0) and Threats (0.6 vs 0.1) compensates for the removed components.

**Prediction Algorithms.** Two algorithms are used to predict game outcomes based on FT (applied independently to both the standard and truncated variants):

1. **Per-Ply (Raw):** Examines the raw FT value at each ply after the opening phase (ply 16, approximately 8 moves). If FT exceeds zero for 10 consecutive plies (5 full moves), a WIN is predicted for that player.

2. **Windowed (Smoothed):** Computes a 10-ply rolling average of FT before applying the same streak detection. This smooths out tactical noise and captures sustained trends rather than momentary fluctuations.

Both algorithms predict DRAW if no player achieves the required streak of positional dominance.

**Interpretation.** The FT prediction measures *positional* dominance, which is a necessary but not sufficient condition for winning. A player may dominate positionally yet fail to convert due to:

- Tactical errors (blunders, missed combinations)
- Time pressure in rapid/blitz games
- Tenacious defense by the opponent
- Drawing tendencies in certain structures (opposite-colored bishops, fortresses)

Conversely, a player may win without ever achieving sustained FT dominance through a single tactical strike or opponent error. The FT metric is most predictive in games where positional understanding—rather than pure calculation—determines the outcome.

**For Draws.** When analyzing drawn games, the FT is computed from both players' perspectives. This reveals whether:

- Neither side achieved dominance (a “true draw” with balanced play)
- One side dominated but failed to convert (a “missed win”)
- Both sides dominated at different points (a “mutual chances draw”)